

One Hundred Sixty-eight Candidates for Datasets 51–200

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The database holds 50 schematized and L0–L4-verified *Saccharomyces cerevisiae* datasets, of which 36 are served and 14 are built and waiting on the next full graph rebuild. Reaching 200 needs 150 more. Table 1 ranks 168 candidates and Table 2 gives each one’s citation, link and data location.

The list is ordered for a decision about the next build week rather than for the long-run total, so it carries two near lines as well as the arithmetic one. Rows 1–50 are the recommended next builds, rows 51–70 are the bench behind them, and rows 71–150 complete the set that reaches 200. Ranking puts the rows a planned yeast Perturb-seq is designed against first, then the rows that put two molecular layers on an axis where they can be joined, then scale (Sec. 1). Those two bands now fill wave 1 exactly, at 30 rows and 20. Table 5 names all 116 joins a candidate opens, against a supported dataset or against another candidate, with the shared axis that makes the pair addressable. Table 4 records every rank change against the previous pass. Table 7 records what was considered and dropped, which is where the one hard constraint shows up: a dataset earns a place only if the total genomic content of each strain can be reconstructed.

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1 The selection rule ✗

1.1 Sequence reconstruction is the gate, not a preference ✗

A phenotype is only trainable against a genotype that can be written down. Every row in Table 1 therefore names a **sequence basis**: the route by which the total genomic content of one strain would be reconstructed. The eleven bases in use are

- **S288C-KO** – the reference genome minus one cataloged ORF (BY4741, BY4742, BY4743). Thirty of the 150 that reach 200 sit here, the largest single group.
- **S288C-KO/het** – a heterozygous diploid deletion, so a dosage edit rather than a null.
- **S288C+guide** – the genome is unedited and the perturbation is a designed cassette plus a guide target. Regulatory, not sequence-level.
- **S288C+designed-edit** – a designed single-nucleotide or small-indel library. Designed, and not verified per strain.
- **S288C+tag** – the reference plus a designed tag, degron or promoter cassette.
- **S288C+ORF-plasmid** – the reference plus a barcoded ORF on a known plasmid.
- **S288C+reporter-locus** – the reference plus a designed regulatory sequence driving a reporter, which is what every massively parallel reporter assay perturbs.
- **isolate-WGS** – a published assembly or variant call set per isolate.
- **segregant-WGS** – genotype calls or whole-genome sequence per progeny.
- **engineered-chassis** – a named production strain plus heterologous cassettes.
- **reference-only** – wild type, where the environment carries the perturbation.

What fails the gate is a strain whose genome is unknown even in principle. Genome shuffling and adaptive laboratory evolution without resequencing both land there: the selected progeny differ from their parents at unmapped positions, so there is no genotype to map a phenotype from. Those are dropped in Table 7 rather than ranked low, since no amount of phenotypic value substitutes for a missing genotype.

The **S288C+designed-edit** and **S288C+guide** bases carry a weaker claim than the others and the distinction is worth holding. A designed variant library states what each member was built to be, not what it turned out to be, so its genotypes are designed rather than realized. Ingesting one commits to storing the design and a verification status, not a verified sequence.

1.2 Band comes before tier, and why ✗

Scale is the right default ordering and it is the wrong first question when a specific campaign is being planned. The largest row in this table, de Boer 2020 at 10^8 promoter sequences, tells a Perturb-seq design nothing; a 61,094-guide CRISPR interference library with a released per-guide matrix tells it what guides to order. Each row therefore carries a **band**, applied before tier and before scale:

- **perturb-seq**, 30 rows. A CRISPR interference or activation library whose per-guide matrix is released, a barcoded genotype set read out per cell, a designed single-gene perturbation crossed with a transcriptome, or the strain and noise measurements a pooled per-cell campaign depends on.
- **molecular layers**, 23 rows. Two or more of transcript, translation, protein, phosphosite, metabolite, flux and turnover, joinable on one axis, either inside one study or against a supported dataset. The join is per *strain* when both layers are measured on the same genotypes, and per *gene* when one layer is a single genome-wide coefficient such as a half-life.
- **scale**, 115 rows. Everything else, ordered as before.

The second band was **metabolism x expression** and is widened, because transcript against protein is the same join as expression against flux, and because the rows that say *why* transcript and protein disagree, translation efficiency and turnover, had no band to sit in at all. Its membership is unchanged apart from the rows added with it.

Within a band the order is tier, then measurements. The cost is visible and is not hidden: the two bands now total 53 rows, so they fill wave 1 and three of them spill into wave 2, and no scale row reaches wave 1 at all. de Boer 2020 falls from 2 to 54 and Lee 2014 from 4 to 56. Both are the largest things in the table by instance count and both are now second wave, which is the deliberate consequence of ordering by what a row combines with rather than by how many numbers it holds. A reader who wants scale back reads wave 2 first. Every move is listed in Table 4.

1.3 Tier, and what rows are ranked on ✗

Tier is assigned from stated bars rather than judgment:

- **Tier 1** clears all three of $\geq 10^3$ genotypes, $\geq 10^4$ instances, and a clean sequence basis, and lands in one of the nine classes.
- **Tier 2** clears two of the three, or is the only dataset covering a requested class, or directly de-risks the Perturb-seq proposal.
- **Tier 3** is a modality or reference backbone the substrate needs without itself carrying a large genotype axis.
- **Tier 4** is real but small, coarsely labeled, or expensive to extract.

The bars apply to what a row contains rather than to its reputation. Piotrowski 2017 has the largest compound library in yeast at 13,524 compounds and still fails Tier 1, because 157 strains is not a genotype axis. Turco 2023 has the widest environment axis at 7,536 and fails for a different reason: it is a meta-aggregation, so its instance count re-serves primary screens, several of them already built, and the net-new fraction is unknown until it is de-duplicated against the rest of this table.

Within a tier, rows sort by **measurements**, meaning instances times phenotype dimensionality. Instances alone is the wrong statistic and gets the order visibly wrong: Muenzner 2024 is 796 instances, which ranks it below a mid-sized fitness screen, but each instance is a proteome, so it is roughly 1.6×10^6 numbers. This is the same normalization the supported-datasets table already applies through its gzip-signal column.

1.4 Where the lines fall, and what they cost ✗

The long-run cut at 150 is arithmetic, $200 - 50$, not a quality threshold. Rows on both sides of it are close, so it should be read as a budget rather than a verdict.

The two nearer lines are what a build week is planned against. Wave 1 is rows 1–50 and wave 2 rows 51–70; a wave-2 row is promoted the moment a wave-1 row turns out to have no recoverable per-strain data, which has already happened once to Lee 2014. Wave 1 is 30 Perturb-seq rows and 20 molecular-layer rows, with no scale row in it, and wave 2 is three molecular-layer rows followed by scale, so the two waves differ in kind and not only in rank.

A dataset named explicitly in the scoping request is pinned into the recommended set even if it ranks below the cut, and the pin is reported rather than folded into the score, so the set stays auditable. No pin is currently in force: Jackson 2020, the one requested row, ranks 3 unaided.

Displacement, when a pin does bind, picks the weakest non-requested row by tier first and measurements second. The tier term is load-bearing rather than decorative: ranking on measurements alone selected Puddu 2019 for eviction, and that row is the whole-collection sequencing on which every **S288C-KO** sequence basis in this table rests. Banding has since moved Puddu to 4 on its own merits, because a pooled campaign reads a barcode and infers a genotype and this is the measurement of how often that inference is wrong.

2 The ranked list ✗

Table 1: Ranked candidates to take the database from 50 supported datasets to 200. *Genotypes* and *Env* are the perturbation and condition axes. *Instances* is the number of genotype $\times$ environment records, marked $\dagger$ where it is the product of the two axes rather than a figure the paper reports, and $\ddagger$ where it is an order-of-magnitude estimate. *Meas.* is instances times phenotype dimensionality, which is what a vector-valued panel actually contributes and what rows are ranked on. *Sequence basis* is how the total genomic content of one strain would be reconstructed; a row with no such route is excluded rather than ranked (Table 7). Tier is defined in Sec. 1. *Link* resolves to the source and is clickable; the full citation and the data location are in Table 2. A $\bullet$ marks a row bearing on the Perturb-seq proposal (Table 6). A superscript **B** marks a row already attempted and blocked on data access, and **L** one that already has a loader in flight; neither is an untouched candidate. Rows 1–150 are the recommended set.

#	Dataset	Class	Genotypes	Env	Instances	Meas.	Phenotype	Sequence basis	Link
1	Boocock 2025 (single-cell eQTL mapping) $\bullet$	Expression / single cell	393 segregants, 27,744 cells	glucose / galactose	$5.5 \times 10^4 \dagger$	3.3×10^8	single-cell transcriptome + eQTL	segregant-WGS	doi:10.7554/eLife.95566
2	Hale 2024 (CRISPRi x natural variation) $\bullet$	Natural variation	169 segregants x 1,721 genes	2 induced	$1.4 \times 10^6 \dagger$	1.4×10^6	relative fitness (double-barcode seq)	segregant-WGS	doi:10.1038/s41467-024-48626-1
3	Jackson 2020 (TF-deletion single-cell atlas) $\bullet$	Expression / single cell	12 genotypes (wild type + 11 TF deletions), 72 barcoded strains	11 growth conditions	132 $\dagger$	7.9×10^5	single-cell transcriptome	S288C-KO	doi:10.7554/eLife.51254
4	Puddu 2019 (WGS of the deletion collection)	Modality / backbone	4,800 deletion strains	1	4.8×10^3	4.8×10^3	structural variants, CNV, aneuploidy	S288C-KO	doi:10.1038/s41586-019-1549-9
5	N’Guessan 2025 (segregant scRNA-seq eQTL) $\bullet$	Expression / single cell	4,500 profiled segregants	1	4.5×10^3	2.7×10^7	single-cell transcriptome + eQTL	segregant-WGS	PMC12303567
6	Hackett 2020 (IDEA inducible-TF transcriptome time series) $\bullet$	Expression / single cell	over 200 inducible-TF strains	estradiol induction, about eight time points	$1.6 \times 10^3 \dagger$	9.6×10^6	bulk transcriptome per time point	S288C+designed-edit	doi:10.15252/msb.20199174
7	Hu 2007 (TF deletion expression compendium) $\bullet$	Expression / single cell	269 TF deletion strains	1	269	1.6×10^6	mRNA abundance	S288C-KO	doi:10.1038/ng2012
8	Dong 2021 (MAGIC + SAM biosensor)	CRISPR library screen	100,000 guides (a/i/d)	1	$1.0 \times 10^5 \ddagger$	1.0×10^5	SAM biosensor fluorescence (FACS)	S288C+guide	doi:10.1016/j.ymben.2021.03.005
9	Momen-Roknabadi 2020 (inducible CRISPRi library)	CRISPR library screen	genome-wide guide library	nutrient limitations	$9.0 \times 10^4 \ddagger$	9.0×10^4	per-guide fitness under induction	S288C+guide	doi:10.1038/s42003-020-01452-9
10	McGlinicy 2021 (genome-scale CRISPRi library) $\bullet$	CRISPR library screen	61,094 guides, 10/gene	continuous culture	6.1×10^4	6.1×10^4	per-guide fitness	S288C+guide	doi:10.1186/s12864-021-07518-0
11	Bao 2018 (CHAnGE single-nucleotide library)	CRISPR library screen	tens of thousands of designed variants	inhibitor selection	$6.0 \times 10^4 \ddagger$	6.0×10^4	variant fitness (pooled barcode seq)	S288C+designed-edit	doi:10.1038/nbt.4132
12	Crook 2016 (tunable RNAi, isobutanol + 1-butanol)	Tolerance / robustness	genome-scale tunable RNAi	isobutanol, 1-butanol	$2.0 \times 10^4 \ddagger$	2.0×10^4	dose-graded knockdown fitness under alcohol stress	S288C+guide	PMID 27654654
13	Roy 2018 (multiplexed precision editing)	CRISPR library screen	thousands of designed edits	1	$1.6 \times 10^4 \ddagger$	1.6×10^4	variant fitness	S288C+designed-edit	doi:10.1038/nbt.4137
14	Newman 2006 (protein noise, GFP library) $\bullet$	Modality / backbone	2,500 GFP-tagged strains	rich / minimal	$5.0 \times 10^3 \dagger$	1.0×10^4	protein abundance and cell-to-cell noise	S288C+tag	doi:10.1038/nature04785

Table 1, continued

#	Dataset	Class	Genotypes	Env	Instances	Meas.	Phenotype	Sequence basis	Link
15	Mukherjee 2021 (CRISPRi essential genes x acetic acid)	Tolerance / robustness	1,100 essential genes	acetic acid vs control	2.2×10^3 †	2.2×10^3	fitness under acid stress	S288C+guide	doi:10.1128/mSystems.00410-21
16	Lian 2017 (CRISPR-AID, beta-carotene)	CRISPR library screen	combinatorial a/i/d triples	1	200‡	200	beta-carotene titer	engineered-chassis	doi:10.1038/s41467-017-01695-x
17	Hughes 2000 (compendium of expression profiles)	Expression / single cell	300 mutations and chemical treatments	1	300	1.8×10^6	mRNA log ratio (two-color array)	S288C-KO	doi:10.1016/S0092-8674(00)00015-5
18	Lenstra 2011 (chromatin regulator deletion expression) •	Expression / single cell	165 chromatin-regulator deletions	1	165	9.9×10^5	mRNA abundance	S288C-KO	doi:10.1016/j.molcel.2011.03.026
19	Sun 2013 (mRNA synthesis and decay rates across deletion strains)	Expression / single cell	46 deletion strains of mRNA degradation and metabolism genes	1	46	8.3×10^5	mRNA level, synthesis rate and decay rate	S288C-KO	doi:10.1016/j.molcel.2013.09.010
20	Jariani 2020 (yeast scRNA-seq through lag phase) •	Expression / single cell	isogenic wild type	glucose to maltose shift, time course through lag phase	6‡	3.6×10^4	single-cell transcriptome	reference-only	doi:10.7554/eLife.55320
21	Guo 2018 (CRISPR-Cas9 tiling of essential genes) •	CRISPR library screen	3e4 tiling guides	1	3.0×10^4 ‡	3.0×10^4	per-guide fitness	S288C+guide	doi:10.1038/nbt.4147
22	Urbonaite 2021 (yeastDrop-Seq under drug treatment) •	Expression / single cell	isogenic wild type	mycophenolic acid, guanine, both, untreated	4	2.4×10^4	single-cell transcriptome	reference-only	doi:10.1038/s42003-021-02895-4
23	Nadal-Ribelles 2019 (sensitive yeast scRNA-seq) •	Expression / single cell	clonal wild type	baseline + stress	2‡	1.2×10^4	single-cell transcriptome	reference-only	doi:10.1038/s41564-018-0346-9
24	Gasch 2017 (single-cell stress heterogeneity) •	Expression / single cell	wild type	stress vs unstressed	2‡	1.2×10^4	single-cell transcriptome	reference-only	doi:10.1371/journal.pbio.2004050
25	Jaffe 2019 (multiplexed CRISPR interference epistasis) •	CRISPR library screen	1e4 guide pairs	1	1.0×10^4 ‡	1.0×10^4	fitness / genetic interaction	S288C+guide	doi:10.1534/g3.119.400323
26	Jackson 2023 (wild-type scRNA-seq time course for RNA kinetics) •	Expression / single cell	wild type	one time course, sampled continuously	1	6.0×10^3	single-cell transcriptome (173,361 cells)	reference-only	doi:10.1101/2023.09.21.558277
27	Brettner 2024 (ultra-high-throughput yeast scRNA-seq) •	Expression / single cell	method	method	–‡	–	single-cell transcriptome	reference-only	doi:10.1002/yea.3927

Table 1, continued

#	Dataset	Class	Genotypes	Env	Instances	Meas.	Phenotype	Sequence basis	Link
28	Mulleder 2012 (prototrophic deletion collection)	Modality / backbone	5,185 prototrophic + 839 titratable essential	1	–	–	strain resource (no phenotype)	S288C-KO	doi:10.1038/nbt.2442
29	Su 2023 (single-cell transcriptomes under four stresses)	Expression / single cell	wild type	isosmotic, hypoosmotic, glucose starvation, amino acid starvation	4	2.4×10^4	single-cell transcriptome (117 cells, full length)	reference-only	doi:10.1186/s12864-023-09184-w
30	Wang 2022 (single-cell transcriptomes across replicative aging)	Expression / single cell	wild type	2 h, 16 h, 36 h of aging	3	6.6×10^3	single-cell transcriptome (125 cells)	reference-only	doi:10.1111/accel.13712
31	Albert 2018 (eQTL in 1,012 segregants) •	Natural variation	1,012 sequenced segregants	1	1.0×10^3	5.8×10^6	mRNA abundance (RNA-seq)	segregant-WGS	doi:10.7554/eLife.35471
32	Jakobson 2025 (genome-to-proteome map)	Natural variation	800 F6 segregants	1	800	1.6×10^6	protein abundance, >6,400 pQTL	segregant-WGS	doi:10.1126/science.adu3198
33	Muenzner 2024 (natural-isolate proteome)	Natural variation	796 natural isolates	1	796	1.6×10^6	protein abundance (DIA-MS)	isolate-WGS	doi:10.1038/s41586-024-07442-9
34	Cooper 2010 (CE-MS amino-acid metabolome)	Metabolite / precursor	4,700 YKO	1	4.7×10^3	9.4×10^4	free amino-acid pools (capillary electrophoresis)	S288C-KO	doi:10.1101/gr.105825.110
35	Aulakh 2025 (genome-scale ionome)	Metabolite / precursor	4,800 YKO	1	4.8×10^3	4.8×10^4	intracellular metal-ion content (ICP-MS)	S288C-KO	doi:10.1016/j.cels.2025.101319
36	Teyssonniere 2024 (species-wide proteome against transcriptome)	Natural variation	942 natural isolates	1	942	1.9×10^6	protein abundance, paired with transcript abundance	isolate-WGS	doi:10.1073/pnas.2319211121
37	Grossbach 2022 (BY x RM transcriptome, proteome and phosphoproteome)	Natural variation	112 genomically defined strains (BY x RM panel)	1	112	1.1×10^6	transcript, protein and phosphosite abundance	segregant-WGS	doi:10.15252/msb.202110712
38	Leutert 2023 (phosphoproteome x 101 conditions)	Expression / single cell	wild type	101 conditions	101	5.0×10^5	phosphosite abundance	reference-only	doi:10.1038/s41594-023-01115-3
39	Brauer 2008 (growth-rate-controlled chemostat transcriptome)	Expression / single cell	wild type	36 chemostats, 6 limitations	36	2.2×10^5	mRNA abundance	reference-only	doi:10.1091/mbc.e07-08-0779
40	Hackett 2016 (SIMMER multi-omic flux)	Expression / single cell	wild type	25 chemostat states	25	7.5×10^4	paired metabolome + proteome + flux	reference-only	doi:10.1126/science.aaf2786
41	Zhu 2014 (kinase / phosphatase lipidomics)	Metabolite / precursor	129 kinase + phosphatase KOs	1	129	2.6×10^4	lipid class and species abundance	S288C-KO	PMC4196872
42	Gerke 2017 (urea-cycle mQTL)	Natural variation	147 diploid segregants	1	147	1.5×10^4	untargeted metabolite abundance	segregant-WGS	doi:10.1534/genetics.117.201107

Table 1, continued

#	Dataset	Class	Genotypes	Env	Instances	Meas.	Phenotype	Sequence basis	Link
43	Ambroset 2014 (metabolite QTL)	Natural variation	100 segregants	1	100	7.4×10^3	74 metabolite concentrations (LC-MS/MS)	segregant-WGS	doi:10.1371/journal.pgen.1004142
44	Blank 2005 (13C metabolic flux)	Metabolite / precursor	200 viable null mutants	glucose minimal	200‡	6.0×10^3	intracellular flux distribution (13C-MFA)	S288C-KO	doi:10.1186/gb-2005-6-6-r49
45	Skelly 2013 (expression variation across isolates)	Natural variation	22 sequenced isolates	1	22	1.8×10^5	transcriptome + proteome	isolate-WGS	doi:10.1101/gr.155762.113
46	Airoldi 2016 (nitrogen-limited steady-state and dynamic transcriptome)	Expression / single cell	wild type, prototrophic	nitrogen sources x steady-state growth rates, plus an upshift time course	20‡	1.2×10^5	bulk transcriptome	reference-only	doi:10.1091/mbc.E14-05-1013
47	McManus 2014 (ribosome profiling, allele-specific)	Modality / backbone	S. cerevisiae, S. paradoxus and their F1 hybrid	1	3	3.3×10^4	ribosome occupancy paired with mRNA abundance	reference-only	doi:10.1101/gr.164996.113
48	Martin-Perez 2017 (protein half-lives)	Modality / backbone	wild type, exponential growth	1	1	3.2×10^3	protein turnover rate	reference-only	doi:10.1016/j.cels.2017.08.008
49	Boer 2010 (metabolome across nutrient limitations)	Metabolite / precursor	wild type	nutrient limitations x growth rates	25	2.5×10^3	intracellular metabolite concentrations	reference-only	doi:10.1091/mbc.e09-07-0597
50	Tengolics 2024 (domestication metabolome)	Natural variation	17 S. cerevisiae populations	1	17	1.6×10^3	19 amino acids + 78 other metabolites	isolate-WGS	doi:10.1073/pnas.2313354121
End of wave 1. Rows 1–50 are the recommended next builds.									
51	Eder 2020 (flux QTL)	Natural variation	segregant panel (n unconfirmed)	wine fermentation	100‡	100	modeled intracellular flux	segregant-WGS	PMID 32034164
52	Foss 2007 (BY x RM segregant proteome)	Natural variation	BY x RM segregants; the count is not stated in the abstract and was not confirmed	1	–‡	–	protein abundance (label-free MS)	segregant-WGS	doi:10.1038/ng.2007.22
53	Yu 2021 (proteome and metabolome under nitrogen limitation)	Metabolite / precursor	wild type	nitrogen-limited chemostats	10‡	2.5×10^4	proteome + metabolome	reference-only	doi:10.7554/eLife.65722
54	de Boer 2020 (100M random promoters) •	Regulatory DNA	>100M random promoters	complex + defined	1.0×10^8	1.0×10^8	expression driven (YFP sort-seq)	S288C+reporter-locus	doi:10.1038/s41587-019-0315-8
55	Vaishnav 2022 (regulatory DNA fitness landscape) •	Regulatory DNA	2e7 promoter variants	1	2.0×10^7 ‡	2.0×10^7	expression + inferred fitness	S288C+reporter-locus	doi:10.1038/s41586-022-04506-6

Table 1, continued

#	Dataset	Class	Genotypes	Env	Instances	Meas.	Phenotype	Sequence basis	Link
56	Lee 2014 (HIP-HOP fitness signatures) ^B	Tolerance / robustness	11,000 (het + hom)	3,356 compounds	1.3×10^7	1.3×10^7	chemogenomic fitness score	S288C-KO	doi:10.1126/science.1250217
57	Nguyen Ba 2022 (barcoded bulk QTL, 100k segregants) •	Natural variation	100,000 barcoded segregants	8 conditions	$8.0 \times 10^5 \dagger$	8.0×10^5	fitness (Bar-seq)	segregant-WGS	doi:10.7554/eLife.73983
58	Galardini 2019 (four backgrounds x 38 conditions)	Natural variation	3,786 KOs x 4 backgrounds	38 conditions	$5.8 \times 10^5 \dagger$	5.8×10^5	colony-size fitness (S-score)	isolate-WGS	doi:10.15252/msb.20198831
59	Parsons 2006 (bioactive-compound profiling)	Tolerance / robustness	5,000 viable YKO	82 compounds	$4.1 \times 10^5 \dagger$	4.1×10^5	hypersensitivity score	S288C-KO	doi:10.1016/j.cell.2006.06.040
60	Dutta 2026 (barcoded natural-isolate chemical response)	Natural variation	520 natural isolates	>600 compounds	$3.1 \times 10^5 \dagger$	3.1×10^5	pooled-barcode fitness	isolate-WGS	doi:10.1038/s41467-026-73532-z
61	Li 2016 (tRNA fitness landscape) •	Deep mutational scan	65,000 tRNA variants	1	$6.5 \times 10^4 \ddagger$	6.5×10^4	fitness	S288C+designed-edit	doi:10.1126/science.aae0568
62	Peter 2018 (1,011 isolate genomes + phenome)	Natural variation	1,011 isolates (971 phenotyped)	36 conditions	$3.5 \times 10^4 \dagger$	3.5×10^4	growth fitness per condition	isolate-WGS	doi:10.1038/s41586-018-0030-5
63	Domingo 2018 (tRNA double-mutant landscape) •	Deep mutational scan	23,000 single + double mutants	1	$2.3 \times 10^4 \ddagger$	2.3×10^4	fitness	S288C+designed-edit	doi:10.1038/s41586-018-0170-7
64	Zhang 2021 (YETI titratable overexpression)	Metabolite / precursor	1,022 essential + 4,668 non-essential	beta-estradiol dose series	$2.3 \times 10^4 \ddagger$	2.3×10^4	fitness across an induction gradient	S288C+tag	doi:10.15252/msb.202010321
65	Peeters 2021 (fermentation-trait QTL atlas)	Natural variation	1,125 sequenced segregants	18 traits	$2.0 \times 10^4 \dagger$	2.0×10^4	ethanol / glycerol / isobutanol titer, stress resistance, aroma	segregant-WGS	PMID 34727964
66	Van Leeuwen 2024 (short-chain organic acids)	Tolerance / robustness	4,800 YKO	acetic, butyric, octanoic	$1.4 \times 10^4 \dagger$	1.4×10^4	fitness per acid	S288C-KO	PMC10903034
67	Chica 2026 (AutoDRY autophagy screen)	Modality / backbone	4,760 KO + 1,159 DAmP	2 (nitrogen starvation, replete)	$1.2 \times 10^4 \dagger$	1.2×10^4	autophagy flux (deep-learning image classifier)	S288C-KO	doi:10.1038/s41556-025-01837-0
68	Sopko 2006 (genome-scale overexpression toxicity)	Metabolite / precursor	5,280 ORFs	GAL induced vs glucose	$1.1 \times 10^4 \dagger$	1.1×10^4	growth-rate reduction on induction	S288C+ORF-plasmid	doi:10.1016/j.molcel.2005.12.011
69	Liu 2021 (tryptophan / isobutanol tolerance)	Tolerance / robustness	5,006 YKO	+/- isobutanol	$1.0 \times 10^4 \dagger$	1.0×10^4	colony-size fitness (ScreenMill imaging)	S288C-KO	doi:10.1186/s13068-021-02048-z
70	Kuroda 2019 (isobutanol-specific tolerance)	Tolerance / robustness	4,800 YKO	isobutanol vs ethanol	$9.6 \times 10^3 \dagger$	9.6×10^3	isobutanol-specific fitness	S288C-KO	doi:10.1016/j.cels.2019.10.006

End of wave 2. Rows 51–70 are the bench, promoted as soon as a wave-1 row proves unreachable.

Table 1, continued

#	Dataset	Class	Genotypes	Env	Instances	Meas.	Phenotype	Sequence basis	Link
71	Turco 2023 (Yeast Phenome)^L	Tolerance / robustness	5,000	7,536 environments	3.8×10^7 †	3.8×10^7	growth / fitness per environment	S288C-KO	doi:10.1126/sciadv.adg5702
72	Rossi 2021 (ChIP-exo protein architecture)	Modality / backbone	400 tagged proteins	1	400	2.4×10^6	genome-wide binding location, near-bp resolution	S288C+tag	doi:10.1038/s41586-021-03314-8
73	Piotrowski 2017 (MOSAIC diagnostic panel)	Tolerance / robustness	157 diagnostic	13,524 compounds	2.1×10^6 †	2.1×10^6	chemogenomic fitness score	S288C-KO	doi:10.1038/nchembio.2436
74	Bloom 2025 (global epistasis in a yeast cross)[•]	Natural variation	large sequenced segregant panel	conditions	1.0×10^6 ‡	1.0×10^6	fitness	segregant-WGS	PMID 40679398
75	Gasch 2000 (environmental stress response)	Expression / single cell	wild type	9 stress series, multi-timepoint	150‡	9.0×10^5	mRNA abundance time course	reference-only	doi:10.1091/mbc.11.12.4241
76	Decourty 2021 (RNA-metabolism genetic interactions)	Modality / backbone	700,000 double mutants	1	7.0×10^5	7.0×10^5	genetic-interaction score	S288C-KO	doi:10.1093/nar/gkab680
77	Cuperus 2017 (random 5' UTR library)	Regulatory DNA	500,000 random 5' UTRs	1	5.0×10^5 ‡	5.0×10^5	protein output (translation)	S288C+reporter-locus	doi:10.1101/gr.224964.117
78	Faure 2022 (doubledeepPCA)[•]	Deep mutational scan	5e5 double mutants, 2 domains	1	5.0×10^5 ‡	5.0×10^5	binding and abundance	engineered-chassis	doi:10.1038/s41586-022-04586-4
79	Chong 2015 / CYCLOPs (single-cell proteome dynamics)[•]	Modality / backbone	4,100 GFP-tagged ORFs	cell-cycle / stress	2.5×10^4 ‡	4.2×10^5	abundance + localization class	S288C+tag	doi:10.1016/j.cell.2015.04.051
80	Teyssonniere 2024 (species-wide trait survey)	Natural variation	1,011 isolates	200 traits	2.0×10^5 ‡	2.0×10^5	growth traits, expressivity and complexity	isolate-WGS	doi:10.1371/journal.pgen.1011119
81	Diss 2018 (Fos-Jun interaction double mutants)[•]	Deep mutational scan	120,000 double mutants	1	1.2×10^5 ‡	1.2×10^5	protein-protein interaction strength	engineered-chassis	doi:10.7554/eLife.32472
82	Warringer 2011 (trait variation across the SGRP panel)	Natural variation	200 sequenced isolates	200 conditions	4.0×10^4 ‡	1.2×10^5	growth-curve parameters	isolate-WGS	doi:10.1371/journal.pgen.1002111
83	Keren 2013 (promoter activity across conditions)	Regulatory DNA	900 native promoter reporters	10 conditions	9.0×10^3 †	9.0×10^4	promoter activity, time-resolved	S288C+reporter-locus	doi:10.1038/msb.2013.59
84	Puchta 2016 (snoRNA U3 landscape)[•]	Deep mutational scan	60,000 U3 variants	1	6.0×10^4 ‡	6.0×10^4	fitness	S288C+designed-edit	doi:10.1126/science.aaf0965
85	Yoshikawa 2011 (deletion + overexpression phenome)	Tolerance / robustness	4,800 KO + matched overexpression	carbon sources / stress	5.8×10^4 ‡	5.8×10^4	growth-rate fitness	S288C-KO	doi:10.1534/g3.111.000695
86	Ho 2009 (MoBY-ORF barcoded overexpression)	Metabolite / precursor	5,100 barcoded ORFs	compound panel	5.1×10^4 ‡	5.1×10^4	fitness per compound	S288C+ORF-plasmid	doi:10.1038/nbt.1534

Table 1, continued

#	Dataset	Class	Genotypes	Env	Instances	Meas.	Phenotype	Sequence basis	Link
87	Bloom 2013 (1,008 BYxRM segregants)	Natural variation	1,008 segregants, 11,623 markers	46 traits	4.6×10^4 †	4.6×10^4	quantitative growth traits	segregant-WGS	doi:10.1038/nature11867
88	Xiao 2014 (genome-wide RNAi furfural tolerance)	Tolerance / robustness	genome-scale RNAi library	furfural vs control	2.0×10^4 ‡	2.0×10^4	knockdown enrichment under furfural	S288C+guide	doi:10.1186/1754-6834-7-78
89	Douglas 2012 (barcoded overexpression fitness)	Metabolite / precursor	genome-scale barcoded ORFs	baseline + stress	2.0×10^4 ‡	2.0×10^4	pooled competitive fitness on induction	S288C+ORF-plasmid	doi:10.1534/g3.112.003400
90	She 2018 (natural variant effects across a large isolate panel)	Natural variation	500 isolates	40 conditions	2.0×10^4 ‡	2.0×10^4	growth traits	isolate-WGS	doi:10.1016/j.cell.2017.12.015
91	Costello 2020 (bioreactor Bar-seq)	Tolerance / robustness	pooled YKO library	feed schemes / pH	1.9×10^4 ‡	1.9×10^4	fitness under fed-batch cultivation	S288C-KO	doi:10.1186/s12934-020-01423-z
92	Gurvitz (fatty-acid utilization screen)	Metabolite / precursor	4,800 viable YKO	oleate, myristate, acetate	1.4×10^4 †	1.4×10^4	fatty-acid utilization competence	S288C-KO	embopress.org/doi/pdf/10.1038/msb4100051
93	Mira 2010 (acetic acid tolerance, full collection)	Tolerance / robustness	EUROSCARF collection	70-110 mM acetic acid, pH 4.5	1.4×10^4 †	1.4×10^4	acetic-acid tolerance	S288C-KO	doi:10.1186/1475-2859-9-79
94	Gameiro 2025 (AID2 degron libraries)	Modality / backbone	>5,600 tagged ORFs	baseline + hydroxyurea	1.1×10^4 †	1.1×10^4	degradation efficiency; fitness under stress	S288C+tag	doi:10.1083/jcb.202409007
95	Si 2017 (RAGE RNAi, isobutanol titer)	CRISPR library screen	genome-scale RNAi, iterative rounds	isobutanol selection	1.0×10^4 ‡	1.0×10^4	construct enrichment; isobutanol titer for top strains	S288C+guide	doi:10.1038/ncomms15187
96	Hamedirad 2018 (RAGE xylose utilization)	CRISPR library screen	genome-scale RNAi	xylose-limited	1.0×10^4 ‡	1.0×10^4	enrichment; xylose consumption rate	S288C+guide	doi:10.1002/bit.26570
97	Pereira 2014 (wheat-straw hydrolysate)	Tolerance / robustness	EUROSCARF deletion collection	hydrolysate vs control	9.6×10^3 †	9.6×10^3	tolerance to industrial hydrolysate	S288C-KO	doi:10.1007/s10295-014-1519-z
98	Endo 2008 (vanillin tolerance)	Tolerance / robustness	diploid deletion collection	vanillin vs control	9.6×10^3 †	9.6×10^3	vanillin sensitivity	S288C-KO	doi:10.1128/AEM.01541-08
99	Sharifpoor 2012 (kinome synthetic dosage lethality)	Modality / backbone	92 kinase queries, genome-wide array	1	8.7×10^3	8.7×10^3	genetic interaction / SDL score	S288C-KO	doi:10.1101/gr.129213.111
100	Mavor 2016 (ubiquitin DMS across conditions)	Deep mutational scan	1,400 ubiquitin variants	chemical stresses	7.0×10^3 †	7.0×10^3	fitness	S288C+designed-edit	doi:10.7554/eLife.15802
101	Flynn 2020 (Hsp90 DMS across environments)	Deep mutational scan	1,000 Hsp90 variants	stress conditions	6.0×10^3 †	6.0×10^3	fitness	S288C+designed-edit	doi:10.7554/eLife.53810
102	Ho 2018 (unified absolute protein abundance)	Modality / backbone	wild type	1	5.9×10^3	5.9×10^3	absolute abundance, molecules/cell	reference-only	doi:10.1016/j.cels.2017.12.004
103	Renganaath 2020 (natural promoter-variant MPRA) •	Regulatory DNA	5,832 natural variants, 2,503 promoters	1	5.8×10^3	5.8×10^3	allele-specific promoter activity	S288C+reporter-locus	doi:10.7554/eLife.62669

Table 1, continued

#	Dataset	Class	Genotypes	Env	Instances	Meas.	Phenotype	Sequence basis	Link
104	Chang 2012 (organic-acid production screen)	Metabolite / precursor	4,800 YKO	1	4.8×10^3	4.8×10^3	acetate / pyruvate / succinate halo (pH indicator)	S288C-KO	PMID 22277779
105	Trikka 2015 (carotenogenic heterozygous screen)	Metabolite / precursor	4,700 heterozygous deletions	1	4.7×10^3	4.7×10^3	carotenoid color; sclereol titer for top strains	S288C-KO/het	doi:10.1186/s12934-015-0246-0
106	Sardi 2018 (natural variation in hydrolysate tolerance)	Tolerance / robustness	sequenced isolate / segregant panel	hydrolysate toxins	$2.0 \times 10^3 \ddagger$	2.0×10^3	tolerance	isolate-WGS	doi:10.1371/journal.pgen.1007217
107	Breslow 2008 (DAmP hypomorphic collection)	Modality / backbone	842 haploid + 970 diploid essential	1	1.8×10^3	1.8×10^3	competitive fitness	S288C+tag	doi:10.1038/nmeth.1234
108	Kofoed 2015 (barcoded ts alleles)	Modality / backbone	600+ barcoded ts alleles	permissive / restrictive	$1.2 \times 10^3 \dagger$	1.2×10^3	ts severity (growth ratio)	S288C+designed-edit	doi:10.1534/g3.115.019174
109	Zhang 2022 (GCE-SCRaMbLE recombination outcomes) •	Combinatorial genome	200 sequenced SCRaMbLE genomes	1	200 $\ddagger$	200	rearrangement structure + fitness	isolate-WGS	doi:10.1038/s41467-022-33606-0
110	Cubillos 2017 (nitrogen-consumption QTL)	Natural variation	165 sequenced F12 segregants	nitrogen-limited fermentation	165	165	nitrogen consumption	segregant-WGS	doi:10.1534/g3.117.042127
111	Hillenmeyer 2010 (systematic fitness reanalysis)	Tolerance / robustness	het + hom collections	400 conditions	$2.4 \times 10^6 \ddagger$	2.4×10^6	fitness-defect score	S288C-KO	doi:10.1186/gb-2010-11-3-r30
112	Smith 2008 (gene-environment interaction eQTL) •	Natural variation	109 segregants	glucose / ethanol	218 $\dagger$	1.3×10^6	mRNA abundance	segregant-WGS	doi:10.1371/journal.pbio.0060083
113	Braberg 2020 (point-mutant E-MAP)	Modality / backbone	350 point mutants x genome-wide array	1	5.0×10^5	5.0×10^5	residue-level genetic interaction	S288C+designed-edit	doi:10.1126/science.aaz4910
114	Breker 2013 (GFP library under stress) •	Modality / backbone	4,159 GFP-tagged strains	stress conditions	$1.2 \times 10^4 \dagger$	2.1×10^5	abundance + localization, per cell	S288C+tag	doi:10.1083/jcb.201301120
115	Tkach 2012 (protein relocation under DNA damage) •	Modality / backbone	GFP collection	DNA-damaging agents	$1.2 \times 10^4 \dagger$	2.0×10^5	localization and abundance change	S288C+tag	doi:10.1038/ncb2549
116	Ehrenreich 2010 (X-QTL bulk segregant mapping)	Natural variation	1e6 pooled segregants	17 selective conditions	17	1.9×10^5	allele-frequency shift (pooled)	segregant-WGS	doi:10.1038/nature08923
117	Denervaud 2013 (microfluidic GFP dynamics) •	Modality / backbone	1,000 GFP strains	dynamic environments	$4.0 \times 10^3 \dagger$	6.8×10^4	time-resolved single-cell abundance	S288C+tag	doi:10.1073/pnas.1308265110
118	Brogaard 2012 (chemical nucleosome map)	Modality / backbone	wild type	1	1	6.7×10^4	nucleosome center positions, bp resolution	reference-only	doi:10.1038/nature11142

Table 1, continued

#	Dataset	Class	Genotypes	Env	Instances	Meas.	Phenotype	Sequence basis	Link
119	Xia 2022 (proteome allocation across conditions)	Modality / backbone	wild type	chemostat conditions	30‡	6.0×10^4	absolute proteome, resource allocation	reference-only	doi:10.1038/s41467-022-30513-2
120	Melamed 2013 (PAB1 RRM domain scan)	Deep mutational scan	40,000 PAB1 variants	1	4.0×10^4 ‡	4.0×10^4	fitness	S288C+designed-edit	doi:10.1261/rna.040709.113
121	De Chiara 2022 (domestication and genome instability)	Natural variation	1,011-isolate panel, reanalyzed	conditions	3.6×10^4 ‡	3.6×10^4	growth, aneuploidy, instability	isolate-WGS	doi:10.1038/s41559-022-01671-9
122	Yvert 2013 (single-cell trait variation across isolates) •	Natural variation	37 isolates	1	37	1.9×10^4	single-cell morphology (CalMorph)	isolate-WGS	doi:10.1186/1752-0509-7-54
123	Lahtvee 2017 (absolute proteome across conditions)	Modality / backbone	wild type	9 growth conditions	9	1.8×10^4	absolute protein abundance + turnover	reference-only	doi:10.1016/j.cels.2017.03.003
124	Weill 2018 (SWAT libraries)	Modality / backbone	5,500 acceptor-tagged strains	1	5.5×10^3	1.6×10^4	abundance, localization, topology	S288C+tag	doi:10.1038/s41592-018-0044-9
125	Kinsler 2020 (barcoded pleiotropy across environments)	Natural variation	300 sequenced adaptive clones	45 environments	1.4×10^4 ‡	1.4×10^4	fitness	isolate-WGS	doi:10.7554/eLife.61271
126	Pelechano 2013 (transcript isoform landscape)	Modality / backbone	wild type	glucose / galactose	2	1.2×10^4	transcript isoform boundaries and abundance	reference-only	doi:10.1038/nature12121
127	Deutschbauer 2005 (haploinsufficiency, genome-wide)	Tolerance / robustness	heterozygous diploid collection	rich / minimal	1.2×10^4 ‡	1.2×10^4	haploinsufficiency fitness	S288C-KO/het	doi:10.1534/genetics.104.036871
128	Valenti 2025 (SWAT degron + GFP collection)	Modality / backbone	proteome-wide SWAT library	depleted / not	1.0×10^4 ‡	1.0×10^4	depletion efficiency + abundance/localization	S288C+tag	doi:10.1083/jcb.202409050
129	Sousa 2013 (acetate resistance overexpression screen)	Tolerance / robustness	genome-wide overexpression library	sodium acetate vs control	1.0×10^4 ‡	1.0×10^4	acetate resistance	S288C+ORF-plasmid	PMID 23262128
130	Sadhu 2016 (CRISPR-directed causal variant mapping)	Natural variation	1,000 targeted recombinants	conditions	1.0×10^4 ‡	1.0×10^4	fitness	segregant-WGS	doi:10.1126/science.aaf5124
131	Weinberg 2016 (ribosome profiling, improved)	Modality / backbone	wild type	1	1	6.0×10^3	ribosome occupancy, translation efficiency	reference-only	doi:10.1016/j.celrep.2016.01.043
132	Ghaemmaghami 2003 (absolute protein abundance)	Modality / backbone	4,251 TAP-tagged ORFs	1	4.3×10^3	4.3×10^3	absolute abundance (molecules/cell)	S288C+tag	doi:10.1038/nature02046
133	Huh 2003 (GFP localization atlas)	Modality / backbone	4,156 GFP-tagged ORFs	1	4.2×10^3	4.2×10^3	subcellular localization (22 classes)	S288C+tag	doi:10.1038/nature02026

Table 1, continued

#	Dataset	Class	Genotypes	Env	Instances	Meas.	Phenotype	Sequence basis	Link
134	Michaelis 2023 (protein interactome)	Modality / backbone	proteome-scale AP-MS baits	1	$4.0 \times 10^3 \ddagger$	4.0×10^3	protein-protein interaction + interface	S288C+tag	doi:10.1038/s41586-023-06739-5
135	Christiano 2014 (protein turnover)	Modality / backbone	wild type	1	$3.9 \times 10^3 \ddagger$	3.9×10^3	protein half-life	reference-only	doi:10.1016/j.celrep.2014.10.065
136	Kitzman 2015 (GAL4 DMS)	Deep mutational scan	1,200 GAL4 variants	selective / permissive	$2.4 \times 10^3 \ddagger$	2.4×10^3	transcription-factor activity	S288C+designed-edit	doi:10.1038/nmeth.3223
137	Sharon 2012 (designed promoter library)	Regulatory DNA	2,000 designed promoters	1	$2.0 \times 10^3 \ddagger$	2.0×10^3	expression (fluorescence)	S288C+reporter-locus	doi:10.1038/nbt.2205
138	Sadhu 2018 (CRISPR-directed mitotic recombination)	CRISPR library screen	1,500 recombinant genotypes	1	$1.5 \times 10^3 \ddagger$	1.5×10^3	fitness / trait	segregant-WGS	doi:10.7554/eLife.41090
139	Duveau 2021 (TDH3 promoter allele series)	Regulatory DNA	250 promoter alleles	carbon sources	$1.0 \times 10^3 \ddagger$	1.0×10^3	expression level, noise, fitness	S288C+designed-edit	doi:10.7554/eLife.67806
140	Strope 2015 (100 clinical and natural genomes)	Natural variation	100 sequenced isolates	phenotype panel	$1.0 \times 10^3 \ddagger$	1.0×10^3	growth and clinical-relevant traits	isolate-WGS	doi:10.1101/gr.185538.114
141	Schulte 2023 (mitochondrial complexome)	Modality / backbone	900 mitochondrial proteins	1	900	900	complex assembly profile	reference-only	doi:10.1038/s41586-022-05641-w
142	Metzger 2015 (promoter mutation effects on expression and noise)	Regulatory DNA	235 promoter mutants	1	235 $\ddagger$	470	expression mean and noise	S288C+designed-edit	doi:10.1038/nature14424
143	Venkataram 2016 (barcoded adaptive lineages, sequenced)	Natural variation	400 sequenced adaptive clones	glucose-limited	400 $\ddagger$	400	fitness	isolate-WGS	doi:10.1016/j.cell.2016.08.002
144	Jakociunas 2021 (degron-tuned terpene flux)	Metabolite / precursor	30 degron strains	induction levels	90 $\ddagger$	270	farnesol / geraniol / nerolidol titer	engineered-chassis	doi:10.1002/bit.27735
145	Shen 2016 (SCRaMbLE genome minimization)	Combinatorial genome	100 SCRaMbLEd genomes	selective / permissive	200 $\ddagger$	200	viability and fitness	isolate-WGS	doi:10.1101/gr.193433.115
146	Blount 2018 (SCRaMbLE for phenotype improvement)	Combinatorial genome	100 rearranged genomes	selection conditions	200 $\ddagger$	200	growth and product phenotype	isolate-WGS	doi:10.1038/s41467-018-03143-w
147	Liti 2009 (population genomics of <i>S. cerevisiae</i> and <i>S. paradoxus</i>)	Natural variation	70 sequenced isolates	1	70 $\ddagger$	70	genome sequence and population structure	isolate-WGS	doi:10.1038/nature07743
148	Yue 2017 (wild vs domesticated assemblies)	Natural variation	12 long-read assemblies	n/a	12	12	genome structure, TE and subtelomere dynamics	isolate-WGS	doi:10.1038/ng.3847
149	Zackrisson 2016 (Scan-o-matic growth curves)	Modality / backbone	YKO-compatible platform	panel-dependent	– $\ddagger$	–	full growth curve (rate, lag, yield)	S288C-KO	doi:10.1534/g3.116.032342

Table 1, continued

#	Dataset	Class	Genotypes	Env	Instances	Meas.	Phenotype	Sequence basis	Link
150	Spellman 1998 (cell-cycle transcriptome)	Expression / single cell	wild type, synchronized	4 synchroniza- tion methods	77	4.6×10^5	mRNA abundance time course	reference-only	doi:10.1091/ mbc.9.12.3273
Long-run cut. Rows above are the 150 that reach 200; rows below are the ranked reserve.									
151	Levy 2008 (morphology and phenotypic capacitance) •	Modality / backbone	1,700 deletion strains	1	1.7×10^3 ‡	3.4×10^5	single-cell morphology variability	S288C-KO	doi:10.1371/ journal.pbio.0060264
152	Causton 2001 (environmental change transcriptome)	Expression / single cell	wild type	environmental changes	40‡	2.4×10^5	mRNA abundance	reference-only	doi:10.1091/mbc.12.2.323
153	Gasch 2001 (DNA damage response transcriptome)	Expression / single cell	wild type + checkpoint mutants	damaging agents	24‡	1.4×10^5	mRNA abundance	S288C-KO	doi:10.1091/ mbc.12.10.2987
154	Handfield 2013 (image-based localization change)	Modality / backbone	GFP collection	baseline / perturbed	8.0×10^3 ‡	1.4×10^5	localization change score	S288C+tag	doi:10.1371/ journal.pcbi.1003085
155	Kaplan 2009 (nucleosome sequence preferences)	Modality / backbone	wild type (in vitro + in vivo)	in vitro / in vivo	2	1.3×10^5	nucleosome occupancy	reference-only	doi:10.1038/nature07667
156	Giaever 2004 (chemogenomic drug-target profiling)	Tolerance / robustness	het + hom collections	10 compounds	5.9×10^4 ‡	5.9×10^4	fitness	S288C-KO	doi:10.1073/ pnas.0307490100
157	Paulo 2016 (TMT proteome across carbon sources)	Modality / backbone	wild type	carbon sources	9	2.2×10^4	protein abundance	reference-only	doi:10.1091/ mbc.E16-03-0184
158	Gibney 2013 (osmotic and glucose stress fitness)	Tolerance / robustness	YKO collection	osmotic / glucose	1.9×10^4 ‡	1.9×10^4	fitness	S288C-KO	doi:10.1073/ pnas.1318100110
159	Meurer 2018 (genome-wide GFP library, rebuilt)	Modality / backbone	4,000 tagged strains	1	4.0×10^3 ‡	1.2×10^4	abundance and localization	S288C+tag	doi:10.1038/ s41592-018-0044-9
160	Bennett 2001 (ionizing-radiation resistance)	Tolerance / robustness	4,800 YKO	irradiated vs control	9.6×10^3 ‡	9.6×10^3	radiation sensitivity	S288C-KO	doi:10.1038/ng778
161	Sugiyama (lactic-acid tolerance)	Tolerance / robustness	4,800 YKO	4 percent lactic acid vs control	9.6×10^3 ‡	9.6×10^3	acid tolerance	S288C-KO	nslr.or.jp/wp-content/ uploads/ NISR06sugiyama.pdf
162	Sousa 2013 (acetic-acid programmed cell death screen)	Tolerance / robustness	YKO haploid collection	acetic acid vs control	9.6×10^3 ‡	9.6×10^3	death / survival (regulated cell death)	S288C-KO	PMID 24286259
163	Nagalakshmi 2008 (RNA-seq transcriptome annotation)	Modality / backbone	wild type	1	1	6.0×10^3	transcript structure and abundance	reference-only	doi:10.1126/ science.1158441
164	Yofe 2016 (SWAT N-terminal library)	Modality / backbone	1,800 N-terminally tagged strains	1	1.8×10^3 ‡	5.4×10^3	localization and abundance	S288C+tag	doi:10.1038/nmeth.3795

Table 1, continued

#	Dataset	Class	Genotypes	Env	Instances	Meas.	Phenotype	Sequence basis	Link
165	Hénault 2023 (hybrid and introgression panel)	Natural variation	hybrids and introgression lines	conditions	$5.0 \times 10^3 \ddagger$	5.0×10^3	fitness	isolate-WGS	doi:10.1038/s41467-024-46155-5
166	Cubillos 2013 (F12 recombinant panel construction)	Natural variation	192 sequenced F12 segregants	conditions	$3.8 \times 10^3 \ddagger$	3.8×10^3	growth traits	segregant-WGS	doi:10.1534/genetics.113.155515
167	Kessi-Perez 2019 (nitrogen-consumption QTL panel)	Natural variation	sequenced segregants / isolates	nitrogen conditions	200 $\ddagger$	200	nitrogen consumption, fermentation rate	segregant-WGS	doi:10.1186/s13068-019-1440-9
168	Lian 2016 (combinatorial xylose pathway optimization)	Metabolite / precursor	combinatorial pathway variants	xylose	100 $\ddagger$	100	ethanol titer and rate on xylose	engineered-chassis	doi:10.1016/j.ymben.2015.11.001

Table 2: Sources for Table 1, in the same order. *Why* states what the row buys that the built set does not. *Data* is where the per-record values live; entries marked unconfirmed were not fetched live and must be checked before a loader is written. Every link is clickable.

#	Dataset, citation and link	Why	Data
1	Boocock 2025 (single-cell eQTL mapping) Boocock J, Alexander N, Alamo Tapia L, Walter-McNeill L, Patel SP, Munugala C, Bloom JS, Kruglyak L. eLife 2025;13:e95566. doi:10.7554/eLife.95566	27,744 single cells drawn from 393 sequenced recombinants in two carbon sources, a median of 17 cells per segregant. High-dimensional on both axes at once, which almost nothing in yeast is, and the closest existing template for the proposed design.	mirrored as boocockSinglecellEQTLMapping2025; GEO (unconfirmed)
2	Hale 2024 (CRISPRi x natural variation) Hale JJ, Matsui T, Goldstein I, et al., Kruglyak L. Nat Commun 2024;15:4234. doi:10.1038/s41467-024-48626-1	Directly measures how a genetic perturbation’s effect changes with genetic background. The single best dataset for asking whether a model trained on one background transfers to another.	SRA PRJNA986287 + Figshare
3	Jackson 2020 (TF-deletion single-cell atlas) Jackson CA, Castro DM, Saldi GA, Bonneau R, Gresham D. eLife 2020;9:e51254. doi:10.7554/eLife.51254	The closest thing to a yeast Perturb-seq that exists. A transcribed barcode in the 3’ UTR of the marker cassette labels every cell with its genotype, so 72 strains, 12 genotypes at 6 independently constructed replicates, are pooled and read out per cell; every strain appears in every one of the 11 conditions. The perturbations are homozygous diploid whole-ORF deletions in a prototrophic FY4/FY5 background rather than the Giaever collection, which is what permits the minimal and nitrogen-limited media. The DNA-only barcodes of the Giaever collection are the stated reason it could not be used, and that is the design constraint any campaign here inherits.	GEO GSE125162; also eLife Source code 2 (103118_SS_Data.tsv.gz). Confirmed from the mirrored paper. The released matrix carries 38,225 cells; the abstract says 38,285 and the discussion 38,255, so the deposit governs. The authoritative 72-strain genotype list is Supplementary file 1 Table S2, an Excel file not held in the mirror.
4	Puddu 2019 (WGS of the deletion collection) Puddu F, et al., Jackson SP. Nature 2019;573:416-420. doi:10.1038/s41586-019-1549-9	Whole-genome sequence for every strain in the deletion collection. This is what turns the S288C-KO sequence basis from an assumption into a measurement, and it applies to roughly half the rows in this table at once.	ENA/SRA (accession unconfirmed)
5	N’Guessan 2025 (segregant scRNA-seq eQTL) N’Guessan A, Boocock J, Kruglyak L, Albert FW. eLife 2025. PMC12303567. PMC12303567	Single-cell transcriptomes indexed by recombinant natural genotype rather than by an engineered knockout. The closest existing analog to the proposed Perturb-seq design, and the only one that already pairs per-cell expression with sequenced genotypes at scale.	eLife data availability; segregant panel from Bloom lineage
6	Hackett 2020 (IDEA inducible-TF transcriptome time series) Hackett SR, Baltz EA, Coram M, Wranik BJ, Kim G, Baker A, Fan M, Hendrickson DG, Berndt M, McIsaac RS. Mol Syst Biol 2020;16:e9174. doi:10.15252/msb.20199174	Each TF is individually induced from a designed synthetic cassette and the whole transcriptome is followed over time, which is the only yeast dataset where a single-gene perturbation is crossed with a dense time axis. scYeast used it as the target for its time-series perturbation-response task (Fan 2027, Sec. 4.1.7). The instance count treats each strain-by-time-point as a record; per strain it is about 200.	Published with the paper (Mol Syst Biol SI); accession unconfirmed from the mirror
7	Hu 2007 (TF deletion expression compendium) Hu Z, Killion PJ, Iyer VR. Nat Genet 2007;39:683-687. doi:10.1038/ng2012	Transcriptomes for the regulators specifically, predating and complementing the already-built Kemmeren compendium. Deleting a transcription factor is the perturbation whose transcriptome response is most interpretable.	GEO (unconfirmed)
8	Dong 2021 (MAGIC + SAM biosensor) Dong C, Schultz JC, Liu W, Lian J, Huang L, Xu Z, Zhao H. Metab Eng 2021;66:319-327. doi:10.1016/j.ymben.2021.03.005	MAGIC re-run with a metabolite biosensor instead of a growth selection, so the label is product concentration rather than fitness. The pattern to copy for every precursor we care about, and the sibling of the already-built Lian 2019 furfural screen.	Metab Eng SI (accession unconfirmed)

Table 2, continued

#	Dataset, citation and link	Why	Data
9	Momen-Roknabadi 2020 (inducible CRISPRi library) Momen-Roknabadi A, Oikonomou P, Zegans M, Tavazoie S. Commun Biol 2020;3:723. doi:10.1038/s42003-020-01452-9	A second CRISPRi library on a different vector and guide-design rule. Whether a model trained on one library transfers to the other is a cheap, decisive test of guide-level overfitting. Not a Perturb-seq row: one edit per cell and a scalar fitness readout is low-dimensional on both axes, however large the library.	Addgene + Commun Biol SI
10	McGlinchy 2021 (genome-scale CRISPRi library) McGlinchy NJ, Meacham ZA, Reynaud KK, Muller R, Baum R, Ingolia NT. BMC Genomics 2021;22:205. doi:10.1186/s12864-021-07518-0	Ten guides per gene gives a graded knockdown axis instead of a binary null, and it covers essential genes. This is the library a yeast Perturb-seq would most plausibly be built on.	ingolia-lab.org/yeast-crispri + Addgene + BMC SI
11	Bao 2018 (CHANGE single-nucleotide library) Bao Z, Hamedirad M, Xue P, Xiao H, Tasan I, Chao R, Liang J, Zhao H. Nat Biotechnol 2018;36:505-508. doi:10.1038/nbt.4132	Single-nucleotide resolution rather than whole-gene nulls. The only genotype axis in this table finer than an ORF, and the one that matches how a real strain-design edit is specified.	Nat Biotechnol SI (raw sequencing location unconfirmed)
12	Crook 2016 (tunable RNAi, isobutanol + 1-butanol) Crook N, Sun J, Morse N, Schmitz A, Alper HS. Appl Microbiol Biotechnol 2016;100:10005-10018. PMID 27654654. PMID 27654654	Two related alcohols at graded knockdown strength. The dosage axis is the signal a binary screen throws away, and Hsp70 emerged only because of it.	AMB SI (DOI is an open verification item)
13	Roy 2018 (multiplexed precision editing) Roy KR, Smith JD, Vonesch SC, et al., St Onge RP. Nat Biotechnol 2018;36:512-520. doi:10.1038/nbt.4137	Genomic rather than plasmid barcodes, which removes the barcode-swapping artifact that quietly corrupts pooled fitness data. The methodological complement to CHANGE.	Nat Biotechnol SI (accession unconfirmed)
14	Newman 2006 (protein noise, GFP library) Newman JRS, Ghaemmaghami S, Ihmels J, Breslow DK, Noble M, DeRisi JL, Weissman JS. Nature 2006;441:840-846. doi:10.1038/nature04785	Measures the distribution rather than the mean, per protein, in two media. Noise is the quantity a per-cell readout recovers and a bulk one destroys, so this sets what a single-cell design should expect to see.	Nature SI
15	Mukherjee 2021 (CRISPRi essential genes x acetic acid) Mukherjee V, Lind U, St Onge RP, Blomberg A, Nystrom T. mSystems 2021;6:e00410-21. doi:10.1128/mSystems.00410-21	Acetic acid is the dominant hydrolysate stressor, and this reaches the essential genes a deletion collection structurally cannot.	mSystems open-access SI
16	Lian 2017 (CRISPR-AID, beta-carotene) Lian J, Hamedirad M, Hu S, Zhao H. Nat Commun 2017;8:1688. doi:10.1038/s41467-017-01695-x	The tri-functional chassis MAGIC was built on, read out on beta-carotene. Small, but it is a combinatorial genotype with a measured product titer, which is the exact record type inverse strain design needs.	Nat Commun SI
17	Hughes 2000 (compendium of expression profiles) Hughes TR, Marton MJ, Jones AR, Roberts CJ, Stoughton R, et al., Friend SH. Cell 2000;102:109-126. doi:10.1016/S0092-8674(00)00015-5	The original deletion expression compendium, and the oldest independent measurement of the phenotype the built Kemmeren 2014 measures. Ingesting it makes cross-laboratory replication of the deletion transcriptome testable on gene-by-gene profiles fourteen years apart and on different array platforms, which is a check the built set cannot run against itself. The 300 profiles mix deletions, titratable alleles and compound treatments, so the compound arm carries a reference genotype and the mutant arm an S288C-KO one; the split was not confirmed this pass.	Rosetta compendium, distributed with the paper; not fetched

Table 2, continued

#	Dataset, citation and link	Why	Data
18	Lenstra 2011 (chromatin regulator deletion expression) Lenstra TL, Benschop JJ, Kim T, et al., Holstege FCP. Mol Cell 2011;42:536-549. doi:10.1016/j.molcel.2011.03.026	Transcriptomes for the chromatin machinery specifically, from the group behind the built Kemmeren compendium and in the same format. Chromatin regulators are the class whose deletion most reshapes the expression landscape.	GEO (unconfirmed)
19	Sun 2013 (mRNA synthesis and decay rates across deletion strains) Sun M, Schwallb B, Pirkel N, Maier KC, Schenk A, Failmezger H, Tresch A, Cramer P. Mol Cell 2013;52:52-62. doi:10.1016/j.molcel.2013.09.010	Comparative dynamic transcriptome analysis decomposes a transcript level into the synthesis rate and the decay rate that produce it, across 46 single deletions. That is the transcript-side counterpart of Martin-Perez 2017's protein half-lives, so the two together give both turnover terms. Its finding also warns what a steady-state compendium hides: a change in degradation rate is generally compensated by a change in synthesis rate, so mRNA level is buffered and two strains with the same level can have different kinetics. The 46 strains are single deletions and overlap the Kemmeren set by gene identity; the abstract does not say they came from that collection, and that provenance was not confirmed.	GEO, series not fetched this pass
20	Jariani 2020 (yeast scRNA-seq through lag phase) Jariani A, Vermeersch L, Cerulus B, Perez-Samper G, Voordeckers K, Van Brussel T, Thienpont B, Lambrechts D, Verstrepen KJ. eLife 2020;9:e55320. doi:10.7554/eLife.55320	A yeast-adapted 10x protocol that resolves the transcriptional states of individual cells during a carbon-source shift, so the readout is the distribution across a population that is not synchronized rather than its mean. One genotype, so the value is the protocol and the per-cell variance structure a campaign has to budget against.	GEO GSE144820 and GSE116246; from the collection sweep, not fetched
21	Guo 2018 (CRISPR-Cas9 tiling of essential genes) Guo X, Chavez A, Tung A, et al., Church GM, Shalem O. Nat Biotechnol 2018;36:540-546. doi:10.1038/nbt.4147	Tiles guides across genes rather than sampling a few per gene, so the readout resolves within-gene functional regions. A finer input axis than a gene-level library.	Nat Biotechnol SI (unconfirmed)
22	Urbonaite 2021 (yeastDrop-Seq under drug treatment) Urbonaite G, Lee JTH, Liu P, Parras GG, Hemberg M, Acar M. Commun Biol 2021;4:1372. doi:10.1038/s42003-021-02895-4	A yeast-optimized Drop-seq with a small chemical condition axis, including a two-compound arm. One genotype, so it enters as a platform reference: the pair of single-drug arms against their combination is a per-cell readout of a two-factor environment, which is the environmental analog of the combinatorial genetic perturbation a Perturb-seq needs.	GEO GSE165686; Zenodo 10.5281/zenodo.4767298 and 10.5281/zenodo.4762526; from the collection sweep, not fetched
23	Nadal-Ribelles 2019 (sensitive yeast scRNA-seq) Nadal-Ribelles M, Islam S, Wei W, et al., Posas F, Steinmetz LM. Nat Microbiol 2019;4:683-692. doi:10.1038/s41564-018-0346-9	Establishes within-clone transcript correlation, which is the noise floor any per-strain Perturb-seq mean has to clear. Directly sizes the cells-per-guide question.	Nat Microbiol SI /GEO
24	Gasch 2017 (single-cell stress heterogeneity) Gasch AP, Yu FB, Hose J, et al., Quake SR. PLoS Biol 2017;15:e2004050. doi:10.1371/journal.pbio.2004050	Separates intrinsic from extrinsic expression heterogeneity under stress. Sets what fraction of Perturb-seq variance is recoverable signal rather than cell-state noise.	PLoS Biology SI /GEO
25	Jaffe 2019 (multiplexed CRISPR interference epistasis) Jaffe M, Sherlock G, Levy SF. G3 2019;9:2843-2852. doi:10.1534/g3.119.400323	Two guides per cell, so the perturbation space is combinatorial rather than one edit at a time. This is the shape a Perturb-seq input axis should have, measured here against fitness only.	G3 open-access SI (unconfirmed)

Table 2, continued

#	Dataset, citation and link	Why	Data
26	Jackson 2023 (wild-type scRNA-seq time course for RNA kinetics) Jackson CA, Beheler-Amass M, Tjarnberg A, Suresh I, Hickey AS, Bonneau R, Gresham D. bioRxiv 2023. 10.1101/2023.09.21.558277. doi:10.1101/2023.09.21.558277	The second scYeast pre-training set: 173,361 cells from one wild-type strain sampled continuously through a time course without metabolic labeling, so the strain-by-condition count is one and the value is the per-cell distribution of transcriptional states, not a genotype axis. A reference backbone for any single-cell expression head, and the companion of Jackson 2020 on the same platform.	GEO GSE242556 (from the scYeast Methods, Sec. 4.1.3)
27	Brettner 2024 (ultra-high-throughput yeast scRNA-seq) Brettner L, et al. Yeast 2024. doi:10.1002/yea.3927. doi:10.1002/yea.3927	The platform the costing model in experiments/024 is built on. Not a dataset to ingest; the reason it is here is that a yeast Perturb-seq proposal has to name its chemistry.	Yeast (Wiley) SI
28	Mulleder 2012 (prototrophic deletion collection) Mulleder M, Capuano F, Pir P, et al., Ralser M. Nat Biotechnol 2012;30:1176-1178. doi:10.1038/nbt.2442	The background the already-built Mulleder 2016 and Messner 2023 were measured in, and mulleder2016.py already cites it. Listed not as an ingestion target but because that loader records the prototrophy-restoring markers as unmodeled: they are a GeneAddition this schema does not yet carry, and auxotrophy markers distort metabolite pools.	EUROSCARF; Addgene kit 40276
29	Su 2023 (single-cell transcriptomes under four stresses) Su Y, Xu C, Shea J, DeStephanis D, Su Z. BMC Genomics 2023;24:88. doi:10.1186/s12864-023-09184-w	Full-length single-cell RNA-seq of 117 cells across four stress conditions in one strain; scYeast’s stress-state classification benchmark (Fan 2027, Sec. 4.1.6). Tiny, but it is a full-length protocol rather than 3’ tag counting, which matters for isoform-level heads.	GEO GSE201386 (from the scYeast Methods)
30	Wang 2022 (single-cell transcriptomes across replicative aging) Wang J, Sang Y, Jin S, Wang X, Azad GK, McCormick MA, Kennedy BK, Li Q, Wang J, Zhang X, et al. Aging Cell 2022;21:e13712. doi:10.1111/ace1.13712	Three age groups of one strain, 37, 43 and 45 cells each, about 2,200 genes detected per cell. Small, but it is the only yeast single-cell set with age as the condition axis, and scYeast used it to infer age-specific regulatory relationships (Fan 2027, Sec. 4.1.4).	GEO GSE210032 (from the scYeast Methods)
31	Albert 2018 (eQTL in 1,012 segregants) Albert FW, Bloom JS, Siegel J, Day L, Kruglyak L. eLife 2018;7:e35471. doi:10.7554/eLife.35471	Transcriptomes on a thousand sequenced recombinants, so genotype maps to a genome-wide expression vector rather than one trait. The single best existing analog to a Perturb-seq readout on a high-dimensional genotype axis.	GEO /eLife data availability (unconfirmed)
32	Jakobson 2025 (genome-to-proteome map) Jakobson CM, et al. Science 2025;390:eadu3198. doi:10.1126/science.adu3198	Natural alleles mapped to enzyme abundance. A lower-risk engineering lever than a heterologous construct, and the proteome counterpart to the metabolite-QTL panels.	Science data-availability statement (accession unconfirmed)
33	Muenzner 2024 (natural-isolate proteome) Muenzner J, et al., Ralser M. Nature 2024;630:149-157. doi:10.1038/s41586-024-07442-9	Proteomes on the sequenced isolate panel whose transcriptomes Caudal 2024 already supplies. That overlap is the paired anchor the RNA-to-protein inference thesis needs.	PRIDE PXD048219
34	Cooper 2010 (CE-MS amino-acid metabolome) Cooper SJ, Finney GL, Brown SL, Nelson SI, Hesse J, MacCoss MJ, Fields S. Genome Res 2010;20:1288-1296. doi:10.1101/gr.105825.110	An independent platform measuring the same trait class as the already-built Mulleder 2016 on an overlapping genotype axis. Two independent measurements of one phenotype is the cleanest available test of whether a model has learned biology or a batch.	Genome Research SI tables
35	Aulakh 2025 (genome-scale ionome) Aulakh SK, et al. Cell Syst 2025;16:101319. doi:10.1016/j.cels.2025.101319	The only gene-indexed ionome. Metal-cofactor supply gates iron-sulfur and zinc-dependent pathway enzymes, and no built dataset measures it.	Cell Systems SI (accession unconfirmed)

Table 2, continued

# Dataset, citation and link	Why	Data
36 Teyssonniere 2024 (species-wide proteome against transcriptome) Teyssonniere EM, Trebulle P, Muenzner J, Loegler V, Ludwig D, Amari F, Mulleder M, Friedrich A, Hou J, Ralser M, Schacherer J. Proc Natl Acad Sci USA 2024;121:e2319211121. doi:10.1073/pnas.2319211121	Quantitative proteomes for 942 sequenced isolates, the same panel whose transcriptomes the built Caudal 2024 supplies, so the strain-level pairing is one to one rather than an overlap. It is also the published answer to what that pairing shows: mRNA and protein correlate weakly at the population gene level, and the variants associated with protein levels overlap those associated with transcript levels by 3 percent. That makes it the calibration for any RNA-to-protein inference rather than only another proteome. DE-DUPLICATION OPEN: it shares authors and a panel with Muenzner 2024 at 796 isolates, and whether the two are independent acquisitions or one re-served is not established here and must be settled before both are built.	PNAS SI; PMC11087752. PRIDE identifier not fetched this pass
37 Grossbach 2022 (BY x RM transcriptome, proteome and phosphoproteome) Grossbach J, Gillet L, Clement-Ziza M, Schmalohr CL, Schubert OT, Schutter M, Mawer JSP, Barnes CA, Bludau I, Weith M, Tessarz P, Graef M, Aebersold R, Beyer A. Mol Syst Biol 2022;18:e10712. doi:10.15252/msb.202110712	The only row that measures transcript, protein and protein phosphorylation on one set of sequenced strains, 1,862 proteins and 2,116 phosphopeptides over 988 proteins. The pairing it yields is per strain rather than per gene: for each of the 112 genotypes the mRNA level and the protein level of the same gene are both observed, which is what turns the RNA-to-protein question from a population correlation into a within-strain residual. It also reports that genetic variants changing a protein's phosphorylation are mostly different from those changing its abundance, so the phosphosite layer is not a proxy for the protein layer.	Mol Syst Biol SI + PRIDE, identifier not fetched this pass
38 Leutert 2023 (phosphoproteome x 101 conditions) Leutert M, et al., Villen J. Nat Struct Mol Biol 2023;30:1761-1773. doi:10.1038/s41594-023-01115-3	Post-translational enzyme control acts faster than transcription and is invisible to every other layer here. Condition axis only.	PRIDE PXD034997
39 Brauer 2008 (growth-rate-controlled chemostat transcriptome) Brauer MJ, Huttenhower C, Airoidi EM, et al., Botstein D. Mol Biol Cell 2008;19:352-367. doi:10.1091/mbc.e07-08-0779	Separates growth rate from nutrient identity by design, which no batch-culture dataset can. A large share of any expression response is a growth-rate confound, and this is the reference for removing it.	GEO (unconfirmed)
40 Hackett 2016 (SIMMER multi-omic flux) Hackett SR, Zanutelli VRT, Xu W, et al., Rabinowitz JD. Science 2016;354:aaf2786. doi:10.1126/science.aaf2786	The only jointly measured metabolome, proteome and fluxome. No genotype axis, so it is a mechanism prior rather than training data, but it is the only place the three layers are measured on the same cells.	Science SI (accession unconfirmed)
41 Zhu 2014 (kinase / phosphatase lipidomics) Zhu Z, Loewen CJR. Mol Biol Cell 2014. PMC4196872. PMC4196872	A gene-indexed lipidome on a regulatory gene set. Complements the already-built da Silveira 2014 lipidomics, which has no kinase axis, and reads out the malonyl-CoA sink directly.	Mol Biol Cell SI Table S4 (Excel)
42 Gerke 2017 (urea-cycle mQTL) Gerke J, et al., Fay JC. Genetics 2017;206:2199. doi:10.1534/genetics.117.201107	A second mQTL panel from a different cross (oak x wine), so cross-background generalization of metabolite QTL can be tested rather than assumed.	Genetics (GSA) SI
43 Ambroset 2014 (metabolite QTL) Ambroset C, et al., Fay JC. PLoS Genet 2014;10:e1004142. doi:10.1371/journal.pgen.1004142	The founding metabolite-QTL panel. Pairs with Cooper and Mulleder as the natural-variation view of a phenotype space the deletion collection covers by engineering.	PLOS Genetics SI

Table 2, continued

#	Dataset, citation and link	Why	Data
44	Blank 2005 (13C metabolic flux) Blank LM, Kuepfer L, Sauer U. Genome Biol 2005;6:R49. doi:10.1186/gb-2005-6-6-r49	The only real flux measurement tied to single-gene deletions. Everything else in the built set is a concentration or a titer, which is a proxy for flux rather than flux.	Genome Biology open-access SI
45	Skelly 2013 (expression variation across isolates) Skelly DA, Merrihew GE, Riffle M, et al., MacCoss MJ, Akey JM. Genome Res 2013;23:1496-1504. doi:10.1101/gr.155762.113	Transcriptome and proteome on the same isolates, which is the paired anchor the RNA-to-protein inference argument needs. Few strains, but the pairing is measured rather than joined across studies.	Genome Research SI (unconfirmed)
46	Airoidi 2016 (nitrogen-limited steady-state and dynamic transcriptome) Airoidi EM, Miller D, Athanasiadou R, Brandt N, Abdul-Rahman F, Neymotin B, Hashimoto T, Bahmani T, Gresham D. Mol Biol Cell 2016;27:1383-1396. doi:10.1091/mbc.E14-05-1013	Chemostat transcriptomes under controlled nitrogen limitation from the same laboratory, and on the same media axis, as Jackson 2020's NLIM-GLN, NLIM-PRO, NLIM-NH4 and NLIM-UREA conditions. That shared axis is what makes the wild-type baseline of the single-cell atlas separable from the deletion effect, and it is the nitrogen counterpart of Brauer 2008's carbon-limited growth-rate series. Counts are from the citation rather than the released series and need confirming.	GEO, series not confirmed this pass
47	McManus 2014 (ribosome profiling, allele-specific) McManus CJ, May GE, Spealman P, Shteyman A. Genome Res 2014;24:422-430. doi:10.1101/gr.164996.113	Translation efficiency measured gene by gene alongside the mRNA abundance of the same gene, which is the first of the two terms that make protein level differ from transcript level. Its result is the reason the term is needed: ribosome occupancy is more conserved than transcript abundance, so translation buffers divergence in mRNA rather than tracking it. The <i>S. paradoxus</i> arm is out of species and would be stored separately or dropped; the <i>cerevisiae</i> arm and the hybrid's <i>cerevisiae</i> alleles are in scope.	GEO, series not fetched this pass
48	Martin-Perez 2017 (protein half-lives) Martin-Perez M, Villen J. Cell Syst 2017;5:283-294.e5. doi:10.1016/j.cels.2017.08.008	Turnover rates for 3,160 proteins, the second term that makes protein level differ from transcript level. Half-lives span three orders of magnitude and are not normally distributed, so a single global scaling from mRNA to protein cannot be right, and this is the per-gene coefficient that replaces it. The paper also reports that localization, complex membership and connectivity predict turnover better than sequence features do, which names the covariates such a model needs.	Cell Systems SI; PRIDE identifier not fetched this pass
49	Boer 2010 (metabolome across nutrient limitations) Boer VM, Crutchfield CA, Bradley PH, Botstein D, Rabinowitz JD. Mol Biol Cell 2010;21:198-211. doi:10.1091/mbc.e09-07-0597	The metabolite counterpart to Brauer on the same chemostat design, so metabolite pools are separated from growth rate the same way. Covers central-carbon nodes the deletion metabolomics rows measure at low power.	MBoC SI (unconfirmed)
50	Tengolics 2024 (domestication metabolome) Tengolics R, Szappanos B, Mulleder M, et al., Ralser M, Papp B. PNAS 2024;121:e2313354121. doi:10.1073/pnas.2313354121	Shows which metabolic traits already moved under domestication, which is a prior on which are tractable to engineer further. Small n.	PNAS Dataset S11
51	Eder 2020 (flux QTL) Eder M, Nidelet T, Sanchez I, Camarasa C, Legras JL, Dequin S. PMID 32034164. PMID 32034164	Flux as a QTL phenotype, with PDB1 and VID30 validated. The label is modeled rather than measured, which caps how far it can be trusted.	Journal SI (venue and accession unconfirmed)

Table 2, continued

# Dataset, citation and link	Why	Data
52 Foss 2007 (BY x RM segregant proteome) Foss EJ, Radulovic D, Shaffer SA, Ruderfer DM, Bedalov A, Goodlett DR, Kruglyak L. Nat Genet 2007;39:1369-1375. doi:10.1038/ng.2007.22	The first segregant proteome, and the independent replicate of Grossbach 2022 on the same cross fifteen years earlier and on a different platform. Its finding is the one this whole group turns on: the loci that influence protein abundance differ from those that influence transcript levels. Ranked below Grossbach because Grossbach measures the transcript layer on the same strains and this does not, and because neither the segregant count nor the proteome depth was confirmed this pass.	Nat Genet supplementary tables; not fetched
53 Yu 2021 (proteome and metabolome under nitrogen limitation) Yu R, Vorontsov E, Sihlbom C, Nielsen J. eLife 2021;10:e65722. doi:10.7554/eLife.65722	Nitrogen limitation is the standard trigger for redirecting carbon to product, and this measures both layers under it. Condition axis only, and small.	eLife data availability (unconfirmed)
54 de Boer 2020 (100M random promoters) de Boer CG, Vaishnav ED, Sadeh R, Abeyta EL, Friedman N, Regev A. Nat Biotechnol 2020;38:56-65. doi:10.1038/s41587-019-0315-8	The largest sequence-to-phenotype dataset in yeast by orders of magnitude, and every genotype is an exactly known 80mer. The perturbation space is continuous rather than a gene list, which is what a model of regulatory grammar needs and no deletion collection can supply.	GEO (de Boer 2020); DREAM 2022 release of 6,739,258 promoters
55 Vaishnav 2022 (regulatory DNA fitness landscape) Vaishnav ED, de Boer CG, Molinet J, Yassour M, Fan L, Adiconis X, Thompson DA, Levin JZ, Cubillos FA, Regev A. Nature 2022;603:455-463. doi:10.1038/s41586-022-04506-6	Extends de Boer from measurement to a fitness landscape, and tests native and natural-isolate promoter variants against the random library. The bridge between synthetic sequence space and the natural variation the isolate panels carry.	GEO per the Nature data-availability statement (unconfirmed)
56 Lee 2014 (HIP-HOP fitness signatures) Lee AY, St Onge RP, Proctor MJ, et al., Giaever G, Nislow C. Science 2014;344:208-211. doi:10.1126/science.1250217	Largest chemical-genetic matrix in existence, on the exact YKO genotype axis Costanzo and Kuzmin already use. Biggest single instance-count gain available. BLOCKED: WS15 already attempted it and it stands as awaiting author matrices, so the Science SI and the lab portal did not yield per-strain values. Unblocking is an author request, not a loader.	Science SI + Nislow/Giaever portal, neither of which yielded per-strain matrices; awaiting author matrices per the WS15 roadmap
57 Nguyen Ba 2022 (barcoded bulk QTL, 100k segregants) Nguyen Ba AN, Lawrence KR, Rego-Costa A, Gopalakrishnan S, Temko D, Michor F, Desai MM. eLife 2022;11:e73983. doi:10.7554/eLife.73983	The largest genotyped recombinant panel in yeast, each segregant barcoded and low-coverage sequenced against known parents. Two orders of magnitude more genotypes than Bloom 2013 on the same kind of cross, which is what makes small-effect and epistatic loci detectable.	eLife data availability; SRA (unconfirmed)
58 Galardini 2019 (four backgrounds x 38 conditions) Galardini M, Busby BP, Vieitez C, Dunham AS, Typas A, Beltrao P. Mol Syst Biol 2019;15:e8831. doi:10.15252/msb.20198831	The same knockout built independently in four sequenced backgrounds. Quantifies the 18.5 percent of deletion phenotypes that do not transfer, which is the generalization risk the whole substrate is exposed to.	GEO GSE123118 + github.com/mgalardini/2018koyeast
59 Parsons 2006 (bioactive-compound profiling) Parsons AB, Lopez A, Givoni IE, et al., Boone C. Cell 2006;126:611-625. doi:10.1016/j.cell.2006.06.040	Full-collection strain coverage against a modest compound panel, the complement to the diagnostic-panel screens above.	Cell SI tables (accession unconfirmed)

Table 2, continued

# Dataset, citation and link	Why	Data
60 Dutta 2026 (barcoded natural-isolate chemical response) Dutta A, Garin M, Loegler V, Brach G, Friedrich A, Yoshimura M, Hirano H, Osada H, Boone C, Yashiroda Y, Hou J, Schacherer J. Nat Commun 2026. doi:10.1038/s41467-026-73532-z	Natural sequence diversity crossed with a chemogenomic panel, on isolates drawn from the sequenced 1,011 collection. The same strains already carry Caudal 2024 transcriptomes and Muenzner 2024 proteomes, so it is a three-modality join on one genotype axis.	Nat Commun SI; isolates from the 1,011 panel (ENA)
61 Li 2016 (tRNA fitness landscape) Li C, Qian W, Maclean CJ, Zhang J. Science 2016;352:837-840. doi:10.1126/science.aae0568	A near-complete fitness landscape of one native yeast gene. Every genotype is a known sequence and the sampling is exhaustive rather than one allele per gene, which is the opposite of what a deletion collection provides.	Science SI (unconfirmed)
62 Peter 2018 (1,011 isolate genomes + phenome) Peter J, De Chiara M, Friedrich A, et al., Schacherer J. Nature 2018;556:339-344. doi:10.1038/s41586-018-0030-5	The genome backbone every other natural-variation row resolves against, and it carries its own phenotype panel. 1,625,809 high-quality SNPs at 232-fold mean coverage.	ENA/SRA; phenotype tables via the paper and France Genomique
63 Domingo 2018 (tRNA double-mutant landscape) Domingo J, Diss G, Lehner B. Nature 2018;558:117-121. doi:10.1038/s41586-018-0170-7	Pairwise epistasis at nucleotide resolution inside one gene. The higher-order structure Costanzo and Kuzmin map between genes, measured within a gene, which is the resolution inverse design actually needs.	Nature SI (unconfirmed)
64 Zhang 2021 (YETI titratable overexpression) Zhang Y, Ho Yee Chow M, Ling Y, et al., Andrews BJ. Mol Syst Biol 2021;17:e10321. doi:10.15252/msb.202010321	Native-locus, continuously titratable overexpression. The dosage axis pathway engineering actually uses, and the only gain-of-function library that is not binary.	Mol Syst Biol SI (repository accession unconfirmed)
65 Peeters 2021 (fermentation-trait QTL atlas) Peeters B, Reijbroek F, Verbaet J, et al., Jarosz DF, Verstrepen KJ. 2021. PMID 34727964. PMID 34727964	The only large sequenced panel with direct isobutanol, ethanol and glycerol titers. Hits the natural-isolate and the isobutanol asks at once.	Journal SI; raw reads at SRA/ENA (accession unconfirmed)
66 Van Leeuwen 2024 (short-chain organic acids) Shared and specific genetic determinants of tolerance to acetic, butyric and octanoic acid. PMC10903034. PMC10903034	Three chemically homologous acids on one strain panel, so shared and acid-specific mechanisms separate within a single study rather than across incomparable screens.	Journal SI (open-access PMC)
67 Chica 2026 (AutoDRY autophagy screen) Chica N, et al. Nat Cell Biol 2026;28:465-479. doi:10.1038/s41556-025-01837-0	Largest single-condition imaging screen with a live Dryad deposit, and it covers essential-adjacent genes via DAmP rather than stopping at the non-essential set.	Dryad 10.5061/dryad.cfxpvnxdh
68 Sopko 2006 (genome-scale overexpression toxicity) Sopko R, Huang D, Preston N, et al., Boone C, Andrews B. Mol Cell 2006;21:319-330. doi:10.1016/j.molcel.2005.12.011	769 genes are growth-inhibitory when overexpressed. Overexpression toxicity is the constraint that caps titer when a pathway enzyme is pushed, and nothing in the built set measures it.	Mol Cell SI tables
69 Liu 2021 (tryptophan / isobutanol tolerance) Liu H-L, Wang CH-T, Chiang EP-I, Huang C-C, Li W-H. Biotechnol Biofuels 2021;14:200. doi:10.1186/s13068-021-02048-z	A second, independent full-collection isobutanol screen with a different readout. Paired with Kuroda it gives a same-phenotype, same-genotype, different-method replicate.	GEO GSE175794 (companion RNA-seq); screen in BMC SI

Table 2, continued

# Dataset, citation and link	Why	Data
70 Kuroda 2019 (isobutanol-specific tolerance) Kuroda K, Hammer SK, Watanabe Y, Montano Lopez J, Fink GR, Stephanopoulos G, Ueda M, Avalos JL. Cell Syst 2019;9:534-547.e5. doi:10.1016/j.cels.2019.10.006	The isobutanol screen, designed against an ethanol comparator so the alcohol-general response subtracts out. Its GLN3 hit raised production 4.9-fold, a rare tolerance-to-titer translation.	ArrayExpress E-MTAB-8175 (RNA-seq); screen data in Cell Syst SI
71 Turco 2023 (Yeast Phenome) Turco G, Chang C, Wang RY, et al., Boone C, Andrews BJ, Roth FP. Sci Adv 2023;9:eadg5702. doi:10.1126/sciadv.adg5702	The widest environment axis in yeast, 96 percent chemical. Held out of tier 1 because it is a meta-aggregation: its instance count re-serves primary screens, several already built, so the net-new fraction is unknown until it is de-duplicated against the rest of this table. A loader already exists and retains 49 growth screens, so this is in flight rather than net-new work; Lee 2014 is not among those screens, so YeastPhenome does not backfill row 1.	yeastphenome.org (Zenodo 10.5281/zenodo.7714347); loader torchcell/datasets/scerevisiae/yeastphenome.py, 49 screens, not yet in the built set
72 Rossi 2021 (ChIP-exo protein architecture) Rossi MJ, Kuntala PK, Lai WKM, et al., Pugh BF. Nature 2021;592:309-314. doi:10.1038/s41586-021-03314-8	Where roughly 400 regulatory proteins sit on the genome at near-base-pair resolution. This is the wiring prior that makes a regulatory-DNA model mechanistic instead of correlational, and it pairs directly with the MPRA rows.	yeastepigenome.org; github.com/CEGRcode/2021-Rossi_Nature
73 Piotrowski 2017 (MOSAIC diagnostic panel) Piotrowski JS, Li SC, Deshpande R, et al., Boone C, Myers CL. Nat Chem Biol 2017;13:982-993. doi:10.1038/nchembio.2436	Highest compound count of any yeast screen. Held out of tier 1 on the genotype bar: 157 strains buys condition breadth, not genotype breadth, so it trains a compound embedding rather than a genotype model.	MOSAIC portal (mosaic.cs.umn.edu) + Nat Chem Biol SI
74 Bloom 2025 (global epistasis in a yeast cross) Bloom JS, et al., Kruglyak L. 2025. PMID 40679398. PMID 40679398	Reports that many natural variants have effects that scale with background fitness, which is a specific, testable claim about how genotype effects compose. Directly relevant to whether a model trained on one background transfers.	journal SI (unconfirmed; recent)
75 Gasch 2000 (environmental stress response) Gasch AP, Spellman PT, Kao CM, et al., Brown PO. Mol Biol Cell 2000;11:4241-4257. doi:10.1091/mbc.11.12.4241	The canonical stress-response signature. Every genotype-indexed transcriptome in the built set needs it as the background any generic stress response is subtracted against.	SPELL /SGD Expression Connection; Stanford legacy mirror
76 Decourty 2021 (RNA-metabolism genetic interactions) Decourty L, et al., Saveanu C. Nucleic Acids Res 2021;49:8535-8555. doi:10.1093/nar/gkab680	Structurally identical to Costanzo and Kuzmin, so the existing loader pattern applies directly, and it extends the interaction network into RNA processing.	NAR open-access SI (accession unconfirmed)
77 Cuperus 2017 (random 5' UTR library) Cuperus JT, Groves B, Kuchina A, Rosenberg AB, Jovic N, Fields S, Seelig G. Genome Res 2017;27:2015-2024. doi:10.1101/gr.224964.117	The translational layer of the same idea at half a million exactly known sequences. Expression is set after transcription as well as at it, and nothing else here measures that axis.	GEO (unconfirmed)
78 Faure 2022 (doubledeepPCA) Faure AJ, Domingo J, Schmiedel JM, Hidalgo-Carcedo C, Diss G, Lehner B. Nature 2022;604:175-183. doi:10.1038/s41586-022-04586-4	Separates a variant's effect on folding from its effect on binding by measuring both in the same library. That decomposition is what turns a fitness landscape into a mechanistic one.	Nature data availability (unconfirmed)
79 Chong 2015 / CYCLOPs (single-cell proteome dynamics) Chong YT, Koh JLY, Friesen H, et al., Andrews BJ, Boone C. Cell 2015;161:1413-1424. doi:10.1016/j.cell.2015.04.051	Per-cell rather than per-strain-mean labels, over 20 million cells. The distributional readout a Perturb-seq design has to be scored against.	thecellvision.org/cyclops (bulk CSV)

Table 2, continued

#	Dataset, citation and link	Why	Data
80	Teyssonniere 2024 (species-wide trait survey) Teyssonniere EM, Shichino Y, Mito M, Friedrich A, Iwasaki S, Schacherer J. PLoS Genet 2024;20:e1011119. doi:10.1371/journal.pgen.1011119	Extends the phenotype axis of the sequenced panel far past Peter 2018's 36 conditions, and classifies traits by how monogenic or polygenic they are. That classification is a prior on which phenotypes a model can be expected to predict at all.	PLoS Genetics SI (unconfirmed)
81	Diss 2018 (Fos-Jun interaction double mutants) Diss G, Lehner B. eLife 2018;7:e32472. doi:10.7554/eLife.32472	Exhaustive pairwise mutation of an interaction interface, read out in yeast. The genotype is a heterologous domain rather than a yeast gene, so it enters as an engineered chassis, but the epistasis structure is the deepest in the table.	eLife data availability (unconfirmed)
82	Warringer 2011 (trait variation across the SGRP panel) Warringer J, Zorgo E, Cubillos FA, et al., Blomberg A, Liti G. PLoS Genet 2011;7:e1002111. doi:10.1371/journal.pgen.1002111	A very wide condition panel on sequenced isolates, with growth-curve parameters rather than endpoint fitness. Pairs with Scan-o-matic as an actually-released panel rather than a platform.	PLoS Genetics SI (unconfirmed)
83	Keren 2013 (promoter activity across conditions) Keren L, Zackay O, Lotan-Pompan M, Barenholz U, Dekel E, Sasson V, Aidelberg G, Bren A, Zeevi D, Weinberger A, Alon U, Milo R, Segal E. Mol Syst Biol 2013;9:701. doi:10.1038/msb.2013.59	Native promoter strength measured directly, across conditions and over time, rather than inferred from mRNA. This is the closest published answer to the question of how strong every promoter actually is.	Mol Syst Biol SI (unconfirmed)
84	Puchta 2016 (snoRNA U3 landscape) Puchta O, Cseke B, Czaja H, Tollervey D, Sanguinetti G, Kudla G. Science 2016;352:840-844. doi:10.1126/science.aaf0965	The companion landscape on a non-coding RNA, published alongside Li 2016. Two exhaustive landscapes on different molecule classes is a stronger test of a sequence model than either alone.	Science SI (unconfirmed)
85	Yoshikawa 2011 (deletion + overexpression phenome) Yoshikawa K, Tanaka T, Furusawa C, Nagahisa K, Hirasawa T, Shimizu H. G3 2011;1:247-267. doi:10.1534/g3.111.000695	The same genes phenotyped in both loss- and gain-of-function side by side. A matched over/under pair on one platform is what a dosage-aware model needs and nothing else here provides.	G3 open-access SI
86	Ho 2009 (MoBY-ORF barcoded overexpression) Ho CH, Magtanong L, Barker SL, et al., Boone C. Nat Biotechnol 2009;27:369-377. doi:10.1038/nbt.1534	Native-promoter overexpression against a compound panel, giving a third dosage state alongside deletion and knockdown on the same chemogenomic readout.	moby.ccbr.utoronto.ca + Andrews lab portal
87	Bloom 2013 (1,008 BYxRM segregants) Bloom JS, Ehrenreich IM, Loo WT, Lite TL, Kruglyak L. Nature 2013;494:234-237. doi:10.1038/nature11867	The reference two-parent panel the field calibrates against, and the direct predecessor of Bloom 2019 and Nguyen Ba 2022. Ingesting all three gives one cross at three panel sizes, which is a clean test of how much genotype coverage a model actually needs.	Nature SI
88	Xiao 2014 (genome-wide RNAi furfural tolerance) Xiao H, Zhao H. Biotechnol Biofuels 2014;7:78. doi:10.1186/1754-6834-7-78	Furfural via a knockdown rather than a deletion axis, so it reaches essential genes and gives a dose-graded rather than binary perturbation on the flagship hydrolysate inhibitor.	BMC open-access SI
89	Douglas 2012 (barcoded overexpression fitness) Douglas AC, Smith AM, Sharifpoor S, et al., Andrews BJ, Boone C, Nislow C. G3 2012;2:1279-1289. doi:10.1534/g3.112.003400	The same barcode-sequencing pipeline as the already-built Hillenmeyer FitDb, run in the opposite dosage direction. Lowest harmonization cost of any overexpression row.	G3 open-access SI
90	She 2018 (natural variant effects across a large isolate panel) She R, Jarosz DF. Cell 2018;172:478-490. doi:10.1016/j.cell.2017.12.015	Maps how often a natural variant's effect depends on the strain it sits in, across a wide condition panel. The natural-variation counterpart to Galardini's four-background deletion tensor.	Cell SI (unconfirmed)

Table 2, continued

#	Dataset, citation and link	Why	Data
91	Costello 2020 (bioreactor Bar-seq) Costello Z, Wehrs M, Mukhopadhyay A, et al. Microb Cell Fact 2020;19:167. doi:10.1186/s12934-020-01423-z	The only fitness measurement at bioreactor scale. Shake-flask ranking is known to diverge from fed-batch, and nothing else here tests that.	Microb Cell Fact open-access SI
92	Gurvitz (fatty-acid utilization screen) Gurvitz A, et al. Mol Syst Biol (EMBO Press). embopress.org/doi/pdf/10.1038/msb4100051	Whole-collection coverage against two fatty-acid substrates. Relevant to the lipid and malonyl-CoA branch, at the cost of manual extraction from an embedded table.	Mol Syst Biol Table I (PDF-embedded)
93	Mira 2010 (acetic acid tolerance, full collection) Mira NP, Palma M, Guerreiro JF, Sa-Correia I. Microb Cell Fact 2010;9:79. doi:10.1186/1475-2859-9-79	About 650 tolerance determinants on the full deletion collection at controlled pH. The loss-of-function counterpart to the already-built Mormino biosensor screen and to Mukherjee's essential-gene CRISPRi, so the same phenotype is covered by deletion, knockdown and biosensor.	Microb Cell Fact SI (PMC2972246)
94	Gameiro 2025 (AID2 degron libraries) Gameiro E, Juarez-Nunez KA, Fung JJ, Shankar S, Luke B, Khmelinskii A. J Cell Biol 2025;224:e202409007. doi:10.1083/jcb.202409007	Acute depletion on a minutes-to-hours timescale, covering essential genes without the pre-adaptation a deletion strain has already done.	J Cell Biol SI; strains via IMB Mainz / EUROSCARF
95	Si 2017 (RAGE RNAi, isobutanol titer) Si T, Chao R, Min Y, Wu Y, Ren W, Zhao H. Nat Commun 2017;8:15187. doi:10.1038/ncomms15187	One of very few library screens whose endpoint is a titer in g/L rather than a fitness proxy, and it is iterative, so it carries combinatorial rather than single-locus effects.	Nat Commun SI (accession unconfirmed)
96	Hamedirad 2018 (RAGE xylose utilization) Hamedirad M, Lian J, Li H, Zhao H. Biotechnol Bioeng 2018;115:1552-1560. doi:10.1002/bit.26570	Pentose utilization is the other half of the recalcitrant-biomass problem, and no built dataset covers a xylose selection.	Biotechnol Bioeng SI (accession unconfirmed)
97	Pereira 2014 (wheat-straw hydrolysate) Pereira FB, Guimaraes PMR, Gomes DG, et al., Domingues L. J Ind Microbiol Biotechnol 2014;41:1753-1761. doi:10.1007/s10295-014-1519-z	Real industrial wheat-straw hydrolysate rather than reconstituted single toxins. The recalcitrant-biomass condition as a process actually presents it.	Journal SI
98	Endo 2008 (vanillin tolerance) Endo A, Nakamura T, Ando A, Tokuyasu K, Shima J. Appl Environ Microbiol 2008;74:7175-7185. doi:10.1128/AEM.01541-08	Vanillin is the one major hydrolysate phenolic with no dedicated screen in the built set; it is currently only folded into broad compound panels.	AEM SI (PMC2375868)
99	Sharifpoor 2012 (kinome synthetic dosage lethality) Sharifpoor S, van Dyk D, Costanzo M, et al., Boone C, Andrews BJ. Genome Res 2012;22:791-801. doi:10.1101/gr.129213.111	The overexpression-by-deletion axis, which neither the built SynLethDB nor Costanzo and Kuzmin carries: SynLethDB models only synthetic lethality and synthetic rescue, both two-loss-of-function, so synthetic dosage lethality is out of its scope by construction and this paper's PMID is absent from both its layers. Kinases are also the regulatory nodes flux redirection actually targets.	Genome Research SI
100	Mavor 2016 (ubiquitin DMS across conditions) Mavor D, Barlow K, Thompson S, et al., Bolon DNA, Fraser JS. eLife 2016;5:e15802. doi:10.7554/eLife.15802	A complete single-mutant scan of an essential yeast gene, repeated across stresses. Variant effects are conditional, and this is the cleanest demonstration of that in the table.	eLife data availability (unconfirmed)

Table 2, continued

#	Dataset, citation and link	Why	Data
101	Flynn 2020 (Hsp90 DMS across environments) Flynn JM, Rossouw A, Cote-Hammarlof P, et al., Bolon DNA. eLife 2020;9:e53810. doi:10.7554/eLife.53810	A chaperone hub scanned under multiple stresses, so the environment-dependence of a variant effect is measured rather than assumed. Hsp90 buffers exactly the variation the natural-isolate rows carry.	eLife data availability (unconfirmed)
102	Ho 2018 (unified absolute protein abundance) Ho B, Baryshnikova A, Brown GW. Cell Syst 2018;6:192-205. doi:10.1016/j.cels.2017.12.004	Reconciles 21 independent abundance datasets into one unified estimate per protein. A better absolute anchor than any single study, including Ghaemmaghani, and the reference frame every relative proteomics row needs.	Cell Systems SI
103	Renganaath 2020 (natural promoter-variant MPRA) Renganaath K, Cheung R, Day L, Kosuri S, Kruglyak L, Albert FW. eLife 2020;9:e62669. doi:10.7554/eLife.62669	Native promoter variation measured allele by allele rather than sampled randomly, with 451 variants called causal. The row that joins the synthetic MPRA libraries to the natural-isolate panels: same readout, real alleles.	eLife data availability; GEO (unconfirmed)
104	Chang 2012 (organic-acid production screen) Chang HJ, Suga H, et al. 2012. PMID 22277779. PMID 22277779	A whole-collection screen whose label is production rather than fitness. Coarse and PDF-only, but organic-acid overproduction at collection scale exists nowhere else.	PDF tables only; manual transcription required
105	Trikka 2015 (carotenogenic heterozygous screen) Trikka FA, Nikolaidis A, Athanasakoglou A, et al., Kampranis SC, Makris AM. Microb Cell Fact 2015;14:60. doi:10.1186/s12934-015-0246-0	The second carotenoid deletion-collection screen, and the only haploinsufficiency design here. Halving dosage finds genes a null would kill, which is where the isoprenoid flux control sits. Currently listed as figure-only, so confirm Additional file 1 carries per-strain scores.	BMC Additional files 1-2
106	Sardi 2018 (natural variation in hydrolysate tolerance) Sardi M, Paithane V, Place M, Robinson DE, Hose J, Wohlbach DJ, Gasch AP. PLoS Genet 2018;14:e1007217. doi:10.1371/journal.pgen.1007217	Natural variation in exactly the bioprocess phenotype the deletion screens cover by engineering. Pairs with Vanacloig-Pedros and Pereira to ask whether tolerance loci found by deletion recur as natural alleles.	PLoS Genetics SI (unconfirmed)
107	Breslow 2008 (DAmP hypomorphic collection) Breslow DK, Cameron DM, Collins SR, et al., Weissman JS. Nat Methods 2008;5:711-718. doi:10.1038/nmeth.1234	Partial loss of function for essential genes at the native locus, about 82 percent of the essential genome. The built set has no essential-gene hypomorph axis at all.	Nat Methods SI; Horizon YSC5050/5090/5093/5094
108	Kofoed 2015 (barcoded ts alleles) Kofoed M, Milbury KL, Chiang JH, et al., Hieter P, Stirling PC. G3 2015;5:1879-1887. doi:10.1534/g3.115.019174	Conditional essential-gene loss of function as point mutants at native loci, barcoded so it runs through the same pooled pipeline as the deletion collection.	G3 open-access SI; strains via Dharmacon/Horizon
109	Zhang 2022 (GCE-SCRaMbLE recombination outcomes) Zhang W, Lazar-Stefanita L, Yamashita H, et al., Boeke JD. Nat Commun 2022;13:5836. doi:10.1038/s41467-022-33606-0	Each cell carries tens of simultaneous structural edits, sequenced afterwards. The deepest per-cell perturbation in the table: a genotype here is a rearranged chromosome, not an allele, which is what inverse strain design eventually has to predict on.	mirrored as zhangSystematicDissectionKey2022; SRA (unconfirmed)
110	Cubillos 2017 (nitrogen-consumption QTL) Cubillos FA, Brice C, Molinet J, et al., Martinez C. G3 2017;7:1693-1705. doi:10.1534/g3.117.042127	One of the few QTL panels whose genotype location is confirmed rather than assumed, and nitrogen assimilation sets higher-alcohol and ester formation.	BioProject PRJNA379146 (RNA-seq); genotypes in Cubillos 2013 Table S1
111	Hillenmeyer 2010 (systematic fitness reanalysis) Hillenmeyer ME, Ericson E, Davis RW, Nislow C, Koller D, Giaever G. Genome Biol 2010;11:R30. doi:10.1186/gb-2010-11-3-r30	The reprocessed and normalized form of the FitDb compendium, with per-condition significance. Ingesting the reanalysis rather than the raw release avoids redoing the normalization the authors already validated.	Genome Biology SI (unconfirmed)

Table 2, continued

#	Dataset, citation and link	Why	Data
112	Smith 2008 (gene-environment interaction eQTL) Smith EN, Kruglyak L. PLoS Biol 2008;6:e83. doi:10.1371/journal.pbio.0060083	Transcriptomes on the same segregants in two carbon sources, so an expression QTL can be shown to be condition-dependent. Small, and it is the design Boocock 2025 scales up.	GEO (unconfirmed)
113	Braberg 2020 (point-mutant E-MAP) Braberg H, et al., Krogan NJ. Science 2020;370:eaaz4910. doi:10.1126/science.aaz4910	Sub-gene resolution genetic interactions. Narrow in gene scope, but it is the only demonstration that the interaction formalism extends below the ORF.	Science SI (accession unconfirmed)
114	Breker 2013 (GFP library under stress) Breker M, Gymrek M, Schuldiner M. J Cell Biol 2013;200:839-850. doi:10.1083/jcb.201301120	The GFP collection re-imaged under stress rather than at baseline, so relocalization is measured as a response. Feeds LoQAtE, and adds the condition axis Huh 2003 lacks.	LoQAtE (weizmann.ac.il/molgen/loqate)
115	Tkach 2012 (protein relocalization under DNA damage) Tkach JM, Yimit A, Lee AY, et al., Andrews BJ, Brown GW. Nat Cell Biol 2012;14:966-976. doi:10.1038/ncb2549	The GFP collection imaged under damage, giving a second condition axis on the imaging layer alongside Breker's stress panel. Relocalization is a response no abundance measurement captures.	TheCellVision /Nat Cell Biol SI (unconfirmed)
116	Ehrenreich 2010 (X-QTL bulk segregant mapping) Ehrenreich IM, Torabi N, Jia Y, Kent J, Martis S, Shapiro JA, Gresham D, Caudy AA, Kruglyak L. Nature 2010;464:1039-1042. doi:10.1038/nature08923	Millions of segregants assayed in bulk rather than individually, so power comes from pool size instead of strain construction. The phenotype is per-pool rather than per-strain, which is a different record type and needs a pooled-genotype schema.	Nature SI (unconfirmed)
117	Denervaud 2013 (microfluidic GFP dynamics) Denervaud N, Becker J, Delgado-Gonzalo R, et al., Maerkl SJ. PNAS 2013;110:15842-15847. doi:10.1073/pnas.1308265110	Single-cell protein dynamics under controlled environmental switches, rather than endpoint snapshots. The temporal axis every other imaging row lacks.	PNAS SI (unconfirmed)
118	Brogaard 2012 (chemical nucleosome map) Brogaard K, Xi L, Wang JP, Widom J. Nature 2012;486:496-501. doi:10.1038/nature11142	Nucleosome positioning at base-pair resolution genome-wide. Promoter accessibility is what the MPRA rows are implicitly measuring, and this is the structural map underneath it.	GEO (unconfirmed)
119	Xia 2022 (proteome allocation across conditions) Xia J, Sanchez BJ, Chen Y, Campbell K, Kasvandik S, Nielsen J. Nat Commun 2022;13:2819. doi:10.1038/s41467-022-30513-2	Proteome allocation measured across a designed condition grid and used to parameterize enzyme-constrained models. The bridge between this table's proteomics rows and the Yeast9 modeling scaffold.	Nat Commun SI (unconfirmed)
120	Melamed 2013 (PAB1 RRM domain scan) Melamed D, Young DL, Gamble CE, Miller CR, Fields S. RNA 2013;19:1537-1551. doi:10.1261/rna.040709.113	Dense coverage of one RNA-binding domain including double mutants, on a native essential yeast gene.	journal SI (unconfirmed)
121	De Chiara 2022 (domestication and genome instability) De Chiara M, Barre BP, Persson K, et al., Liti G. Nat Ecol Evol 2022;6:761-773. doi:10.1038/s41559-022-01671-9	Ties genome instability and aneuploidy in the isolate panel to phenotype. Industrial strains acquire aneuploidy under selection, so this is the natural-variation view of a failure mode long fermentations create.	Nat Ecol Evol SI (unconfirmed)
122	Yvert 2013 (single-cell trait variation across isolates) Yvert G, Ohnuki S, Nogami S, Imanaga Y, Fehrmann S, Schacherer J, Ohya Y. BMC Syst Biol 2013;7:54. doi:10.1186/1752-0509-7-54	CalMorph morphology on natural isolates rather than deletion strains, in the same feature space as the already-built Ohya and Ohnuki datasets. Joins the morphology layer to the natural-variation axis without harmonization.	BMC SI (unconfirmed)

Table 2, continued

#	Dataset, citation and link	Why	Data
123	Lahtvee 2017 (absolute proteome across conditions) Lahtvee PJ, Sanchez BJ, Smialowska A, et al., Nielsen J. Cell Syst 2017;4:495-504. doi:10.1016/j.cels.2017.03.003	Absolute enzyme copy numbers across conditions, which is the quantity enzyme-constrained metabolic models consume. Ties this table's proteomics to the Yeast9 modeling scaffold.	Cell Systems SI (unconfirmed)
124	Weill 2018 (SWAT libraries) Weill U, Yofe I, Sass E, et al., Schuldiner M. Nat Methods 2018;15:617-622. doi:10.1038/s41592-018-0044-9	The library the degron rows are built from. Ingest for its abundance and localization tables; its main value is as the cassette definition those rows' sequences resolve against.	Nat Methods SI; strains via EUROSCARF
125	Kinsler 2020 (barcoded pleiotropy across environments) Kinsler G, Geiler-Samerotte K, Petrov DA. eLife 2020;9:e61271. doi:10.7554/eLife.61271	Sequenced adaptive clones assayed across dozens of environments, so one genotype yields a fitness vector rather than a point. Included because the clones were sequenced; the parent barcoding study was not, and is excluded.	eLife data availability (unconfirmed)
126	Pelechano 2013 (transcript isoform landscape) Pelechano V, Wei W, Steinmetz LM. Nature 2013;497:127-131. doi:10.1038/nature12121	Shows a single gene produces many transcript isoforms with distinct 5' and 3' ends. Every expression row in this table collapses that to one number per gene, and this is the map of what that collapse discards.	ArrayExpress /GEO (unconfirmed)
127	Deutschbauer 2005 (haploinsufficiency, genome-wide) Deutschbauer AM, Jaramillo DF, Proctor M, Kumm J, Hillenmeyer ME, Davis RW, Nislow C, Giaever G. Genetics 2005;169:1915-1925. doi:10.1534/genetics.104.036871	Gene-dosage sensitivity genome-wide, from halving rather than removing. Dosage-sensitive genes are exactly the ones whose overexpression in a pathway will misbehave.	Genetics SI (unconfirmed)
128	Valenti 2025 (SWAT degron + GFP collection) Valenti M, et al. J Cell Biol 2025;224:e202409050. doi:10.1083/jcb.202409050	Depletion and its abundance consequence measured in the same strain, so enzyme dosage and its phenotype are not joined across two experiments.	J Cell Biol SI
129	Sousa 2013 (acetate resistance overexpression screen) Sousa M, Duarte AM, Fernandes TR, et al. 2013. PMID 23262128. PMID 23262128	Gain-of-function on the same stress the deletion and CRISPRi rows cover by loss. Over and under the same condition is what makes a dosage-aware model trainable rather than sign-blind.	journal SI (unconfirmed)
130	Sadhu 2016 (CRISPR-directed causal variant mapping) Sadhu MJ, Bloom JS, Day L, Kruglyak L. Science 2016;352:1113-1116. doi:10.1126/science.aaf5124	Drives recombination to a chosen point, so a locus can be swapped against a fixed background rather than inherited with everything linked to it. Turns correlational QTL into causal tests.	Science SI (unconfirmed)
131	Weinberg 2016 (ribosome profiling, improved) Weinberg DE, Shah P, Eichhorn SW, Hussmann JA, Plotkin JB, Bartel DP. Cell Rep 2016;14:1787-1799. doi:10.1016/j.celrep.2016.01.043	The corrected reference translation-efficiency dataset, after earlier profiling runs were shown to carry a drug-induced artifact. Translation efficiency is the step between the transcriptome and proteome layers this table otherwise joins by assumption.	GEO (unconfirmed)
132	Ghaemmaghani 2003 (absolute protein abundance) Ghaemmaghani S, Huh WK, Bower K, et al., Weissman JS. Nature 2003;425:737-741. doi:10.1038/nature02046	Converts every relative proteomics row here into molecules per cell. Without an absolute anchor, enzyme-cost and flux-capacity arguments have no units.	Nature SI (Excel)
133	Huh 2003 (GFP localization atlas) Huh WK, Falvo JV, Gerke LC, et al., O'Shea EK. Nature 2003;425:686-691. doi:10.1038/nature02026	The wild-type localization baseline every perturbation-imaging dataset measures change against. Compartment identity is a prerequisite for compartmentalized pathway engineering.	SGD /YeastGFP mirror

Table 2, continued

#	Dataset, citation and link	Why	Data
134	Michaelis 2023 (protein interactome) Michaelis AC, et al., Mann M. Nature 2023;624:192-200. doi:10.1038/s41586-023-06739-5	A physical-interaction layer to constrain which genetic interactions reflect complex membership. A graph prior rather than a phenotype.	yeast-interactome.biochem.mpg.de + ProteomeXchange
135	Christiano 2014 (protein turnover) Christiano R, Nagaraj N, Frohlich F, Walther TC. Cell Rep 2014;9:1959-1965. doi:10.1016/j.celrep.2014.10.065	Half-lives genome-wide. Abundance is production over degradation, and every other proteomics row here measures only the product, so turnover is the missing half.	Cell Reports SI (unconfirmed)
136	Kitzman 2015 (GAL4 DMS) Kitzman JO, Starita LM, Lo RS, Fields S, Shendure J. Nat Methods 2015;12:203-206. doi:10.1038/nmeth.3223	A transcription factor scanned residue by residue, which pairs with the regulatory-DNA rows: one measures the site, the other the protein that reads it.	Nat Methods SI (unconfirmed)
137	Sharon 2012 (designed promoter library) Sharon E, Kalma Y, Sharp A, et al., Segal E. Nat Biotechnol 2012;30:521-530. doi:10.1038/nbt.2205	Designed rather than random, so site number, affinity, spacing and orientation vary one at a time. Small, but it is the controlled counterpart the random libraries lack.	Nat Biotechnol SI (unconfirmed)
138	Sadhu 2018 (CRISPR-directed mitotic recombination) Sadhu MJ, Bloom JS, Day L, Siegel JJ, Kosuri S, Kruglyak L. eLife 2018;7:e41090. doi:10.7554/eLife.41090	Uses targeted recombination to build a genotype series with a single locus swapped against a fixed background, isolating causal effects the way a QTL panel cannot.	eLife data availability (unconfirmed)
139	Duveau 2021 (TDH3 promoter allele series) Duveau F, Vande Zande P, Metzger BP, et al., Wittkopp PJ. eLife 2021;10:e67806. doi:10.7554/eLife.67806	One native promoter dissected at its own locus, with expression and fitness on the same alleles across environments. The rare case tying a regulatory change to a fitness consequence rather than a reporter number.	eLife data availability (unconfirmed)
140	Strope 2015 (100 clinical and natural genomes) Strope PK, Skelly DA, Kozmin SG, Mahadevan G, Stone EA, Magwene PM, Dietrich FS, McCusker JH. Genome Res 2015;25:762-774. doi:10.1101/gr.185538.114	A sequenced panel enriched for clinical isolates, which sit outside the vineyard and laboratory backgrounds most panels oversample. Diversity of origin is what makes a natural-variation training set generalize.	Genome Research SI; SRA (unconfirmed)
141	Schulte 2023 (mitochondrial complexome) Schulte U, et al. Nature 2023;614:153-159. doi:10.1038/s41586-022-05641-w	Mitochondrial compartmentalization is a live strategy for sequestering toxic intermediates, and this maps the import and assembly bottlenecks that would cap it.	complexomics.org + ProteomeXchange
142	Metzger 2015 (promoter mutation effects on expression and noise) Metzger BPH, Yuan DC, Gruber JD, Duveau F, Wittkopp PJ. Nature 2015;521:344-346. doi:10.1038/nature14424	Single mutations in one native promoter, each with expression and noise measured. The mutational spectrum underlying the natural promoter variation the MPRA rows survey.	Nature SI (unconfirmed)
143	Venkataram 2016 (barcoded adaptive lineages, sequenced) Venkataram S, Dunn B, Li Y, et al., Petrov DA, Kryazhimskiy S, Sherlock G. Cell 2016;166:1585-1596. doi:10.1016/j.cell.2016.08.002	Only the sequenced subset qualifies. The half-million barcoded lineages behind it do not, because their mutations were never determined, and that boundary is the sequence gate doing its job.	Cell SI; SRA (unconfirmed)

Table 2, continued

#	Dataset, citation and link	Why	Data
144	Jakociunas 2021 (degron-tuned terpene flux) Jakociunas T, Klitgaard AK, Kristensen M, Jensen MK, Keasling JD. 2021. doi:10.1002/bit.27735	Erg20p is FPP synthase, so degron-tuning it reports FPP pathway throughput directly. Small, but it is the only row measuring the mevalonate branch, which the precursor accounting lists as unmeasured.	journal SI (unconfirmed)
145	Shen 2016 (SCRaMbLE genome minimization) Shen Y, Stracquadanio G, Wang Y, et al., Boeke JD, Bader JS. Genome Res 2016;26:36-49. doi:10.1101/gr.193433.115	Combinatorial deletion and inversion across a synthetic chromosome, with the surviving genotypes sequenced. Maps which gene combinations are jointly dispensable, which single deletions cannot.	Genome Research SI (unconfirmed)
146	Blount 2018 (SCRaMbLE for phenotype improvement) Blount BA, Gowers GF, Ho JCH, Ledesma-Amaro R, Jovicevic D, McKiernan RM, Xie ZX, Li BZ, Yuan YJ, Ellis T. Nat Commun 2018;9:1932. doi:10.1038/s41467-018-03143-w	SCRaMbLE driven toward an industrial phenotype rather than toward minimization, so the genotype-to-product link is measured on multiply rearranged genomes.	Nat Commun SI (unconfirmed)
147	Liti 2009 (population genomics of <i>S. cerevisiae</i> and <i>S. paradoxus</i>) Liti G, Carter DM, Moses AM, et al., Louis EJ. Nature 2009;458:337-341. doi:10.1038/nature07743	The founding sequenced isolate panel and the source of the SGRP strains many later crosses use as parents. Historical, and it is where several parental genomes in the segregant rows trace to.	SGRP /ENA (unconfirmed)
148	Yue 2017 (wild vs domesticated assemblies) Yue JX, Li J, Aigrain L, et al., Liti G. Nat Genet 2017;49:913-924. doi:10.1038/ng.3847	Structural variation short reads miss. Relevant because industrial strains differ from S288C structurally, not only at SNPs.	ENA/GenBank (accession unconfirmed)
149	Zackrisson 2016 (Scan-o-matic growth curves) Zackrisson M, Hallin J, Ottosson LG, et al., Warringer J, Blomberg A. G3 2016;6:3003-3014. doi:10.1534/g3.116.032342	Time-series rather than endpoint fitness. A platform, so a specific released condition panel has to be chosen before this becomes a row rather than a plan.	Open-source platform; per-study datasets
150	Spellman 1998 (cell-cycle transcriptome) Spellman PT, Sherlock G, Zhang MQ, Iyer VR, Anders K, Eisen MB, Brown PO, Botstein D, Futcher B. Mol Biol Cell 1998;9:3273-3297. doi:10.1091/mbc.9.12.3273	The cell-cycle expression reference. O'Duibhir showed a large share of deletion expression response is a cell-cycle-distribution artifact, and this is the signature used to detect and remove it.	SGD /SPELL
151	Levy 2008 (morphology and phenotypic capacitance) Levy SF, Siegal ML. PLoS Biol 2008;6:e264. doi:10.1371/journal.pbio.0060264	Scores how much a deletion increases cell-to-cell variability rather than shifting the mean, identifying capacitor genes. Variance as the phenotype is a readout only per-cell data supports, which is the argument the Perturb-seq proposal rests on.	PLoS Biology SI (unconfirmed)
152	Causton 2001 (environmental change transcriptome) Causton HC, Ren B, Koh SS, et al., Young RA. Mol Biol Cell 2001;12:323-337. doi:10.1091/mbc.12.2.323	An independently generated environmental-response compendium contemporaneous with Gasch 2000. Two independent measurements of the same response class is a platform-effect control.	SGD /SPELL
153	Gasch 2001 (DNA damage response transcriptome) Gasch AP, Huang M, Metzner S, Botstein D, Elledge SJ, Brown PO. Mol Biol Cell 2001;12:2987-3003. doi:10.1091/mbc.12.10.2987	Extends the stress-response backbone to genotoxic stress, and includes checkpoint mutants, so it carries a small genotype axis rather than being wild-type only.	SGD /SPELL

Table 2, continued

#	Dataset, citation and link	Why	Data
154	Handfield 2013 (image-based localization change) Handfield LF, Chong YT, Simmons J, Andrews BJ, Moses AM. PLoS Comput Biol 2013;9:e1003085. doi:10.1371/journal.pcbi.1003085	An unsupervised treatment of the same imaging collection, which matters because the supervised class labels in CYCLOPs constrain what can be found.	PLoS Comput Biol SI (unconfirmed)
155	Kaplan 2009 (nucleosome sequence preferences) Kaplan N, Moore IK, Fondufe-Mittendorf Y, et al., Segal E. Nature 2009;458:362-366. doi:10.1038/nature07667	Separates the sequence-intrinsic component of nucleosome positioning from the cellular one by reconstituting in vitro. That separation is what lets a promoter model attribute an effect to sequence rather than to context.	GEO (unconfirmed)
156	Giaever 2004 (chemogenomic drug-target profiling) Giaever G, Flaherty P, Kumm J, et al., Davis RW. PNAS 2004;101:793-798. doi:10.1073/pnas.0307490100	An early HIP profiling set from the group that built the collection. Superseded in scale by Lee and Hoepfner, retained as a low-cost cross-validation panel.	PNAS SI (unconfirmed)
157	Paulo 2016 (TMT proteome across carbon sources) Paulo JA, O’Connell JD, Gaun A, Gygi SP. Mol Biol Cell 2016;27:2823-2838. doi:10.1091/mbc.E16-03-0184	Proteome remodeling across carbon sources, which is the substrate switch every bioprocess makes. Condition axis only.	ProteomeXchange (unconfirmed)
158	Gibney 2013 (osmotic and glucose stress fitness) Gibney PA, Lu C, Caudy AA, Hess DC, Botstein D. PNAS 2013;110:E4393-E4402. doi:10.1073/pnas.1318100110	High-gravity fermentation is an osmotic problem, and this covers the osmotic axis on the deletion collection with a growth readout rather than a transcriptome.	PNAS SI (unconfirmed)
159	Meurer 2018 (genome-wide GFP library, rebuilt) Meurer M, Duan Y, Sass E, Kats I, Herbst K, Buchmuller BC, Dederer V, Huber F, Kirrmaier D, Steff M, Van Laer K, Dick TP, Lemberg MK, Khmelinskii A, Levy ED, Knop M. Nat Methods 2018;15:617-620. doi:10.1038/s41592-018-0044-9	A rebuilt and sequence-verified tagging collection, correcting errors accumulated in the original. Verify the DOI before ingestion; it collides with the Weill 2018 SWAT row in this table.	Nat Methods SI (DOI collision with Weill 2018 UNRESOLVED)
160	Bennett 2001 (ionizing-radiation resistance) Bennett CB, et al. Nat Genet 2001;29:426-434. doi:10.1038/ng778	Genome-stability tolerance. Only indirectly relevant, through strain robustness over long fed-batch runs where instability erodes titer.	Nat Genet SI (likely PDF)
161	Sugiyama (lactic-acid tolerance) Sugiyama M, Kaneko Y, et al. National Research Institute of Brewing technical report. nissr.or.jp/wp-content/uploads/NISR06sugiyama.pdf	Lactic acid is a major platform chemical, but this is a non-peer-reviewed technical report with PDF-only tables. Lowest provenance confidence in the table; treat as a hypothesis source.	NRIB technical report PDF
162	Sousa 2013 (acetic-acid programmed cell death screen) Sousa M, Duarte AM, Fernandes TR, et al. 2013. PMID 24286259. PMID 24286259	Survival rather than growth rate under the same stressor. A different phenotype from fitness, and the one that matters when a fed-batch culture crashes rather than slows.	journal SI (unconfirmed)
163	Nagalakshmi 2008 (RNA-seq transcriptome annotation) Nagalakshmi U, Wang Z, Waern K, Shou C, Raha D, Gerstein M, Snyder M. Science 2008;320:1344-1349. doi:10.1126/science.1158441	The first RNA-seq transcriptome, and the annotation correction underneath the microarray-era expression rows. Historical, and it is why gene models used by later datasets differ.	SRA /GEO (unconfirmed)

Table 2, continued

#	Dataset, citation and link	Why	Data
164	Yofe 2016 (SWAT N-terminal library) Yofe I, Weill U, Meurer M, et al., Knop M, Schuldiner M. Nat Methods 2016;13:371-378. doi:10.1038/nmeth.3795	N-terminal tagging, which preserves C-terminal signals the GFP collection destroys. Complements Huh 2003 on exactly the proteins that collection mislocalizes.	Nat Methods SI; EUROSCARF (unconfirmed)
165	Hénault 2023 (hybrid and introgression panel) Henault M, Marsit S, Charron G, Landry CR. Nat Commun / Landry lab hybrid panels. doi:10.1038/s41467-024-46155-5	Hybrid genomes rather than recombinants of one cross, so the genotype axis includes whole-subgenome combinations. Confirm the exact release before ingestion.	journal SI (unconfirmed)
166	Cubillos 2013 (F12 recombinant panel construction) Cubillos FA, Parts L, Salinas F, et al., Liti G. Genetics 2013;195:1141-1155. doi:10.1534/genetics.113.155515	The panel and genotype table the already-listed Cubillos 2017 nitrogen study resolves against. Ingesting it is what makes that row's genotype axis self-contained.	Genetics SI Table S1
167	Kessi-Perez 2019 (nitrogen-consumption QTL panel) Kessi-Perez EI, Molinet J, Martinez C, et al. 2019/2020 nitrogen-utilization QTL work. doi:10.1186/s13068-019-1440-9	A second nitrogen-utilization mapping panel alongside Cubillos 2017. Nitrogen sets higher-alcohol and ester formation, so it is fermentation-relevant, and the exact release needs pinning.	journal SI (unconfirmed)
168	Lian 2016 (combinatorial xylose pathway optimization) Lian J, Chao R, Zhao H. Metab Eng 2016;33:139-148. doi:10.1016/j.ymben.2015.11.001	Combinatorial promoter and copy-number variants of one pathway with titers measured per combination. Small, and it is the record type inverse design produces.	Metab Eng SI (unconfirmed)

3 What the list adds up to x

Table 3: Candidates by class and by band, split at the two wave lines. *Genotypes* and *Instances* sum the per-row axes over all waves; rows with no count contribute nothing, so both totals are lower bounds.

	Wave 1	Wave 2	Reserve	Genotypes	Instances
Natural variation	10	7	19	1,519,412	4.5×10^6
Tolerance / robustness	2	5	17	128,063	5.6×10^7
CRISPR library screen	8	0	3	328,794	3.9×10^5
Expression / single cell	20	0	4	5,903	6.3×10^4
Metabolite / precursor	5	3	7	45,331	1.4×10^5
Modality / backbone	5	1	32	789,985	1.4×10^6
Regulatory DNA	0	2	6	120,509,217	1.2×10^8
Deep mutational scan	0	2	7	811,600	8.2×10^5
Combinatorial genome	0	0	3	400	600
<i>The same rows, by band</i>					
perturb-seq	30	0	0	628,459	1.8×10^6
molecular layers	20	3	0	13,887	1.4×10^4
scale	0	17	98	123,496,359	1.8×10^8
Total	50	20	98	124,138,705	1.8×10^8

All 168 rows carry roughly 1.8×10^8 genotype-by-environment instances and 5.8×10^8 individual measurements. Both are lower bounds: rows whose axes could not be pinned down contribute nothing to either total, and the instance figures marked † are products of the two axes rather than counts the papers report.

The totals are dominated by a handful of rows, so quoting them without naming those rows would mislead. de Boer 2020 alone supplies 10^8 instances, Turco 2023 3.8×10^7 and Lee 2014 1.3×10^7 . Of those three, Turco’s contribution is not net new, since it is a meta-aggregation of screens this table lists separately, and Lee’s is not currently reachable (Sec. 8.1). de Boer’s stands, and it is a different kind of instance from the rest: an 80-base promoter with an expression readout, not a strain with a phenotype.

The honest summary is that this list contains one dataset larger than everything already built, one larger than everything else here, and about 2×10^7 instances spread across the remaining 157 rows.

Wave 1 reads differently from the list as a whole, which is the point of banding it. Its class mix is 20 expression and single-cell rows, 10 natural variation, 8 CRISPR library screens, 5 modality rows and 5 metabolite rows, against a full-list mix in which tolerance screens are the largest group; only 2 tolerance rows reach wave 1. Its sequence bases shift the same way: 15 of the 50 are **reference-only**, meaning the environment rather than a strain edit carries the perturbation, which is what a condition-crossed expression layer looks like.

Instance counts invert across the two waves and the inversion is worth stating before someone reads it as an error. Wave 1 holds about 1.8×10^6 instances against 3.9×10^8 measurements, because its rows are vector-valued: a few hundred strains each carrying a transcriptome or a proteome. Wave 2 holds far more instances and fewer numbers per instance. Ranking on instances would therefore have ordered these two waves the other way round, which is the same argument the measurements column exists to settle.

4 Rank changes against the previous pass ✗

Table 4: Rank changes between the previous pass and this one, for every row in either pass’s first 70, every row added, and every row that left. *Was* is recomputed under the previous rule, tier then measurements, over the rows that pass held. A dash means the row is new or has gone.

Dataset	Was	Now	Reason
Hale 2024 (CRISPRi x natural variation)	8	2	The one released yeast library that crosses a CRISPR interference guide set with a sequenced genetic background, which is the input axis a campaign has to plan against.
Jackson 2020 (TF-deletion single-cell atlas)	10	3	Transcribed genotype barcodes with a per-cell transcriptome readout, fully crossed over 11 conditions. The design a yeast Perturb-seq is specified against.
Puddu 2019 (WGS of the deletion collection)	27	4	Whole-genome sequence for every strain in the deletion collection. A pooled campaign reads a barcode and infers a genotype; this is the measurement of how often that inference is wrong, and Jackson 2020 built its own strains rather than use the collection for related reasons.
N’Guessan 2025 (segregant scRNA-seq eQTL)	29	5	About 4,500 sequenced segregants profiled by single-cell RNA-seq, so genotype and transcriptome are measured in the same cell.
Hackett 2020 (IDEA inducible-TF transcriptome time series)	30	6	Induction rather than deletion, with a transcriptome followed over time. Jackson 2020 names transient induction as the perturbation modality most likely to produce a detectable expression response.
Hu 2007 (TF deletion expression compendium)	33	7	A designed single-gene perturbation crossed with a transcriptome over 269 transcription-factor deletions, which is the bulk form of what the campaign measures per cell, and the widest regulator coverage available.
Dong 2021 (MAGIC + SAM biosensor)	45	8	A genome-wide CRISPR activation, interference and deletion library sorted on a metabolite biosensor, so the label is product concentration rather than fitness. The pattern a sorted Perturb-seq would reuse.
Momen-Roknabadi 2020 (inducible CRISPRi library)	46	9	A genome-scale inducible CRISPR interference library with a released per-guide matrix. Induction control is what lets a knockdown be applied after a cell is captured rather than during outgrowth.
McGlincy 2021 (genome-scale CRISPRi library)	49	10	The second independently designed genome-scale yeast CRISPR interference library with a released per-guide matrix, so guide design and gene effect can be separated.
Bao 2018 (CHAnGE single-nucleotide library)	50	11	A guide-indexed genome-wide library whose edits sit below the open reading frame, so the perturbation is an allele rather than a null.
Crook 2016 (tunable RNAi, isobutanol + 1-butanol)	57	12	A dose-graded knockdown axis. Graded rather than binary perturbation is what turns a per-cell readout into a dose-response curve.
Roy 2018 (multiplexed precision editing)	61	13	Multiplexed designed edits per cell, which is combinatorial input in the sense the Perturb-seq specification requires.
Newman 2006 (protein noise, GFP library)	68	14	Per-cell protein abundance distributions across a genome-scale tagged collection. Expression noise per gene is what sets how large a perturbation effect has to be before a per-cell readout can resolve it.
Mukherjee 2021 (CRISPRi essential genes x acetic acid)	80	15	Knockdown reaches the essential genes a deletion collection cannot hold, which is a third of the genome a deletion-based campaign would miss.
Lian 2017 (CRISPR-AID, beta-carotene)	84	16	Three perturbation modalities in one cell, activation, interference and deletion, which is combinatorial input by modality rather than by gene.
Hughes 2000 (compendium of expression profiles)	–	17	New this pass. The original designed-perturbation expression compendium, and the independent measurement that makes cross-laboratory replication of the deletion transcriptome testable against the built Kemmeren 2014.
Lenstra 2011 (chromatin regulator deletion expression)	89	18	The same design over chromatin regulators rather than sequence-specific factors, so the two together cover both halves of transcriptional control by deletion.
Sun 2013 (mRNA synthesis and decay rates across deletion strains)	–	19	New this pass. A designed single-gene perturbation crossed with a transcriptome, and the only one that resolves the level into a synthesis rate and a decay rate. Rates are what a per-cell snapshot cannot recover on its own.
Jariani 2020 (yeast scRNA-seq through lag phase)	–	20	New this pass. Per-cell states through a carbon-source shift, so the readout is the distribution across an unsynchronized population rather than its mean.
Guo 2018 (CRISPR-Cas9 tiling of essential genes)	100	21	Guide tiling produces a graded allelic series across one gene, so guide position rather than gene identity carries the effect.
Urbonaite 2021 (yeastDrop-Seq under drug treatment)	–	22	New this pass. Single-drug arms against their combination with a per-cell readout, the environmental analog of a combinatorial genetic perturbation.

Dataset	Was	Now	Reason
Nadal-Ribelles 2019 (sensitive yeast scRNA-seq)	105	23	The high-sensitivity yeast protocol, strand and isoform aware. Sensitivity per cell is the constraint on how small a perturbation effect a campaign can resolve.
Gasch 2017 (single-cell stress heterogeneity)	106	24	Separates intrinsic from extrinsic per-cell variation in an isogenic population under stress, which is the null a perturbation effect is measured against.
Jaffe 2019 (multiplexed CRISPR interference epistasis)	111	25	Two knockdowns in the same cell with a measured interaction, which is the only combinatorial CRISPR interference design located in yeast.
Jackson 2023 (wild-type scRNA-seq time course for RNA kinetics)	114	26	The platform companion of Jackson 2020: 173,361 cells of one strain, which is what sets the per-cell variance a perturbation has to exceed.
Brettner 2024 (ultra-high-throughput yeast scRNA-seq)	135	27	Combinatorial barcoding in yeast at 96 to 384 multiplexed genotypes or environments per run. The throughput route by which a campaign becomes affordable.
Mulleder 2012 (prototrophic deletion collection)	136	28	The prototrophic deletion collection. Jackson 2020 used a prototrophic background because auxotrophy blocks minimal and nitrogen-limited media, so this is the strain resource a campaign in defined media needs.
Su 2023 (single-cell transcriptomes under four stresses)	146	29	Full-length rather than three-prime tag counting, so isoform-level readout is on the table for a campaign that needs it.
Wang 2022 (single-cell transcriptomes across replicative aging)	153	30	The only yeast single-cell set with age as the condition axis, which is a covariate any pooled campaign carries whether or not it measures it.
Albert 2018 (eQTL in 1,012 segregants)	5	31	The transcriptome half of the segregant panel that Jakobson 2025, Gerke 2017 and Eder 2020 measure protein, metabolite and flux on.
Jakobson 2025 (genome-to-proteome map)	6	32	Protein abundance across the same sequenced segregant panel Albert 2018 profiles by transcriptome, so protein and transcript quantitative trait loci are measurable on one genotype set.
Muenzner 2024 (natural-isolate proteome)	7	33	796 proteomes drawn from the sequenced 1,011-isolate panel that the supported Caudal 2024 transcriptomes also come from.
Cooper 2010 (CE-MS amino-acid metabolome)	15	34	Amino-acid pools on the deletion collection by capillary electrophoresis, the same trait class the supported Mulleder 2016 measures by mass spectrometry and the same strains Kemmeren 2014 profiles.
Aulakh 2025 (genome-scale ionome)	17	35	A metabolic readout on the whole deletion collection, which is the genotype axis the supported expression compendium already covers.
Teyssonniere 2024 (species-wide proteome against transcriptome)	–	36	New this pass. Proteomes for 942 isolates, the panel whose transcriptomes the built Caudal 2024 supplies, so the strain-level pairing is one to one.
Grossbach 2022 (BY x RM transcriptome, proteome and phosphoproteome)	–	37	New this pass. Three layers on one set of 112 sequenced strains, so the transcript and the protein of a gene are observed in the same genotype and the RNA to protein question becomes a within-strain residual.
Leutert 2023 (phosphoproteome x 101 conditions)	37	38	Enzyme regulation acts faster than transcription, so a phosphosite layer over 101 conditions is what explains flux changes an expression table cannot.
Brauer 2008 (growth-rate-controlled chemostat transcriptome)	41	39	Expression at controlled growth rate under each nutrient limitation, which is the condition axis Boer 2010 and Hackett 2016 measure metabolites and flux on.
Hackett 2016 (SIMMER multi-omic flux)	48	40	Flux, metabolite and transcript measured in one study under the same nutrient limitations, so the join is internal rather than across papers.
Zhu 2014 (kinase / phosphatase lipidomics)	55	41	A lipidome over 129 signaling deletions, most of which carry an expression profile in the supported compendium.
Gerke 2017 (urea-cycle mQTL)	62	42	Metabolite quantitative trait loci on a segregant cross that also has an expression map, so a metabolite locus can be read through its transcript.
Ambroset 2014 (metabolite QTL)	72	43	A 74-metabolite panel on a sequenced segregant panel, the widest metabolite vector available on a recombinant genotype axis.
Blank 2005 (13C metabolic flux)	74	44	The only row measuring flux rather than a concentration, on deletion mutants that also have a transcriptome in the supported set.
Skelly 2013 (expression variation across isolates)	94	45	Transcriptome and proteome on the same 22 strains, which is the only row where the cheap and expensive layers are paired within one experiment rather than joined across two.

Dataset	Was	Now	Reason
Airoidi 2016 (nitrogen-limited steady-state and dynamic transcriptome)	–	46	New this pass. Expression under the exact nitrogen-limited conditions that Jackson 2020 profiles single cells in, so the deletion effect separates from the medium effect.
McManus 2014 (ribosome profiling, allele-specific)	–	47	New this pass. Translation efficiency per gene alongside the mRNA abundance of that gene, which is the first of the two terms that separate protein level from transcript level.
Martin-Perez 2017 (protein half-lives)	–	48	New this pass. Protein turnover per gene, the second of those two terms, spanning three orders of magnitude and therefore not replaceable by a global constant.
Boer 2010 (metabolome across nutrient limitations)	119	49	The metabolite half of the chemostat nutrient-limitation series whose transcriptome half is Brauer 2008, same laboratory and same conditions.
Tengolics 2024 (domestication metabolome)	122	50	Metabolite levels across sequenced isolates, the panel the supported Caudal 2024 transcriptomes and Muenzner 2024 proteomes also sit on.
Eder 2020 (flux QTL)	132	51	Flux mapped to segregant genotypes, which is the flux counterpart of the expression and protein maps on the same panel type.
Foss 2007 (BY x RM segregant proteome)	–	52	New this pass. The independent replicate of Grossbach 2022 on the same cross, and the first measurement that protein-level loci differ from transcript-level loci.
Yu 2021 (proteome and metabolome under nitrogen limitation)	145	53	Both layers measured in one study under nitrogen limitation, so it calibrates the protein-to-metabolite step that cross-study joins assume.
de Boer 2020 (100M random promoters)	2	54	Unchanged criteria; moved by 52 as rows around it were promoted into a band.
Vaishnav 2022 (regulatory DNA fitness landscape)	3	55	Unchanged criteria; moved by 52 as rows around it were promoted into a band.
Lee 2014 (HIP-HOP fitness signatures)	4	56	Unchanged criteria; moved by 52 as rows around it were promoted into a band.
Nguyen Ba 2022 (barcoded bulk QTL, 100k segregants)	9	57	Unchanged criteria; moved by 48 as rows around it were promoted into a band.
Galardini 2019 (four backgrounds x 38 conditions)	11	58	Unchanged criteria; moved by 47 as rows around it were promoted into a band.
Parsons 2006 (bioactive-compound profiling)	13	59	Unchanged criteria; moved by 46 as rows around it were promoted into a band.
Dutta 2026 (barcoded natural-isolate chemical response)	14	60	Unchanged criteria; moved by 46 as rows around it were promoted into a band.
Li 2016 (tRNA fitness landscape)	16	61	Unchanged criteria; moved by 45 as rows around it were promoted into a band.
Peter 2018 (1,011 isolate genomes + phenome)	18	62	Unchanged criteria; moved by 44 as rows around it were promoted into a band.
Domingo 2018 (tRNA double-mutant landscape)	19	63	Unchanged criteria; moved by 44 as rows around it were promoted into a band.
Zhang 2021 (YETI titratable overexpression)	20	64	Unchanged criteria; moved by 44 as rows around it were promoted into a band.
Peeters 2021 (fermentation-trait QTL atlas)	21	65	Unchanged criteria; moved by 44 as rows around it were promoted into a band.
Van Leeuwen 2024 (short-chain organic acids)	22	66	Unchanged criteria; moved by 44 as rows around it were promoted into a band.
Chica 2026 (AutoDRY autophagy screen)	23	67	Unchanged criteria; moved by 44 as rows around it were promoted into a band.
Sopko 2006 (genome-scale overexpression toxicity)	24	68	Unchanged criteria; moved by 44 as rows around it were promoted into a band.
Liu 2021 (tryptophan / isobutanol tolerance)	25	69	Unchanged criteria; moved by 44 as rows around it were promoted into a band.
Kuroda 2019 (isobutanol-specific tolerance)	26	70	Unchanged criteria; moved by 44 as rows around it were promoted into a band.
Turco 2023 (Yeast Phenome)	28	71	Unchanged criteria; moved by 43 as rows around it were promoted into a band.

Dataset	Was	Now	Reason
Rossi 2021 (ChIP-exo protein architecture)	31	72	Unchanged criteria; moved by 41 as rows around it were promoted into a band.
Piotrowski 2017 (MOSAIC diagnostic panel)	32	73	Unchanged criteria; moved by 41 as rows around it were promoted into a band.
Bloom 2025 (global epistasis in a yeast cross)	34	74	Unchanged criteria; moved by 40 as rows around it were promoted into a band.
Gasch 2000 (environmental stress response)	35	75	Unchanged criteria; moved by 40 as rows around it were promoted into a band.
Decourty 2021 (RNA-metabolism genetic interactions)	36	76	Unchanged criteria; moved by 40 as rows around it were promoted into a band.
Cuperus 2017 (random 5' UTR library)	38	77	Unchanged criteria; moved by 39 as rows around it were promoted into a band.
Faure 2022 (doubleddeepPCA)	39	78	Unchanged criteria; moved by 39 as rows around it were promoted into a band.
Chong 2015 / CYCLOPs (single-cell proteome dynamics)	40	79	Unchanged criteria; moved by 39 as rows around it were promoted into a band.
Teyssonniere 2024 (species-wide trait survey)	42	80	Unchanged criteria; moved by 38 as rows around it were promoted into a band.
Diss 2018 (Fos-Jun interaction double mutants)	43	81	Unchanged criteria; moved by 38 as rows around it were promoted into a band.
Warringer 2011 (trait variation across the SGRP panel)	44	82	Unchanged criteria; moved by 38 as rows around it were promoted into a band.
Keren 2013 (promoter activity across conditions)	47	83	Unchanged criteria; moved by 36 as rows around it were promoted into a band.
Puchta 2016 (snoRNA U3 landscape)	51	84	Unchanged criteria; moved by 33 as rows around it were promoted into a band.
Yoshikawa 2011 (deletion + overexpression phenome)	52	85	Unchanged criteria; moved by 33 as rows around it were promoted into a band.
Ho 2009 (MoBY-ORF barcoded overexpression)	53	86	Unchanged criteria; moved by 33 as rows around it were promoted into a band.
Bloom 2013 (1,008 BYxRM segregants)	54	87	Unchanged criteria; moved by 33 as rows around it were promoted into a band.
Xiao 2014 (genome-wide RNAi furfural tolerance)	56	88	Unchanged criteria; moved by 32 as rows around it were promoted into a band.
Douglas 2012 (barcoded overexpression fitness)	58	89	Unchanged criteria; moved by 31 as rows around it were promoted into a band.
She 2018 (natural variant effects across a large isolate panel)	59	90	Unchanged criteria; moved by 31 as rows around it were promoted into a band.
Costello 2020 (bioreactor Bar-seq)	60	91	Unchanged criteria; moved by 31 as rows around it were promoted into a band.
Gurvitz (fatty-acid utilization screen)	63	92	Unchanged criteria; moved by 29 as rows around it were promoted into a band.
Mira 2010 (acetic acid tolerance, full collection)	64	93	Unchanged criteria; moved by 29 as rows around it were promoted into a band.
Gameiro 2025 (AID2 degron libraries)	65	94	Unchanged criteria; moved by 29 as rows around it were promoted into a band.
Si 2017 (RAGE RNAi, isobutanol titer)	66	95	Unchanged criteria; moved by 29 as rows around it were promoted into a band.
HamediRad 2018 (RAGE xylose utilization)	67	96	Unchanged criteria; moved by 29 as rows around it were promoted into a band.
Pereira 2014 (wheat-straw hydrolysate)	69	97	Unchanged criteria; moved by 28 as rows around it were promoted into a band.
Endo 2008 (vanillin tolerance)	70	98	Unchanged criteria; moved by 28 as rows around it were promoted into a band.

Dataset	Was	Now	Reason
Bloom 2019 (16-parent cross)	12	–	Built since the previous pass as the 50th supported dataset, 13,950 segregants over 38 traits, L0-L4 verified. It is now a join partner rather than a candidate.

5 What each candidate joins to ✗

A dataset’s value in a substrate is not only what it measures. It is also what becomes measurable once it sits beside what is already held. Table 5 names 116 such pairs across 66 candidates, each with the axis the two share.

5.1 A join needs a shared axis, not a shared theme ✗

The **join key** column is required rather than decorative. Two metabolomics papers are not joinable because both measure metabolites; they are joinable when they measure overlapping analytes on overlapping strains. The keys in use are the deletion collection, the sequenced 1,011-isolate panel, a segregant panel from one cross, a guide library over one gene set, a media formulation, a compound set, and a single gene. A pair with none of these is a theme and is not listed.

Sixty-two of the 116 pairs have a partner that is already built, so that join becomes runnable on the day the candidate is ingested rather than after two ingestions. Those are the cheapest experiments in the table and none of them has been run.

Two kinds of key are distinguished in the table and the difference decides what a join can do. A *strain-level* key means both layers were measured on the same genotypes, so the join is a paired observation and the residual between the layers is per strain. A *gene-level* key means one side is a single genome-wide coefficient, a half-life or a translation rate, that applies to every strain alike. A gene-level layer cannot explain why two strains differ; it explains why two genes differ, and the two are not interchangeable.

5.2 The joins that motivate the ordering ✗

Four groups carry most of the weight, and each is a specific question rather than an accumulation.

Transcript against protein, and the two terms between them. Steady-state transcript and steady-state protein are both well covered, by the built Kemmeren 2014 and Messner 2023 and by several rows here, and they disagree. Four rows make the disagreement something a model can carry a term for rather than absorb as error. Grossbach 2022 measures transcript, protein and phosphosite on one set of 112 sequenced strains, so the residual between transcript and protein is observed within a genotype instead of across a population. Teyssonniere 2024 does the strain-level version at species scale, 942 isolate proteomes against the built Caudal 2024 transcriptomes on the same panel, and reports what the pairing shows: the two correlate weakly and the variants associated with each overlap by 3 percent. The remaining two are gene-level. McManus 2014 gives translation efficiency beside mRNA abundance for 5,474 genes and finds ribosome occupancy more conserved than transcript abundance, so translation buffers rather than tracks. Martin-Perez 2017 gives turnover for 3,160 proteins, spanning three orders of magnitude, which is why a single global mRNA-to-protein scaling cannot be right. Sun 2013 supplies the transcript-side counterpart, decomposing mRNA level into synthesis and decay across 46 deletion strains.

Cross-laboratory replication of the deletion transcriptome. Kemmeren 2014 is built and is the largest genotype-indexed transcriptome with designed perturbations, and nothing in the built set can check it. Hughes 2000, Hu 2007 and Sun 2013 all profile deletion strains by expression, in three other laboratories and across fourteen years of array and sequencing platforms, so agreement can be measured gene by gene on shared strains rather than assumed. That check matters here specifically: the built Nadal-Ribelles 2025 does not replicate its own genotype assignment across batches, so a second opinion on what a deletion does to a transcriptome is not a formality.

Metabolism against expression on one genotype axis. Blank 2005 measures flux through central carbon metabolism on deletion mutants that already carry an expression profile in the supported Kemmeren 2014 compendium, which is the only place in this table where the transcript-to-flux step can be fitted rather than assumed. Cooper 2010 measures the same amino-acid pools as the supported Mulleder 2016 by a different platform on an overlapping genotype set, which separates biology from batch. Aulakh 2025 adds a second metabolic panel, the ionome, on the whole collection. Above them sit Jakobson 2025 and Albert 2018, protein and transcript variation on one sequenced segregant panel, and Muenzner 2024 against the supported Caudal 2024, proteome and transcriptome on one isolate panel.

A condition axis measured on more than one layer. Brauer 2008 and Boer 2010 are the transcript and metabolite halves of one chemostat series, same laboratory and same limitations, and Hackett 2016 adds flux over the same conditions. Airolidi 2016, new to this pass, supplies the nitrogen-limited counterpart, and its media are the media Jackson 2020 profiles single cells in. That last pairing is what makes a deletion effect in the single-cell atlas a contrast against a wild-type baseline in the same medium rather than an absolute.

One perturbation, two modalities. Liu 2021 against Kuroda 2019 is isobutanol tolerance on one collection by two readouts. Mukherjee 2021 against the supported Mormino 2022 is acetic acid by knockdown at genome scale against twelve genes. Dong 2021 against the supported Lian 2019 is one tri-functional library read out by growth selection and by a metabolite biosensor, which is how a tolerance screen becomes a production screen. McGlincy 2021 against Momen-Roknabadi 2020 is two independently designed CRISPR interference libraries over the same genes, which is the cheapest decisive test of whether a model has learned gene function or guide design.

The same genotype read out in bulk and per cell. Jackson 2020 against the supported Kemmeren 2014 is the same transcription-factor deletions as a population average and as a per-cell distribution, which is the only available calibration of what pseudobulk loses. Hu 2007 extends the bulk side to 269 regulators. Boocock 2025 against Albert 2018 is the same cross measured in bulk and per cell.

None of these transfers has been measured. Each is a hypothesis about what a model could learn, and each becomes a run once both halves are held.

Table 5: Named joins between a candidate and either a supported dataset or another candidate. # is the candidate’s rank in Table 1. A partner in **bold** is already built, so that row’s join becomes runnable the moment the candidate is ingested; an unbolded partner is itself a candidate, so the join costs two ingestions. *Join key* is the shared axis that makes the pair addressable; a pair with no such axis is a theme and is not listed. None of these transfers has been measured.

#	Candidate	Partner	Join key	What the join yields
1	Boocock 2025 (single-cell eQTL mapping)	Nadal-Ribelles 2025 (Perturb-seq)	single-cell transcriptome readout, one genotype per cell	One dataset carries genotypes made by recombination and the other genotypes made by deletion, on the same readout, which is the test of whether a per-cell expression model transfers between the two.
		Albert 2018 (eQTL in 1,012 segregants)	same BY by RM cross, bulk against single cell	The same expression quantitative trait loci measured in bulk and per cell, so the cell-to-cell variance a bulk average hides is recoverable.
		Jakobson 2025 (genome-to-proteome map)	segregant genotype class	Transcript and protein variation on one recombinant panel.
2	Hale 2024 (CRISPRi x natural variation)	Smith 2016 (CRISPRi chem-genetic)	CRISPR interference knockdown in S288C	One knockdown effect measured in the reference background and across 169 segregants, which separates a gene effect from a background effect.
		McGlinchey 2021 (genome-scale CRISPRi library)	guide library design	Whether a knockdown effect is a property of the gene or of the guide.
		Bloom 2019 (16-parent cross)	segregant genotype class, growth traits	Knockdown effect against natural allele effect on comparable panels.
3	Jackson 2020 (TF-deletion single-cell atlas)	Kemmeren 2014	single deletions of the same transcription factors, bulk against single cell	The same knockout read out as a population average and as a per-cell distribution, which is the only available calibration of what pseudobulk loses.
		Nadal-Ribelles 2025 (Perturb-seq)	deletion perturbation with a single-cell transcriptome readout	Twelve genotypes over eleven conditions against about 3,500 genotypes over two, so genotype breadth and condition breadth are separable for once.
		Airoldi 2016 (nitrogen-limited steady-state and dynamic transcriptome)	the same nitrogen-limited media formulations	A wild-type expression baseline in the same medium, which is what makes the deletion effect a contrast rather than an absolute.
		Hackett 2020 (IDEA inducible-TF transcriptome time series)	the same transcription factors, deletion against induction	Loss of function against gain of function on one regulator set.
4	Puddu 2019 (WGS of the deletion collection)	Costanzo 2016 dmf	the same deletion collection	Converts the sequence basis of every S288C-KO row from an assumption into a measurement, including the largest built datasets.
		Nadal-Ribelles 2025 (Perturb-seq)	pooled deletion strains identified by barcode	Which pooled strains are not what their barcode says, which bears directly on the reported genotype-assignment impurity.
5	N’Guessan 2025 (segregant scRNA-seq eQTL)	Boocock 2025 (single-cell eQTL mapping)	segregant genotype class, single-cell transcriptome	Two independent single-cell expression quantitative trait loci maps, which is the replication neither has on its own.
		Bloom 2019 (16-parent cross)	segregant genotype class, 16 founders against 2	Whether an expression effect measured in a two-parent cross holds across sixteen founders.
6	Hackett 2020 (IDEA inducible-TF transcriptome time series)	Kemmeren 2014	the same transcription factors, induction against deletion	Whether a regulator’s targets are the same set whether it is removed or over-induced.
		Sameith 2015 dm	transcription-factor pairs	Induction dynamics for the single factors whose double deletions carry a measured expression epistasis.
7	Hu 2007 (TF deletion expression compendium)	Kemmeren 2014	single deletions with a bulk expression readout	Two independently produced deletion expression compendia over overlapping regulators, which is the reproducibility check neither has alone and the largest such pair in yeast.

Table 5, continued

# Candidate	Partner	Join key	What the join yields
8 Dong 2021 (MAGIC + SAM biosensor)	Jackson 2020 (TF-deletion single-cell atlas)	transcription-factor deletions	269 regulators in bulk against 11 per cell, so the regulators the single-cell atlas cannot reach still carry a profile.
	Lian 2019 (MAGIC CRISPR-AID) Cachera 2023 (CRI-SPA betaxanthin)	the same tri-functional library, fitness against biosensor readout metabolite biosensor or colorimetric product readout on a genome-scale library	One library, two labels: growth selection and product concentration, which is how a tolerance screen is converted into a production screen. Two product-labeled genome-scale screens on different chemistry.
9 Momen-Roknabadi 2020 (inducible CRISPRi library)	McGlinchy 2021 (genome-scale CRISPRi library) SGD essentiality	independently designed guide libraries over the same genes essential genes, knockdown against deletion	The paired half of the library-transfer test. A graded phenotype where a deletion collection records only absence.
10 McGlinchy 2021 (genome-scale CRISPRi library)	Momen-Roknabadi 2020 (inducible CRISPRi library) Smith 2016 (CRISPRi chem-genetic)	independently designed guide libraries over the same genes CRISPR interference over the same gene set	Library-to-library transfer, which is the cheapest decisive test of whether a model learned gene function or guide-design idiosyncrasy. A knockdown fitness prior for the chemical-genetic conditions already built.
11 Bao 2018 (CHANGE single-nucleotide library)	Lian 2019 (MAGIC CRISPR-AID) Li 2016 (tRNA fitness landscape)	guide-indexed genome-wide library from the same laboratory designed single-nucleotide variants with a fitness label	Sub-gene alleles against whole-gene activation, interference and deletion on one platform lineage. Genome-wide shallow variant coverage against single-gene exhaustive coverage.
12 Crook 2016 (tunable RNAi, isobutanol + 1-butanol)	Lopez 2024 (isobutanol screen, private) Kuroda 2019 (isobutanol-specific tolerance)	isobutanol, knockdown against deletion isobutanol tolerance on the same collection	A graded knockdown axis over the phenotype the built biosensor screen measures as a titer proxy. Dose-graded knockdown against whole-gene deletion for one phenotype.
13 Roy 2018 (multiplexed precision editing)	Bao 2018 (CHANGE single-nucleotide library)	designed edits, single against multiplexed	Whether multiplexed edit effects are the sum of their single-edit effects.
14 Newman 2006 (protein noise, GFP library)	Messner 2023 (proteome) Gasch 2017 (single-cell stress heterogeneity)	protein abundance across the genome, per cell against population per-cell variation in an isogenic population	Which proteins have a population mean that no single cell is near, which decides where a mean-level model is the wrong object. Noise at the protein and transcript layers on the same question.
15 Mukherjee 2021 (CRISPRi essential genes x acetic acid)	Mormino 2022 (CRISPRi acetic-acid) Mota 2024 (weak-acid screen)	acetic acid, CRISPR interference weak-acid stress on deletion strains	The genome-scale extension of a twelve-gene screen already built. Essential-gene coverage for a phenotype the deletion collection can only sample from the non-essential side.
16 Lian 2017 (CRISPR-AID, beta-carotene)	Ozaydin 2013 (beta-carotene screen)	beta-carotene titer	Three perturbation modalities against a whole-collection deletion screen on the same product.
17 Hughes 2000 (compendium of expression profiles)	Kemmeren 2014 Hu 2007 (TF deletion expression compendium)	deletion strains, gene by gene, fourteen years and two array platforms apart deletion strains with a bulk expression readout	Cross-laboratory replication of the deletion transcriptome, which the built set cannot test against itself and which Nadal-Ribelles 2025 failed cross-batch. A third independent compendium, so agreement can be measured across three laboratories rather than asserted from two.

Table 5, continued

# Candidate	Partner	Join key	What the join yields
18 Lenstra 2011 (chromatin regulator deletion expression)	Kemmeren 2014	single deletions with a bulk expression readout, different gene class	Chromatin regulators beside sequence-specific factors on one expression readout.
	Sameith 2015 dm	regulator deletions, single against double	Whether chromatin-regulator effects combine the way transcription-factor effects do.
19 Sun 2013 (mRNA synthesis and decay rates across deletion strains)	Kemmeren 2014	single deletion strains, gene by gene	A steady-state transcript level beside the synthesis and decay rates that produce it, so two strains with the same level but different kinetics stop looking identical.
	Martin-Perez 2017 (protein half-lives)	turnover, transcript side against protein side, gene by gene	Both half-life terms in one place, which is what an RNA-to-protein model needs in order to have a reason for the two to disagree.
20 Jariani 2020 (yeast scRNA-seq through lag phase)	Jackson 2020 (TF-deletion single-cell atlas)	carbon-source shift with a per-cell readout	A transition sampled densely in time where the atlas samples eleven steady states.
21 Guo 2018 (CRISPR-Cas9 tiling of essential genes)	SGD essentiality	essential genes	A graded allelic series where the built record is binary.
22 Urbonaitė 2021 (yeastDrop-Seq under drug treatment)	Hoepfner 2014 (HIP/HOP atlas)	chemical treatment of yeast	A per-cell readout for compound response where the built atlas has a pooled fitness score.
23 Nadal-Ribelles 2019 (sensitive yeast scRNA-seq)	Nadal-Ribelles 2025 (Perturb-seq)	same laboratory, protocol against application	The sensitivity ceiling of the platform the built genome-scale single-cell dataset was produced on.
	Gasch 2017 (single-cell stress heterogeneity)	isogenic per-cell variation in BY4741	Two independent measurements of how much an unperturbed yeast population varies, which is the floor a perturbation must clear.
24 Gasch 2017 (single-cell stress heterogeneity)	Gasch 2000 (environmental stress response)	the same stress conditions, bulk against single cell	How much of the canonical stress response is a population average of cells that are individually in different states.
25 Jaffe 2019 (multiplexed CRISPR interference epistasis)	Costanzo 2016 dmi	gene pairs, knockdown against deletion	Whether digenic interaction measured between two nulls is recovered between two knockdowns, which decides if the built interaction data can supervise a knockdown campaign.
	Kuzmin 2018 tmi	higher-order gene combinations	The knockdown analog of trigenic interaction.
26 Jackson 2023 (wild-type scRNA-seq time course for RNA kinetics)	Jackson 2020 (TF-deletion single-cell atlas)	same platform and laboratory, wild type against deletions	The unperturbed per-cell variance the perturbed atlas is read against.
27 Brettner 2024 (ultra-high-throughput yeast scRNA-seq)	Jackson 2020 (TF-deletion single-cell atlas)	multiplexed genotypes per single-cell run	The throughput ceiling for a barcoded-genotype design, measured rather than assumed.
28 Mulleder 2012 (prototrophic deletion collection)	Mulleder 2016 (amino-acid metabolome)	the same prototrophic deletion strains	The strain resource the built amino-acid metabolome was measured on, which is what makes defined-medium phenotyping interpretable.
	Jackson 2020 (TF-deletion single-cell atlas)	prototrophy as a precondition for minimal media	A ready-made prototrophic library for a campaign that needs nitrogen or carbon limitation.
29 Su 2023 (single-cell transcriptomes under four stresses)	Gasch 2000 (environmental stress response)	osmotic and starvation stress	Full-length per-cell transcripts against the bulk stress-response reference.
30 Wang 2022 (single-cell transcriptomes across replicative aging)	Jackson 2023 (wild-type scRNA-seq time course for RNA kinetics)	wild-type per-cell states over time	Replicative age as a covariate on a platform that does not measure it.

Table 5, continued

# Candidate	Partner	Join key	What the join yields
31 Albert 2018 (eQTL in 1,012 segregants)	Jakobson 2025 (genome-to-proteome map) Gerke 2017 (urea-cycle mQTL)	the same segregant panel segregant genotype class	The transcript half of a paired transcript and protein map. A metabolite locus read through the transcripts of the pathway that produces it.
32 Jakobson 2025 (genome-to-proteome map)	Albert 2018 (eQTL in 1,012 segregants) Messner 2023 (proteome)	the same sequenced segregant panel protein abundance, natural variation against deletion	Protein and transcript quantitative trait loci on one genotype set, which is the direct measurement of how much of protein variation transcript variation explains. Whether the proteome response to a deleted gene resembles the response to a natural allele of it.
33 Muenzner 2024 (natural-isolate proteome)	Caudal 2024 (pan-transcriptome) Dutta 2026 (barcoded natural-isolate chemical response) Teyssonniere 2024 (species-wide proteome against transcriptome)	the sequenced 1,011-isolate panel the same 1,011-isolate panel the 1,011-isolate panel, protein layer	Transcriptome and proteome on overlapping isolates, which is the cheap-layer against expensive-layer transfer question at natural genetic distance. Genome, transcriptome, proteome and a chemogenomic response surface on one genotype axis. De-duplication before either is built: 796 isolates here against 942 there, shared authors and one panel, and no evidence yet on whether the two acquisitions are independent.
34 Cooper 2010 (CE-MS amino-acid metabolome)	Mulleder 2016 (amino-acid metabolome) Kemmeren 2014	the same amino acids on an overlapping deletion set the same deletion collection	Two platforms measuring one trait class, which is the cleanest available test of whether a model learned biology or a batch effect. Amino-acid pools paired with the expression profile of the same knockout.
35 Aulakh 2025 (genome-scale ionome)	Mulleder 2016 (amino-acid metabolome) Kemmeren 2014	the whole deletion collection, metabolite readout the same deletion collection	Two orthogonal metabolic panels on one genotype axis. Element levels against the transcriptional response of the same knockout.
36 Teyssonniere 2024 (species-wide proteome against transcriptome)	Caudal 2024 (pan-transcriptome) Muenzner 2024 (natural-isolate proteome)	the same sequenced isolates, 942 against 943 the 1,011-isolate panel, protein layer	Transcript and protein for the same strain, one to one rather than by overlap. The published result on this pair is that the two correlate weakly and share 3 percent of their associated variants, so it is the calibration an inference model is scored against. De-duplication before either is built: the two share authors and a panel, and whether they are independent acquisitions is unsettled.
37 Grossbach 2022 (BY x RM transcriptome, proteome and phosphoproteome)	Albert 2018 (eQTL in 1,012 segregants) Jakobson 2025 (genome-to-proteome map) Leutert 2023 (phosphoproteome x 101 conditions)	BY x RM segregants, transcript layer segregant genotype class, protein layer phosphosite identity	112 strains with transcript and protein measured together against 1,012 with transcript alone, so the within-strain RNA-to-protein residual fitted on the small panel can be applied to the large one. Two segregant proteomes on different panels and platforms, which is what separates a protein quantitative trait locus from a batch. Phosphorylation driven by genotype against phosphorylation driven by condition, on one site vocabulary.
38 Leutert 2023 (phosphoproteome x 101 conditions)	Gasch 2000 (environmental stress response) Messner 2023 (proteome)	overlapping stress conditions protein identity	Post-translational regulation against transcriptional regulation for one condition set, on the layer that acts first. Modification state against abundance, which abundance alone cannot separate.
39 Brauer 2008 (growth-rate-controlled chemostat transcriptome)	Boer 2010 (metabolome across nutrient limitations)	the same chemostat limitations	The transcript half of a paired transcript and metabolite series.

Table 5, continued

# Candidate	Partner	Join key	What the join yields
	Airol di 2016 (nitrogen-limited steady-state and dynamic transcriptome)	growth rate under limitation, carbon against nitrogen	Whether the growth-rate expression program is nutrient-general.
40 Hackett 2016 (SIMMER multi-omic flux)	Boer 2010 (metabolome across nutrient limitations) Brauer 2008 (growth-rate-controlled chemostat transcriptome)	the same nutrient-limited chemostats nutrient limitation at controlled growth rate	Metabolite concentration and flux under one condition set, which is what a kinetic model needs and neither supplies alone. Expression, metabolite and flux over one condition axis.
41 Zhu 2014 (kinase / phosphatase lipidomics)	da Silveira 2014 (lipidomics) Kemmeren 2014 Xue 2025 (free fatty acids, private)	lipid species on deletion strains kinase and phosphatase deletions fatty-acid metabolism	A second lipidome on a signaling-focused genotype set. Lipid composition against the transcriptional response of the same signaling mutant. Lipid class distribution against free fatty-acid titer, the two sides of the malonyl-CoA sink.
42 Gerke 2017 (urea-cycle mQTL)	Albert 2018 (eQTL in 1,012 segregants) Mulleder 2016 (amino-acid metabolome)	segregant genotype class nitrogen and amino-acid metabolites	A metabolite locus and the expression loci in the same interval. Natural-allele against deletion effects on one metabolic module.
43 Ambroset 2014 (metabolite QTL)	Peeters 2021 (fermentation-trait QTL atlas) Zelezniak 2018 (metabolome)	segregant panels with fermentation traits metabolite panel	74 metabolites against 18 mapped traits, so a trait locus can be read as a metabolite change. Recombinant against deletion genotypes on a metabolite readout.
44 Blank 2005 (13C metabolic flux)	Kemmeren 2014 Mulleder 2016 (amino-acid metabolome)	deletion strains present in both deletion strains, flux against pool size	Flux against expression for the same knockout, which is the only place the transcript-to-flux step can be fitted rather than assumed. Whether a changed pool reflects changed flux or changed demand.
45 Skelly 2013 (expression variation across isolates)	Muenzner 2024 (natural-isolate proteome)	natural isolates, paired against joined modalities	A within-experiment paired transcript and protein measurement to calibrate the cross-study join.
46 Airol di 2016 (nitrogen-limited steady-state and dynamic transcriptome)	Jackson 2020 (TF-deletion single-cell atlas) Yu 2021 (proteome and metabolome under nitrogen limitation)	the same nitrogen-limited media nitrogen limitation	A wild-type bulk baseline for the medium the single-cell atlas perturbs in. Transcript, protein and metabolite under one limitation.
47 McManus 2014 (ribosome profiling, allele-specific)	Messner 2023 (proteome) Martin-Perez 2017 (protein half-lives)	gene identity, translation rate against protein abundance gene identity, synthesis against degradation	A per-gene translation term to put against measured protein level, which is the coefficient a model otherwise has to learn blind. Both halves of protein turnover, so steady-state abundance can be decomposed rather than only predicted.
48 Martin-Perez 2017 (protein half-lives)	Messner 2023 (proteome) Kemmeren 2014	gene identity, half-life against abundance gene identity, protein half-life against transcript response	Which proteins are abundant because they are made fast and which because they are destroyed slowly, a distinction abundance alone cannot make. Whether a transcript change reaches the protein layer at all, which for a long-lived protein it largely does not.

Table 5, continued

#	Candidate	Partner	Join key	What the join yields
49	Boer 2010 (metabolome across nutrient limitations)	Brauer 2008 (growth-rate-controlled chemostat transcriptome) Zelezniak 2018 (metabolome)	the same chemostat limitations and growth rates metabolite panel	The metabolite and transcript halves of one experimental series. Condition-driven against genotype-driven metabolite variation.
50	Tengolics 2024 (domestication metabolome)	Caudal 2024 (pan-transcriptome)	sequenced natural isolates	Metabolite against transcript variation across the same population.
51	Eder 2020 (flux QTL)	Blank 2005 (13C metabolic flux) Albert 2018 (eQTL in 1,012 segregants)	flux measurement, natural variation against deletion segregant genotype class	Whether flux control points found by deletion are the loci natural variation actually moves. Flux loci against expression loci on comparable panels.
52	Foss 2007 (BY x RM segregant proteome)	Grossbach 2022 (BY x RM transcriptome, proteome and phosphoproteome)	the same cross, protein layer, fifteen years and two platforms apart	Cross-laboratory replication of segregant protein abundance, which neither measurement can establish alone.
53	Yu 2021 (proteome and metabolome under nitrogen limitation)	Airoidi 2016 (nitrogen-limited steady-state and dynamic transcriptome)	nitrogen limitation	The transcript layer for the protein and metabolite layers measured here.
54	de Boer 2020 (100M random promoters)	Vaishnav 2022 (regulatory DNA fitness landscape) Renganaath 2020 (natural promoter-variant MPRA)	the same random promoter library, expression against fitness promoter sequence with a measured expression readout	Two labels on one sequence set, which turns a sequence-to-expression model into a sequence-to-fitness model without new data. Whether a model trained on random sequence predicts the effect of a real allele, the cheapest decisive transfer test in the table.
55	Vaishnav 2022 (regulatory DNA fitness landscape)	de Boer 2020 (100M random promoters)	the same promoter library	Fitness label for sequences that already carry an expression label.
56	Lee 2014 (HIP-HOP fitness signatures)	Hoepfner 2014 (HIP/HOP atlas) Hillenmeyer 2008 het (FitDb HIP)	the same heterozygous and homozygous collections, compound response the same collections and readout	Two chemical-genetic matrices on one genotype axis, which is enough compound overlap to measure cross-screen reproducibility directly. A third independent screen of the same design.
57	Nguyen Ba 2022 (barcoded bulk QTL, 100k segregants)	Bloom 2019 (16-parent cross)	segregant genotype class, panel size	One cross at 100,000 progeny against sixteen founders at 14,000, which separates panel size from allelic diversity.
58	Galardini 2019 (four backgrounds x 38 conditions)	Costanzo 2021 (condition-SGA) Hale 2024 (CRISPRi x natural variation)	deletion strains across conditions perturbation held fixed, background varied	The same conditional fitness question asked in four backgrounds rather than one. Deletion against knockdown for the background-dependence question.
59	Parsons 2006 (bioactive-compound profiling)	Hoepfner 2014 (HIP/HOP atlas)	compound response on deletion strains	Compound overlap across screens run on different platforms.
60	Dutta 2026 (barcoded natural-isolate chemical response)	Caudal 2024 (pan-transcriptome) Hoepfner 2014 (HIP/HOP atlas)	the sequenced 1,011-isolate panel compound response, isolates against deletions	Chemical response and transcriptome on one isolate set. Whether compound sensitivity found by gene deletion predicts sensitivity across natural genetic backgrounds.

Table 5, continued

#	Candidate	Partner	Join key	What the join yields
61	Li 2016 (tRNA fitness landscape)	Domingo 2018 (tRNA double-mutant landscape)	the same tRNA gene, single against double mutants	Whether a single-mutant landscape predicts the double-mutant one, which is the within-gene form of the epistasis question.
62	Peter 2018 (1,011 isolate genomes + phenome)	Caudal 2024 (pan-transcriptome)	the same 1,011 isolates	The reference genome set that every isolate-WGS row in this table resolves its genotypes against.
69	Liu 2021 (tryptophan / isobutanol tolerance)	Kuroda 2019 (isobutanol-specific tolerance)	isobutanol tolerance on the same deletion collection	Same genotypes, same phenotype, different readout, which is a method-effect control no single screen provides.
70	Kuroda 2019 (isobutanol-specific tolerance)	Lopez 2024 (isobutanol screen, private)	isobutanol on the deletion collection	Tolerance against production for one product, on one genotype axis.
71	Turco 2023 (Yeast Phenome)	Hoepfner 2014 (HIP/HOP atlas)	aggregated growth screens against a primary screen	De-duplication, which is the precondition for knowing what the aggregate adds.
105	Trikka 2015 (carotenogenic heterozygous screen)	Ozaydin 2013 (beta-carotene screen)	carotenoid readout, heterozygous against homozygous deletion	Halved dosage against full deletion, which finds flux control points a null removes entirely.

6 The classes ✗

Six classes were requested. Three more were added because the expansion turned up coherent groups that did not fit any of them: regulatory DNA, deep mutational scans and combinatorial genomes, all covered in Sec. 7.2.

6.1 Natural isolates, and where sequence diversity crosses over ✗

Natural variation is the largest class after the backbone rows (Table 3), and the crossover the class was asked for is concentrated in three of them.

Dutta 2026 barcodes 520 natural isolates and profiles them against more than 600 compounds. Those isolates are drawn from the sequenced 1,011 panel, which is the same panel behind the already-built Caudal 2024 pan-transcriptome and behind Muenzner 2024. One genotype axis therefore carries a genome, a transcriptome, a proteome, and now a chemogenomic response surface, which is the join that makes a transfer question askable at all.

Hale 2024 is the sharpest instrument for that question. It crosses 8,046 CRISPRi guides over 1,721 genes into 169 sequenced segregants, so it measures directly how much a perturbation's effect depends on the background it is made in. Galardini 2019 asks the same question with deletions instead of knockdowns, building 3,786 knockouts independently in four sequenced backgrounds across 38 conditions, and reports that about 18.5 percent of deletion phenotypes change between backgrounds.

That 18.5 percent is the reason this class matters beyond breadth. Nearly every fitness and metabolite dataset in the built set is S288C, and a model trained only there has no way to know which of its predictions survive a change of chassis.

6.2 Tolerance on recalcitrant biomass, and product tolerance ✗

Lignocellulosic inhibitor coverage in the built set currently runs through broad compound panels rather than dedicated screens, so three rows close named gaps: Pereira 2014 screens the deletion collection against real industrial wheat-straw hydrolysate rather than reconstituted single toxins, Endo 2008 covers vanillin, and Xiao 2014 covers furfural through a knockdown axis that reaches essential genes a deletion collection cannot.

Product tolerance is answered mainly on isobutanol, and the earlier assumption that no genome-wide isobutanol screen exists is wrong. Kuroda 2019 screens the deletion collection for isobutanol-specific fitness against an ethanol comparator, so the alcohol-general stress response subtracts out; its *GLN3* hit raised production 4.9-fold, which is a rare case of a tolerance screen translating into a titer. Liu 2021 is an independent full-collection screen of the same phenotype with a different readout, giving a same-genotype, same-phenotype, different-method replicate. Crook 2016 adds a dose-graded knockdown axis over isobutanol and 1-butanol together, and Peeters 2021 supplies isobutanol titer as a mapped trait across 1,125 sequenced segregants.

Van Leeuwen 2024 runs acetic, butyric and octanoic acid on one strain panel, which is what lets shared and acid-specific mechanisms separate inside a single study instead of across screens that cannot be compared.

Acetic acid, production as well as tolerance. Acetic acid is now covered on both sides, which is unusual. On tolerance: Mira 2010 screens the full deletion collection at controlled pH and reports about 650 determinants, Mukherjee 2021 covers the essential genes by CRISPRi, Sousa 2013 covers gain-of-function by overexpression, and the already-built Mormino 2022 covers it by biosensor. Four perturbation types, one phenotype.

Production is thinner but real. Peeters 2021 measures acetic acid titer as one of its 18 mapped traits across 1,125 sequenced segregants, which is the only quantitative genotype-to-acetate-titer map at scale. Chang 2012 screens the whole deletion collection for altered organic-acid output including acetate, though the readout is an ordinal halo rather than a titer. No genome-wide screen was found whose primary label is a quantitative acetate titer per strain, so the pairing that exists is Peeters for production against the four tolerance screens, not a matched production screen against them.

6.3 CRISPR library screens ✗

The already-built Lian 2019 MAGIC screen reads out furfural tolerance by growth selection; Dong 2021 is the sibling that swaps the growth selection for an S-adenosyl-methionine biosensor and sorts by fluorescence, so the label becomes product concentration rather than fitness. That is the pattern worth copying for every precursor in Sec. 6.6, and it is the biosensor MAGIC screen the scoping request asked after.

Bao 2018 and Roy 2018 both move the genotype axis below the ORF, to designed single-nucleotide and small-indel variants. Both are designed rather than per-strain-verified, so both would enter as designed genotypes with a verification status. McGlincy 2021 and Momen-Roknabadi 2020 are two independently designed CRISPRi libraries; ingesting both makes library-to-library transfer a cheap and decisive test of whether a model has learned gene function or guide-design idiosyncrasy.

Eight of this class sit in wave 1, which is the largest change the banding makes. Knockdown is the modality a pooled per-cell campaign can actually deliver: it needs no strain construction per gene, it is inducible so the perturbation can be timed, and it reaches the roughly one gene in five that a deletion collection cannot hold at all. That last point is Mukherjee 2021's, which covers the essential genes under acetic acid by CRISPRi. Two rows go further than one knockdown per cell: Jaffe 2019 measures an interaction between two simultaneous knockdowns, and Crook 2016 grades the knockdown by dose rather than applying it as a switch, which is the difference between a point and a curve per gene. Hale 2024 supplies the third dimension by building 8,046 guides into 169 sequenced backgrounds.

6.4 Expression across conditions ✗

This class is genuinely thin in a way worth stating rather than papering over. Nadal-Ribelles 2025 is the largest genotype-indexed expression dataset in yeast at roughly 3,500 knockouts under osmotic stress, and it is already built. The candidates here supply what it lacks rather than more of it: Gasch 2000 is the canonical wild-type stress-response baseline that any genotype-indexed transcriptome has to be read against, N'Guessan 2025 profiles roughly 4,500 sequenced segregants by single-cell RNA-seq, and Leutert 2023 covers 101 conditions at the phosphosite level, where enzyme regulation acts faster than transcription.

All 42 items in the microbe-perturb-seq topic collection were read this pass, and the earlier reading that they carry little data over large genotype numbers holds. Seven have a real genotype axis in a single-cell host: Kemmeren 2014, Sameith 2015, Lian 2019 and Nadal-Ribelles 2025 are built, Jackson 2020 and Boocock 2025 are rows 3 and 1, and the seventh is a preprint with no deposited accession (Sec. 7.1). The other 35 divide into platform papers, mammalian Perturb-seq, bacterial condition atlases and single-cell statistics, all recorded in Table 7. Two small yeast platform rows were added from the sweep, Jariani 2020 and Urbonaite 2021, for the per-cell variance structure they measure rather than for a genotype axis they do not have.

Two rows already on the list moved a long way up it for the same reason. Hu 2007 profiles 269 transcription-factor deletions and Lenstra 2011 profiles chromatin regulators, so between them and the built Kemmeren 2014 the bulk deletion-expression coverage of both halves of transcriptional control is reachable; they were ranked 33 and 89 on scale alone and are 7 and 18 once the quadrant they sit in is what orders the list. Airolidi 2016, new from the Jackson 2020 citation list, supplies the wild-type nitrogen-limited baseline for the media that atlas works in.

Two more deletion-expression rows were added after that. Hughes 2000 is the original compendium, 300 mutations and chemical treatments on two-color arrays, and Sun 2013 decomposes the transcript level of 46 deletion strains into the synthesis and decay rates that produce it. With Hu 2007 they make cross-laboratory replication of the built Kemmeren 2014 testable in three independent directions, and Sun 2013 adds the rate layer none of the others has.

6.5 Transcript, translation, protein and turnover ✗

Steady-state transcript and steady-state protein were both already well covered and the list had no row for what sits between them, which is where a model asked to infer protein from RNA loses its footing. Five rows close that.

Grossbach 2022 measures transcript, protein and phosphosite on one panel of 112 sequenced strains, which is the only row where the RNA-to-protein residual is observable within a genotype. Teyssonniere 2024 runs the same comparison at species scale, 942 isolate proteomes against the built Caudal 2024 transcriptomes on the same panel, and it is the published calibration for that pairing: weak correlation, and 3 percent overlap between the variants associated with protein and with transcript. Foss 2007 is the same measurement on the same cross as Grossbach fifteen years earlier and on a different platform, so it is the replicate rather than the primary.

The other two are gene-level rather than strain-level. McManus 2014 supplies translation efficiency against mRNA abundance for 5,474 genes, and Martin-Perez 2017 supplies turnover for 3,160 proteins. Those two coefficients are the reason transcript and protein disagree at all, and neither varies by strain, so they enter as a per-gene layer over every genotype rather than as a genotype axis of their own.

6.6 Metabolite and precursor coverage ✗

The precursor accounting in `notes/metabolism.central-carbon-precursors.md` sets the target: of the twelve central carbon precursors, acetyl-CoA is measured on 19 strains and succinyl-CoA and malonyl-CoA on none, so identity coverage is good and statistical power is not.

Blank 2005 is the only dataset in this table that measures flux rather than a concentration or a titer, across roughly 200 deletion mutants, and it is the only real handle on acetyl-CoA turnover under genetic perturbation. Malonyl-CoA still has no direct measurement anywhere; the nearest handles remain its dominant sinks, which the already-built Xue 2025 free fatty acids and da Silveira 2014 lipidomics supply, and Zhu 2014 extends with a lipidome across 129 kinase and phosphatase deletions.

Cooper 2010 is the highest-value row in the class for a reason unrelated to novelty. It measures the same amino-acid pools as the already-built Mulleder 2016, on an overlapping genotype axis, by capillary electrophoresis instead of mass spectrometry. Two independent platforms measuring one trait class is the cleanest available test of whether a model has learned biology or a batch effect, and nothing else in the built set offers it.

On carotenoids, Ozaydin 2013 is built. Triikka 2015 is the other high-throughput carotenoid screen: 4,700 heterozygous deletions scored by color, then carried through iterative rounds to a sclareol titer. Halving gene dosage rather than deleting outright is what finds the isoprenoid flux control points a null would kill, and no other row in this table uses that design. It is currently recorded as figure-only, so confirming that Additional file 1 carries per-strain scores is a precondition, not a formality.

6.7 Regulatory DNA, deep scans and combinatorial genomes ✗

Three new classes, all with exactly known sequences and all going deep on a locus rather than shallow across a genome. They are covered in Sec. 7.2, because their value is best stated against the Perturb-seq specification rather than as another phenotype class.

6.8 Perturbation modality and reference backbone ✗

None of these is a headline dataset and two are load-bearing. Puddu 2019 is whole-genome sequence for every strain in the deletion collection. Thirty-one rows inside the long-run cut declare an **S288C-KO** sequence basis, and without Puddu that basis is an assumption that each strain differs from the reference at exactly the intended locus. Puddu shows it frequently does not. That matters more for a pooled campaign than for an arrayed screen, because a pooled assay reads a barcode and infers a genotype, so it is row 4 rather than row 27.

Mulleder 2012 is the second. Jackson 2020 built its own prototrophic strains rather than use the deletion collection, because auxotrophic markers rule out minimal and nitrogen-limited media, and Mulleder 2012 is the prototrophic collection that removes that constraint. It is also the strain resource behind the built Mulleder 2016 amino-acid metabolome.

7 Rows that bear on a yeast Perturb-seq ✗

A Perturb-seq is defined by two dimensionalities, and a dataset earns a place here only by being high on at least one of them.

- **Input dimensionality** is the size of the perturbation space per cell: combinatorial edits, or a perturbation crossed with genetic background. A large library of single perturbations is not high-dimensional input. It is one edit per cell, sampled widely, and a model fit to it never sees two edits interact.
- **Output dimensionality** is the readout: a transcriptome or a per-cell distribution rather than a scalar fitness or titer.

That distinction removes a row this table previously carried. Momen-Roknabadi 2020 is a genome-wide CRISPRi library, which is large, but it delivers one guide per cell and reads out competitive fitness. It is low-dimensional on both axes and is no longer flagged; its value is as a second library for testing guide-design transfer, not as Perturb-seq evidence.

The axis flag and the **perturb-seq** band of Sec. 1.2 answer different questions and a row can carry one without the other. The flag asks what a dataset *measured*; the band asks what it *supplies to a campaign*. Momen-Roknabadi 2020

measures neither axis at depth and supplies a guide library, so it is banded and unflagged. Puddu 2019 measures no axis at all and supplies the genotype verification a pooled assay depends on.

Table 6: Rows bearing on a yeast Perturb-seq, grouped by axis rather than by rank. *Rank* is the row’s position in Table 1. A large library of single perturbations read out by fitness is not high-dimensional input: it is one edit per cell, sampled widely.

Rank	Dataset	What it supplies
High on BOTH axes: combinatorial or background-crossed perturbation AND a transcriptome-scale readout		
1	Boocock 2025 (single-cell eQTL mapping)	27,744 single cells drawn from 393 sequenced recombinants in two carbon sources, a median of 17 cells per segregant. High-dimensional on both axes at once, which almost nothing in yeast is, and the closest existing template for the proposed design.
5	N’Guessan 2025 (segregant scRNA-seq eQTL)	Single-cell transcriptomes indexed by recombinant natural genotype rather than by an engineered knockout. The closest existing analog to the proposed Perturb-seq design, and the only one that already pairs per-cell expression with sequenced genotypes at scale.
31	Albert 2018 (eQTL in 1,012 segregants)	Transcriptomes on a thousand sequenced recombinants, so genotype maps to a genome-wide expression vector rather than one trait. The single best existing analog to a Perturb-seq readout on a high-dimensional genotype axis.
High-dimensional perturbation space, scalar readout		
2	Hale 2024 (CRISPRi x natural variation)	Directly measures how a genetic perturbation’s effect changes with genetic background. The single best dataset for asking whether a model trained on one background transfers to another.
10	McGlinchy 2021 (genome-scale CRISPRi library)	Ten guides per gene gives a graded knockdown axis instead of a binary null, and it covers essential genes. This is the library a yeast Perturb-seq would most plausibly be built on.
21	Guo 2018 (CRISPR-Cas9 tiling of essential genes)	Tiles guides across genes rather than sampling a few per gene, so the readout resolves within-gene functional regions. A finer input axis than a gene-level library.
25	Jaffe 2019 (multiplexed CRISPR interference epistasis)	Two guides per cell, so the perturbation space is combinatorial rather than one edit at a time. This is the shape a Perturb-seq input axis should have, measured here against fitness only.
54	de Boer 2020 (100M random promoters)	The largest sequence-to-phenotype dataset in yeast by orders of magnitude, and every genotype is an exactly known 80mer. The perturbation space is continuous rather than a gene list, which is what a model of regulatory grammar needs and no deletion collection can supply.
55	Vaishnav 2022 (regulatory DNA fitness landscape)	Extends de Boer from measurement to a fitness landscape, and tests native and natural-isolate promoter variants against the random library. The bridge between synthetic sequence space and the natural variation the isolate panels carry.
57	Nguyen Ba 2022 (barcoded bulk QTL, 100k segregants)	The largest genotyped recombinant panel in yeast, each segregant barcoded and low-coverage sequenced against known parents. Two orders of magnitude more genotypes than Bloom 2013 on the same kind of cross, which is what makes small-effect and epistatic loci detectable.
61	Li 2016 (tRNA fitness landscape)	A near-complete fitness landscape of one native yeast gene. Every genotype is a known sequence and the sampling is exhaustive rather than one allele per gene, which is the opposite of what a deletion collection provides.
63	Domingo 2018 (tRNA double-mutant landscape)	Pairwise epistasis at nucleotide resolution inside one gene. The higher-order structure Costanzo and Kuzmin map between genes, measured within a gene, which is the resolution inverse design actually needs.
74	Bloom 2025 (global epistasis in a yeast cross)	Reports that many natural variants have effects that scale with background fitness, which is a specific, testable claim about how genotype effects compose. Directly relevant to whether a model trained on one background transfers.
78	Faure 2022 (doubledeepPCA)	Separates a variant’s effect on folding from its effect on binding by measuring both in the same library. That decomposition is what turns a fitness landscape into a mechanistic one.
81	Diss 2018 (Fos-Jun interaction double mutants)	Exhaustive pairwise mutation of an interaction interface, read out in yeast. The genotype is a heterologous domain rather than a yeast gene, so it enters as an engineered chassis, but the epistasis structure is the deepest in the table.
84	Puchta 2016 (snoRNA U3 landscape)	The companion landscape on a non-coding RNA, published alongside Li 2016. Two exhaustive landscapes on different molecule classes is a stronger test of a sequence model than either alone.
103	Renganaath 2020 (natural promoter-variant MPRA)	Native promoter variation measured allele by allele rather than sampled randomly, with 451 variants called causal. The row that joins the synthetic MPRA libraries to the natural-isolate panels: same readout, real alleles.

Rank	Dataset	What it supplies
109	Zhang 2022 (GCE-SCRaMbLE recombination outcomes)	Each cell carries tens of simultaneous structural edits, sequenced afterwards. The deepest per-cell perturbation in the table: a genotype here is a rearranged chromosome, not an allele, which is what inverse strain design eventually has to predict on.
Transcriptome-scale or per-cell readout, low-dimensional perturbation		
3	Jackson 2020 (TF-deletion single-cell atlas)	The closest thing to a yeast Perturb-seq that exists. A transcribed barcode in the 3' UTR of the marker cassette labels every cell with its genotype, so 72 strains, 12 genotypes at 6 independently constructed replicates, are pooled and read out per cell; every strain appears in every one of the 11 conditions. The perturbations are homozygous diploid whole-ORF deletions in a prototrophic FY4/FY5 background rather than the Giaever collection, which is what permits the minimal and nitrogen-limited media. The DNA-only barcodes of the Giaever collection are the stated reason it could not be used, and that is the design constraint any campaign here inherits.
6	Hackett 2020 (IDEA inducible-TF transcriptome time series)	Each TF is individually induced from a designed synthetic cassette and the whole transcriptome is followed over time, which is the only yeast dataset where a single-gene perturbation is crossed with a dense time axis. scYeast used it as the target for its time-series perturbation-response task (Fan 2027, Sec. 4.1.7). The instance count treats each strain-by-time-point as a record; per strain it is about 200.
7	Hu 2007 (TF deletion expression compendium)	Transcriptomes for the regulators specifically, predating and complementing the already-built Kemmeren compendium. Deleting a transcription factor is the perturbation whose transcriptome response is most interpretable.
14	Newman 2006 (protein noise, GFP library)	Measures the distribution rather than the mean, per protein, in two media. Noise is the quantity a per-cell readout recovers and a bulk one destroys, so this sets what a single-cell design should expect to see.
18	Lenstra 2011 (chromatin regulator deletion expression)	Transcriptomes for the chromatin machinery specifically, from the group behind the built Kemmeren compendium and in the same format. Chromatin regulators are the class whose deletion most reshapes the expression landscape.
20	Jariani 2020 (yeast scRNA-seq through lag phase)	A yeast-adapted 10x protocol that resolves the transcriptional states of individual cells during a carbon-source shift, so the readout is the distribution across a population that is not synchronized rather than its mean. One genotype, so the value is the protocol and the per-cell variance structure a campaign has to budget against.
22	Urbonaite 2021 (yeastDrop-Seq under drug treatment)	A yeast-optimized Drop-seq with a small chemical condition axis, including a two-compound arm. One genotype, so it enters as a platform reference: the pair of single-drug arms against their combination is a per-cell readout of a two-factor environment, which is the environmental analog of the combinatorial genetic perturbation a Perturb-seq needs.
23	Nadal-Ribelles 2019 (sensitive yeast scRNA-seq)	Establishes within-clone transcript correlation, which is the noise floor any per-strain Perturb-seq mean has to clear. Directly sizes the cells-per-guide question.
24	Gasch 2017 (single-cell stress heterogeneity)	Separates intrinsic from extrinsic expression heterogeneity under stress. Sets what fraction of Perturb-seq variance is recoverable signal rather than cell-state noise.
26	Jackson 2023 (wild-type scRNA-seq time course for RNA kinetics)	The second scYeast pre-training set: 173,361 cells from one wild-type strain sampled continuously through a time course without metabolic labeling, so the strain-by-condition count is one and the value is the per-cell distribution of transcriptional states, not a genotype axis. A reference backbone for any single-cell expression head, and the companion of Jackson 2020 on the same platform.
27	Brettner 2024 (ultra-high-throughput yeast scRNA-seq)	The platform the costing model in experiments/024 is built on. Not a dataset to ingest; the reason it is here is that a yeast Perturb-seq proposal has to name its chemistry.
79	Chong 2015 / CYCLOPs (single-cell proteome dynamics)	Per-cell rather than per-strain-mean labels, over 20 million cells. The distributional readout a Perturb-seq design has to be scored against.
112	Smith 2008 (gene-environment interaction eQTL)	Transcriptomes on the same segregants in two carbon sources, so an expression QTL can be shown to be condition-dependent. Small, and it is the design Boocock 2025 scales up.
114	Breker 2013 (GFP library under stress)	The GFP collection re-imaged under stress rather than at baseline, so relocalization is measured as a response. Feeds LoQAtE, and adds the condition axis Huh 2003 lacks.
115	Tkach 2012 (protein relocalization under DNA damage)	The GFP collection imaged under damage, giving a second condition axis on the imaging layer alongside Breker's stress panel. Relocalization is a response no abundance measurement captures.
117	Denervaud 2013 (microfluidic GFP dynamics)	Single-cell protein dynamics under controlled environmental switches, rather than endpoint snapshots. The temporal axis every other imaging row lacks.

Rank	Dataset	What it supplies
122	Yvert 2013 (single-cell trait variation across isolates)	CalMorph morphology on natural isolates rather than deletion strains, in the same feature space as the already-built Ohya and Ohnuki datasets. Joins the morphology layer to the natural-variation axis without harmonization.
151	Levy 2008 (morphology and phenotypic capacitance)	Scores how much a deletion increases cell-to-cell variability rather than shifting the mean, identifying capacitor genes. Variance as the phenotype is a readout only per-cell data supports, which is the argument the Perturb-seq proposal rests on.

7.1 The quadrant that matters is nearly empty ✗

Of 168 candidates, 15 are high on the input axis, 18 on the output axis, and **three** on both: Boocock 2025, Albert 2018 and N’Guessan 2025. All three reach the both-axes quadrant the same way, by crossing recombinant natural genotypes with a transcriptome, and none of them perturbs a gene deliberately.

So the gap is specific rather than general. Yeast has deep genotype-to-fitness data and deep wild-type transcriptome data, and where the two meet it is through natural variation, not through designed perturbation. The largest genotype-indexed transcriptome with designed perturbations is Kemmeren 2014, already built, and it is 1,484 single deletions read out one gene at a time, which is exactly the shallow case: one knockout, one profile, no interaction measured. Sameith 2015, also built, adds double mutants but only 72 of them.

That is the argument for running the experiment, and it is also the specification. To land in the empty quadrant a design needs a combinatorial perturbation per cell, not a larger single-perturbation library, and a transcriptome readout rather than a growth score.

The full collection was read, and it does not close the gap. All 42 items in the microbe-perturb-seq topic collection were checked this pass. Seven carry a genotype axis in a single-cell host and four of those are built. The one that reaches both axes on purpose is Brandner 2025, which perturbs 52 transcription factors with 118 guides in *Escherichia coli* and *Pseudomonas putida*, includes multi-guide combinations specifically to resolve genetic interactions, and reads out a single-cell transcriptome. *P. putida* is a single-cell bioproduction host and so in scope, but the paper is a preprint and no deposited accession was found, so it is excluded rather than ranked and should be revisited on publication. It is the closest published design template for the campaign.

Three mammalian rows set the format rather than supply data. Replogle 2022 perturbs every expressed gene across more than 2.5 million cells; Zhu 2026 crosses a genome-scale perturbation with stimulation time points across about 22 million cells, which is the perturbation-by-context tensor a yeast campaign is the analog of; and Yao 2024 pools random multi-perturbation composites over 598 genes and decompresses them, which is the one published route to second-order interactions at a tractable cell count. Compression is therefore the lever that makes combinatorial input affordable, and it is worth costing before the library is designed rather than after.

7.2 Depth rather than breadth ✗

Three classes were added to this pass specifically because they go deep on one locus instead of shallow across many, and every one of them carries an exactly known sequence per genotype.

Deep mutational scans, 9 rows. Li 2016 and Puchta 2016 are near-complete fitness landscapes of a single native yeast gene at roughly 65,000 and 60,000 variants. Domingo 2018 and Diss 2018 go further and measure pairwise mutations, so epistasis is resolved inside a gene rather than between genes. Faure 2022 separates a variant’s effect on folding from its effect on binding by measuring both in the same library, which is what turns a landscape mechanistic. These supply the input dimensionality the fitness screens lack.

Regulatory DNA, 8 rows. de Boer 2020 is above 100 million exactly known promoter sequences with a measured expression level each, which is the largest sequence-to-phenotype dataset in yeast by orders of magnitude. Vaishnav 2022 extends it to a fitness landscape and tests natural-isolate promoter variants against the random library, and Renganaath 2020 measures 5,832 real promoter alleles from natural variation. This is the answer to whether promoter strength has been measured at scale: it has, both for random sequence and for native alleles, and the two are on the same readout.

Combinatorial genomes, 3 rows. SCRaMbLE genotypes carry tens of simultaneous structural edits and are sequenced afterwards. A genotype there is a rearranged chromosome rather than an allele, which is the deepest per-cell perturbation available and the hardest thing a genotype-to-phenotype model will eventually be asked to predict on.

7.3 The synergy question is answerable, and not yet answered ✗

Whether one dataset predicts another decides how much new data is worth collecting, and the list contains the pairs to test it before any is collected. All 101 are in Table 5; six kinds recur, and the cheapest to run is the last.

- **Same phenotype, different platform.** Cooper 2010 against the built Mulleder 2016, amino acids by capillary electrophoresis against mass spectrometry on an overlapping genotype axis.
- **Same phenotype, different method.** Liu 2021 against Kuroda 2019, isobutanol tolerance on the same collection by colony imaging against competitive fitness.
- **Same perturbation, different background.** Hale 2024, Galardini 2019 and She 2018, all of which vary the background while holding the perturbation fixed.
- **Different library, same modality.** McGlincy 2021 against Momen-Roknabadi 2020, two CRISPRi libraries with different vectors and design rules.
- **Cheap layer against expensive layer.** Caudal 2024 transcriptomes against Muenzner 2024 proteomes on overlapping isolates, and Skelly 2013, which measures both on the same 22 strains rather than joining across studies.
- **Synthetic sequence against natural allele.** de Boer 2020 and Vaishnav 2022 against Renganaath 2020: whether a model trained on random promoters predicts the effect of a real one.

None of these transfers has been measured. Each is a hypothesis about what a model could learn, and each becomes a run once both halves are ingested.

8 What was dropped ✗

Table 7: Considered and dropped. *no-sequence* is the hard gate: without a route to the strain’s genomic content there is no genotype to map a phenotype from.

Dataset or group	Rule	Reason
Snoek 2015 robot-assisted genome shuffling (ethanol tolerance)	no-sequence	Shuffled progeny are recombinants of unrecorded parental segments with no per-strain sequencing, so the genotype cannot be reconstructed.
Adaptive laboratory evolution / EMS mutagenesis tolerance panels (e.g. farnesol-FPP selection)	no-sequence	Mutations are unmapped, so a strain’s genome is unknown even in principle without whole-genome resequencing the study did not do.
Avalos lab isobutanol-biosensor deletion screen	not-a-dataset	Described only in a PhD thesis and a DOE report. Track for publication; not citable as a dataset.
Kitamoto/Kaneko ester and higher-alcohol screen	not-a-dataset	Citation could not be pinned to a specific paper. A placeholder, not a located dataset.
Fuhrer/Zamboni flow-injection metabolomics	off-species	The validated genome-scale application is in <i>E. coli</i> . No yeast deletion-collection version was located.
Anglada-Giroto 2022 (CRISPRi + metabolomics)	off-species	352 CRISPRi genes against 1,342 drug-induced metabolic changes, but in <i>E. coli</i> . Verified from the mirrored PDF.
Wildenhain 2016 (Sci Data chemical-genetic matrix)	already-built	A data descriptor, not a second experiment. Its own Data Citation 1 is NCBI PubChem BioAssay AID 1159580, which is exactly what torchcell/datasets/scerevisiae/wildenhain2015.py already ingests: 242 strains, 5,518 compounds, 492,126 interaction tests.
O’Duibhir 2014 expression compendium	already-built	The paper’s expression data IS the Kemmeren 2014 compendium plus PCA transforms, already built, as oduibhir2014.py states; only its growth-rate readout was new and that is built too. Caught by check_candidate_overlap.py rather than by inspection, which is the first time the script found a duplicate before a person did.
Ozturk 2022 proteome-effects screen	off-species	3,308 deletions in <i>S. pombe</i> . Out of species scope.
Chen 2024 <i>K. phaffii</i> CRISPR fitness screen; Y1000+ multi-species panel	off-species	Not <i>S. cerevisiae</i> . Revisit if the generalization axis widens to other hosts.
iIsor850 genome-scale model (Issatchenkia orientalis)	not-a-dataset	A reconstruction, not an experiment, and off species. No genome-wide <i>I. orientalis</i> phenotype screen was located; the CABBI knockout library is unpublished.

Dataset or group	Rule	Reason
CABBI 13C-MFA kinetic-model repository	not-a-dataset	16 knockout strains total, and the released fluxes are K-FIT model fits recapitulating 75-77 percent of measurements, not raw measurement. Blank 2005 supersedes it on both counts.
YEASTRACT+ (Teixeira 2023; Monteiro 2020)	not-a-dataset	A literature-curated knowledge base of 304,547 documented TF-target associations, each tagged with the evidence type (DNA binding or expression) and with the environmental condition in which it was observed, so a condition-scoped regulatory graph is recoverable from it. It is a prior over the genome, not a genotype-by-environment measurement, and it belongs in the graph layer beside GO and the metabolic model rather than in this table. Both papers are mirrored; the flat files are shared on request by the maintainers and are not yet held, and scYeast (Fan 2027) built its attention prior from the documented-targets subset (about 170,000 pairs).
Sc2.0 / Sc3.0 synthetic-genome design	not-a-dataset	Genome-engineering programs, not gene-indexed perturbation-phenotype data.
Lee 2014 / Hoepfner 2014 / Vanacloig-Pedros 2022 / Messner 2023 / Mulleder 2016 / Lian 2019 / Mormino 2022 / Nadal-Ribelles 2025 and 12 others	already-built	Named as top candidates by the 2026-07 triage pass and BUILT since. Only Lee 2014 remains outstanding and is kept as row 4.
Bloom 2019 (16-parent cross)	already-built	Ranked 12 in the previous pass and built since as the 50th supported dataset: 13,950 sequenced segregants over 38 traits, L0-L4 verified. Its place in this pass is as a join partner, not a candidate.
TF-target prior networks used by Jackson 2020 (YEASTRACT, ATAC-motif of Castro 2019, Bussemaker affinity of Ward 2008, Tchourine 2018 gold standard)	not-a-dataset	Four regulator-by-gene matrices: 11,486 unsigned YEASTRACT edges over 3,912 genes and 152 TFs, 71,865 signed ATAC-motif edges over 5,551 genes and 138 TFs, a dense 6,516 by 123 affinity matrix, and the 1,403-edge signed gold standard. Each is a prior over the genome with no genotype and no environment, so each belongs in the graph layer beside GO and YEASTRACT+, not in a genotype-by-environment table. All four ship as TSVs in the eLife Source code 1 archive.
Tchourine 2018 bulk expression compendium (2,577 observations)	not-a-dataset	The benchmark Jackson 2020 compares its single-cell network against. It is a re-aggregation of public bulk series, and the strain behind an observation lives in each source series' metadata rather than in the released matrix, so admitting it is a de-duplication and re-curation task against Kemmeren 2014, Gasch 2000 and Brauer 2008 rather than one loader. Revisit if the per-sample genotype table is located.
Albert 2014 (X-pQTL, single-cell protein abundance in large populations)	not-a-dataset	Albert FW, Treusch S, Shockley AH, Bloom JS, Kruglyak L. Nature 2014;506:494-497. Considered as the BY x RM segregant proteome and it is not one: protein level is read from a green fluorescent protein fusion for one gene at a time in a large unsequenced pool, and the mapping comes from sorting that pool rather than from genotyped individuals. There is no strain-by-protein matrix to store, so it cannot enter as a proteome row. The segregant proteome rows here are Grossbach 2022 and Foss 2007.
Scholes 2019 bulk RNA-seq control (GEO GSE135430)	not-a-dataset	Jackson 2020's external bulk reference point. One wild-type BY4741 genotype in one condition, and the study's own variable is the RNA isolation protocol, so there is neither a genotype nor an environment axis to record.
Brandner 2025 (mapSPLiT, CRISPRa and CRISPRi single-cell transcriptomes in E. coli and P. putida)	not-a-dataset	The only microbial dataset found that reaches both Perturb-seq axes on purpose: 52 transcription factors targeted by 118 guides, pooled, with multi-guide combinations included to resolve genetic interactions, read out as a single-cell transcriptome. Pseudomonas putida is a single-cell bioproduction host and so is in scope for the generalization axis. It is a preprint and no deposited accession was found, so there is no scriptable data route and it is excluded rather than ranked. It is the closest existing design template for the campaign and should be revisited on publication.

Dataset or group	Rule	Reason
Mammalian Perturb-seq (Replogle 2022, Zhu 2026, Dixit 2016, Datlinger 2017, Yao 2024)	off-species	The format references, and off species. Replogle perturbs every expressed gene by CRISPR interference across more than 2.5 million cells; Zhu 2026 crosses a genome-scale perturbation with several stimulation time points across about 22 million cells, which is the perturbation-by-context tensor a yeast campaign is an analog of; Yao 2024 pools random multi-perturbation composites over 598 genes and decompresses them, which is the one published route to second-order genetic interactions at a tractable cell count. None supplies yeast data; all four inform the design.
Bacterial single-cell transcriptomics (Wang 2023 M3-seq, Ma 2023 BacDrop, McNulty 2023 ProBac-seq, Kuchina 2021 microSPLiT)	off-species	Off species, and the axis is environmental rather than genetic: hundreds of thousands of cells over stress, antibiotic and growth-stage conditions in isogenic backgrounds. No perturbation-by-readout matrix to ingest.
Fulcher 2024 (nanoSPLITS), Taniguchi 2010, Baronas 2026, Leonaviciene 2023 and 2020, Datlinger 2021, Macosko 2015, Rosenberg 2018	off-species	Platform and paired-modality papers, all off species. nanoSPLITS measures a transcriptome and a mass-spectrometry proteome from the same single cell, which is the precedent for inferring an expensive layer from a cheap one on paired anchors; Taniguchi 2010 is the reference for protein and mRNA copy-number noise in a microbial host. Neither releases a perturbation matrix.
Reviews, protocols and single-cell statistics in the collection (Nadal-Ribelles 2024, Sun 2023, Larson 2013, Zun 2023, Gaisser 2024, Squair 2021, Zhang 2020, Svensson 2020, Grun 2014, Robinson 2008, McCarthy 2012, Hart 2013, Wang 2026)	not-a-dataset	Thirteen items with no perturbation data of their own. They are the sizing and false-discovery discipline for the campaign, not rows: pseudobulk replicate variation, sequencing depth per cell, dispersion estimation, and whether droplet counts are zero-inflated.

The **already-built** row in that table is the largest single correction this pass makes. A July 2026 triage pass named a top ten for immediate ingestion; nine of those ten have since been built, along with roughly a dozen other rows it ranked highly. Only Lee 2014 remains outstanding, and it is row 56 here. Bloom 2019 joined them since the previous pass and is the 50th supported dataset. Any list that still treats that top ten as outstanding work is stale.

8.1 Four rows were not what their names said ✕

Checking a name against the built list is not sufficient. A dataset can carry a loader, a failed retrieval attempt, or an already-ingested deposit under a different paper’s name. Three rows did one of those each, and a fourth was caught by the script in Sec. 8.2: O’Duibhir 2014’s expression data is the Kemmeren compendium plus PCA transforms, already built, as that loader’s own docstring states. It is the first duplicate here found mechanically rather than by a reader, which is the point of the check.

Wildenhain 2016 was already built, under the 2015 name. It is a data descriptor rather than a second experiment, and its own Data Citation 1 is NCBI PubChem BioAssay AID 1159580, which is exactly the accession `torchcell/datasets/scerevisiae/wildenhain2015.py` already ingests. The identity is not a judgment call: the 2016 paper reports 242 sentinel strains, 5,518 compounds and 492,126 interaction tests, and the loader’s own docstring records 242 systematic ORFs and 492,126 datapoints from that AID. The row is removed.

That leaves a citation-fidelity defect in the built dataset worth fixing separately. The data it ingests is the *extended* matrix described by the 2016 release, but it is filed under the 2015 paper, which reports a strictly smaller experiment: 195 sentinels and 4,915 compounds. The loader already half-flags this, noting that the released data covers 242 ORFs rather than the 195-strain panel it quotes. Under the sourcing rule that a value carries the citation it was actually read from, the released matrix should cite the 2016 descriptor, with 2015 as the original subset.

Lee 2014, row 56, is blocked rather than available. WS15 already attempted this ingestion. The roadmap records it as awaiting author matrices, which means the Science supplementary tables and the Nislow and Giaever lab portal did not yield per-strain values. Its rank is still earned: 3,356 compounds against the full heterozygous and homozygous collections really is the largest chemical-genetic matrix in existence, and it really is on the YKO genotype axis the built set already uses. What changes is the nature of the work. Row 56 is an author request, not a loader, and the 1.3×10^7 instances it contributes to the totals in Sec. 3 should not be counted as reachable until that request succeeds.

Turco 2023, row 71, already has a loader. `torchcell/datasets/scerevisiae/yeastphenome.py` consumes the curated YeastPhenome release and retains 49 growth screens, so this row is in flight rather than net new. It is not in

the built set, which is why it appears here at all. Two consequences follow. Its de-duplication problem is partly solved already, since the loader excludes the primaries built directly, along with heterozygous-diploid and expression screens. And Lee 2014 is not among its 49 screens, so ingesting YeastPhenome does not backfill row 56.

8.2 Overlap is now checked by script, and the check is not yet strong ✗

Three rows turned out to be already held and none was caught by inspection, so the check is now `experiments/database/scripts/check_candidate_overlap.py`, run against the candidate JSON. It applies three keys of increasing strength: the candidate DOI cited in a built loader, an accession token shared with one, and the candidate PMID appearing in an aggregator's source-PMID column.

It currently reports one overlap, Turco 2023, which Sec. 8.1 already records as a loader in flight; a second, O'Duibhir 2014's expression compendium, was found and removed during this pass. It reports no overlap with SynLethDB across 3,307 source PMIDs spanning both the lethality and rescue layers. That covers the question of whether the genetic-interaction rows are already re-served: Sharifpoor 2012, Decourty 2021 and Braberg 2020 are all absent. For Sharifpoor the absence is structural rather than a curation gap, since SynLethDB models synthetic lethality and synthetic rescue, both two-loss-of-function events, and synthetic dosage lethality pairs an overexpression with a deletion.

The honest limit is coverage, and the expansion made it worse in relative terms. The script resolves a PMID for 56 of 168 rows and finds a parseable accession in 9. Accession is the strongest key and the one that would have caught Wildenhain 2016, so it is running on about one row in seventeen. Recording each row's deposit identifier is the single highest-value edit available to this list, and it now matters more than it did at 73 rows. It is also the check that would have caught Bloom 2019 leaving the list a pass earlier than a reader did.

8.3 The expansion tranche is less checked than the first 73 ✗

Rows carry a confidence value in the generated JSON. *sourced* means the figures trace to something read this session or to the sourced triage note, and covers 75 rows. *recall* means the row was written from domain knowledge, and covers 93. For a recall row the dataset is real and the description is sound, but the citation details, the exact counts and the accession all need confirming before a loader is written, and a few carry an explicit warning in the accession column where a collision or a species question is open.

The two tranches are not interleaved by accident: ranking put several recall rows at the very top, because de Boer 2020 and Boocock 2025 genuinely are the largest things here. That is the right ranking and the wrong reading if the confidence flag is ignored. Treat the top of the table as a claim to verify first, not a queue to start work from.

Banding does not change that and in one place sharpens it. Hackett 2020 at rank 6, Hu 2007 at 7, Hughes 2000 at 17, Lenstra 2011 at 18 and Sun 2013 at 19 are all *recall* rows, so five of the first twenty carry accessions that have not been confirmed against a deposit.

The ten rows added over the last two passes are a tranche of their own and are weaker again. Their citations, venues and headline counts were verified against PubMed, which is why the author lists and page ranges can be relied on. Their deposits were not: no accession among them was fetched, and none of the ten papers is in the mirror, so none has been read beyond its abstract. Foss 2007 is the extreme case and is recorded as such, with neither a segregant count nor a proteome depth. Teyssonniere 2024 carries a second warning that is not about confidence: it shares authors and an isolate panel with Muenzner 2024, and whether the two proteomes are independent acquisitions has not been established, so building both without settling that would risk ingesting one measurement twice.

9 Before a loader is written ✗

Every figure in Table 1 is an axis count, not a verified record count, and several are weaker than that. Three qualifications apply, in decreasing order of how much damage they would do if ignored.

Instance counts marked † are products, not counts. A screen reported as 4,800 strains against 3 conditions is entered as 14,400 instances. Real screens drop strains per condition, so each such figure is an upper bound. The ones marked ‡ are weaker still: order-of-magnitude estimates for libraries whose exact size was not confirmed this pass. Neither should be quoted as a dataset size without opening the source.

Phenotype dimensionality is estimated where no fixed panel is published. Ranking uses instances times dimensionality, and dimensionality is exact for a stated panel such as Ambroset 2014 at 74 metabolites, but estimated

for open-ended ones. Proteome rows are entered at 2,000 proteins per strain and transcriptome rows at 6,000 genes. Those estimates move rank order, so a row whose position looks surprising should have its dimensionality checked first.

Accessions marked unconfirmed were not fetched. The accession column separates locations verified live this pass from those taken from a paper’s data-availability statement. Confirmed live: GEO GSE123118 with its companion GitHub release for Galardini 2019, SRA PRJNA986287 for Hale 2024, PRIDE PXD048219 for Muenzner 2024, Dryad for Chica 2026, ChemGRID for Wildenhain 2016, and BioProject PRJNA379146 for Cubillos 2017. Everything else is a claim from a publication, and the pattern is well established that a data-availability statement and a reachable file are different things.

Four rows carry a risk specific enough to name. Triikka 2015 is recorded as figure-only, so whether Additional file 1 holds per-strain scores decides whether Triikka is ingestible at all. Chang 2012 and the Gurvitz fatty-acid screen are PDF-table-only and need manual transcription. Sugiyama is a non-peer-reviewed technical report and has the lowest provenance confidence of anything listed; it belongs in the reserve as a source of candidate genes rather than as training data.

Two rows are not single datasets. Turco 2023 is a meta-aggregation and Zackrisson 2016 is a platform. Neither reduces to one loader: Turco needs a de-duplication pass against the rest of this table before its net-new content is known, and Scan-o-matic needs a specific released condition panel chosen before it becomes a row rather than a plan.